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(54) **TRUNCATED RECOMBINANT L1 PROTEIN
OF HUMAN PAPILLOMAVIRUS**

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(57)

ABSTRACT

The disclosure relates to a truncated recombinant L1 protein of Human papillomavirus (HPV), a recombinant polynucleotide encoding the protein, a recombinant vector comprising the polynucleotide, an isolated cell comprising the recombinant vector, a composition comprising the protein, and a method for producing the protein. The disclosure also relates to an HPV virus-like particle (VLP) comprising the protein, an immunogenic composition comprising the protein or the VLP, and a method for inducing an immune response to HPV in an individual.

Specification includes a Sequence Listing.

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(60) Provisional application No. 63/563,554, filed on Mar. 11, 2024.

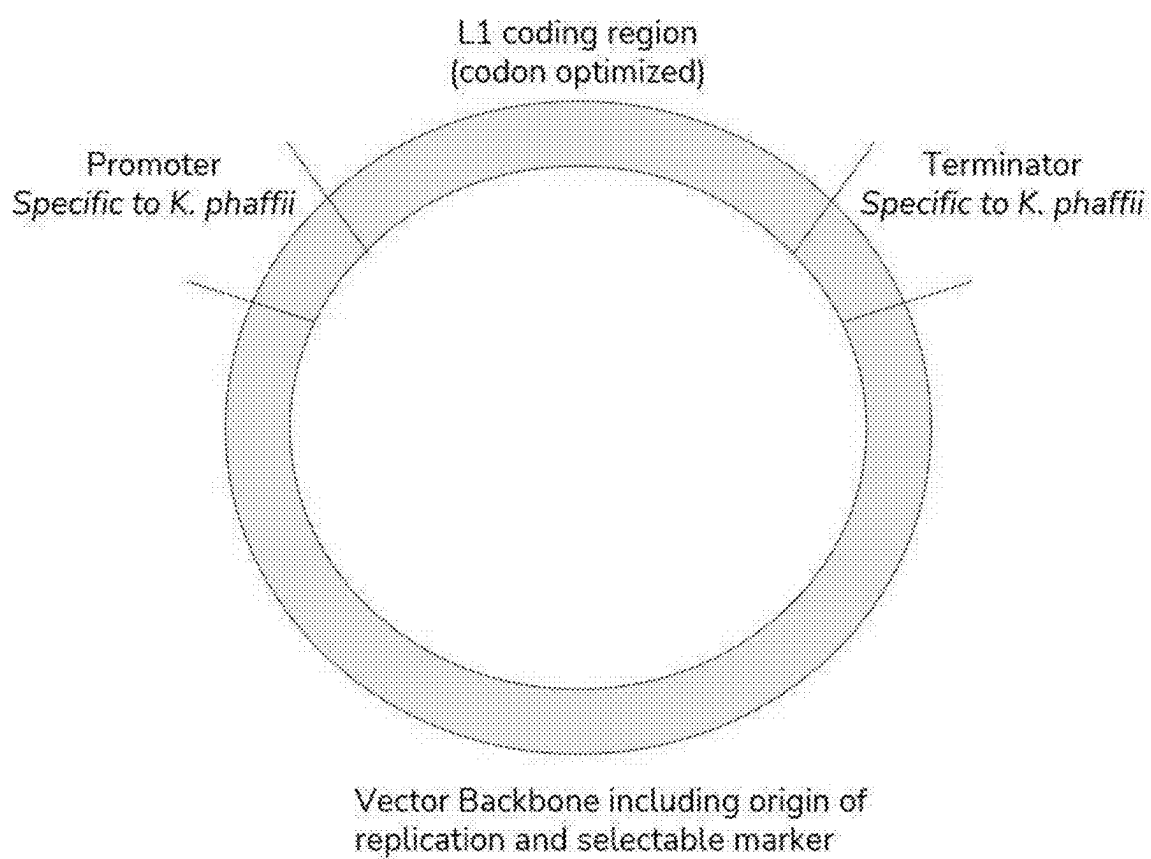
Publication Classification

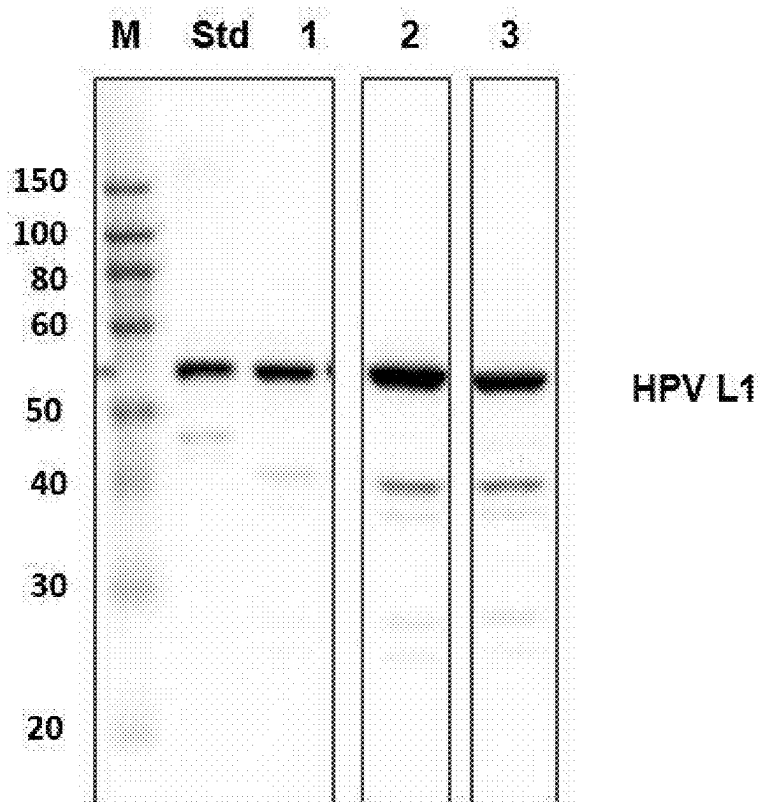
(51) **Int. Cl.**

A61K 39/12 (2006.01)

A61K 39/00 (2006.01)

A61P 31/16 (2006.01)

**Fig. 1**



M: Marker

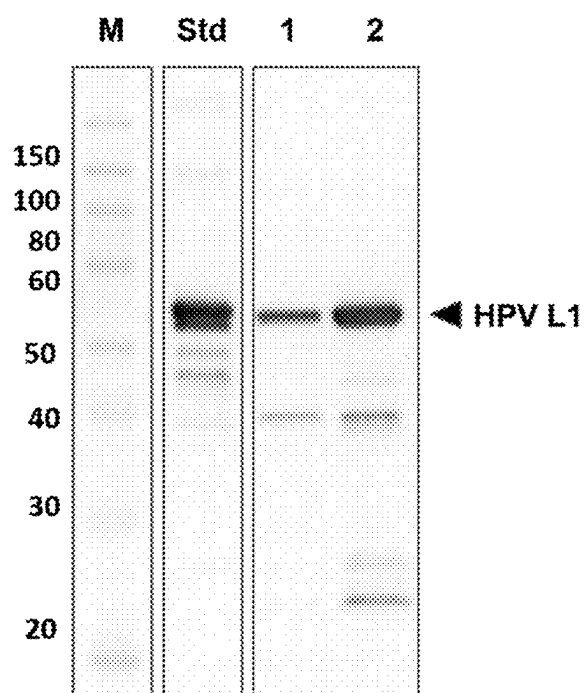
Std: Purified protein standard (HPV 6 L1)

1: *K. phaffii* cell lysate producing HPV 6 L1 original sequence

2: *K. phaffii* cell lysate producing HPV 6 L1 Δ 2-3 AAs @ n-term

3: *K. phaffii* cell lysate producing HPV 6 L1 Δ 2-5 AAs @ n-term

Fig. 2



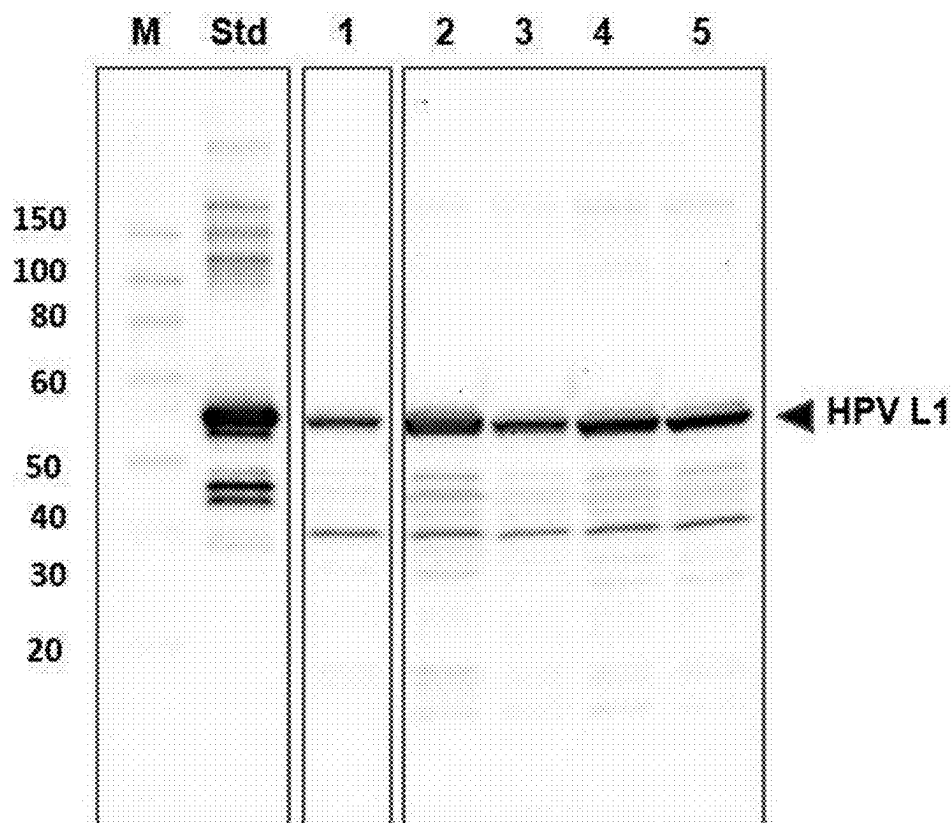
M: Marker

Std: Purified protein standard (HPV 11 L1)

1: *K. phaffii* cell lysate producing HPV 11 L1 original sequence

2: *K. phaffii* cell lysate producing HPV 11 L1 Δ 2-3 AAs @ n-term

FIG. 3



M: Marker

Std: Purified protein standard (HPV 16 L1)

1: *K. phaffii* cell lysate producing HPV 16 L1 original sequence

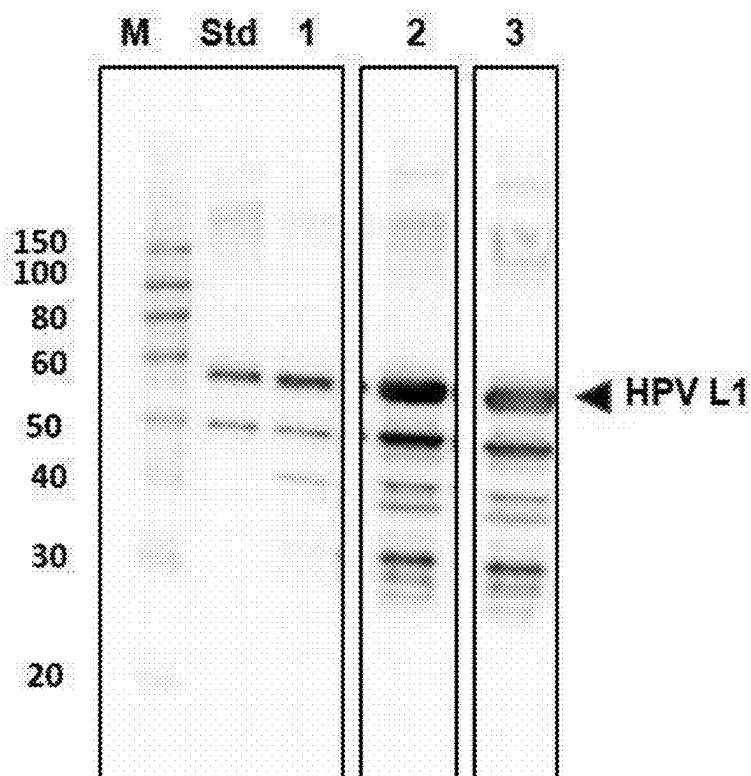
2: *K. phaffii* cell lysate producing HPV 16 L1 Δ 2-5 AAs @ n-term

3: *K. phaffii* cell lysate producing HPV 16 L1 Δ 2-10 AAs @ n-term

4: *K. phaffii* cell lysate producing HPV 16 L1 Δ 2-15 AAs @ n-term

5: *K. phaffii* cell lysate producing HPV 16 L1 Δ 2-20 AAs @ n-term

Fig. 4



M: Marker

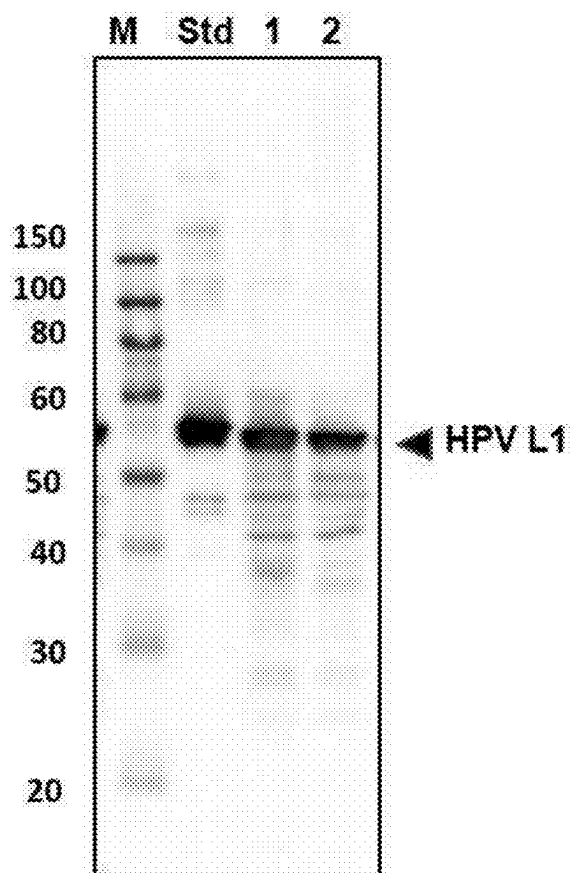
Std: Purified protein standard (HPV 18 L1)

1: *K. phaffii* cell lysate producing HPV 18 L1 original sequence

2: *K. phaffii* cell lysate producing HPV 18 L1 Δ 2-5 AAs @ n-term

3: *K. phaffii* cell lysate producing HPV 18 L1 Δ 2-7 AAs @ n-term

Fig. 5



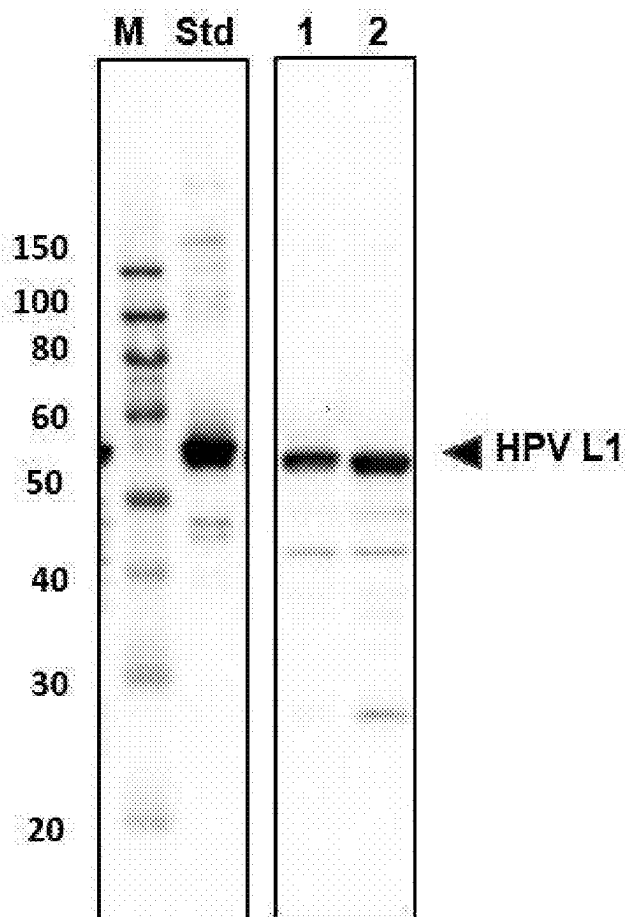
M: Marker

Std: Purified protein standard (HPV 16 L1)

1: *K. phaffii* cell lysate producing HPV 31 L1 original sequence

2: *K. phaffii* cell lysate producing HPV 31 L1 Δ 2-5 AAs @ n-term

Fig. 6



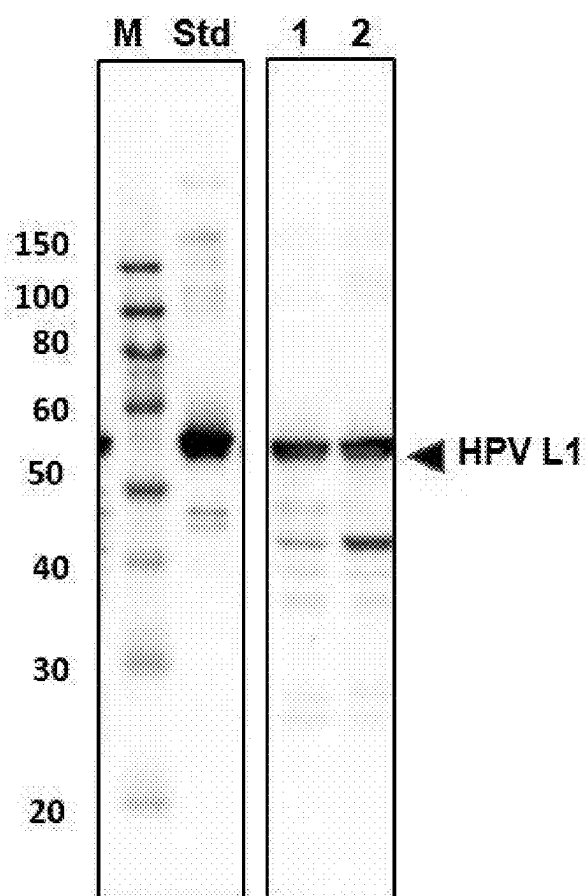
M: Marker

Std: Purified protein standard (HPV 16 L1)

1: *K. phaffii* cell lysate producing HPV 33 L1 original sequence

2: *K. phaffii* cell lysate producing HPV 33 L1 Δ 2-5 AAs @ n-term

Fig. 7



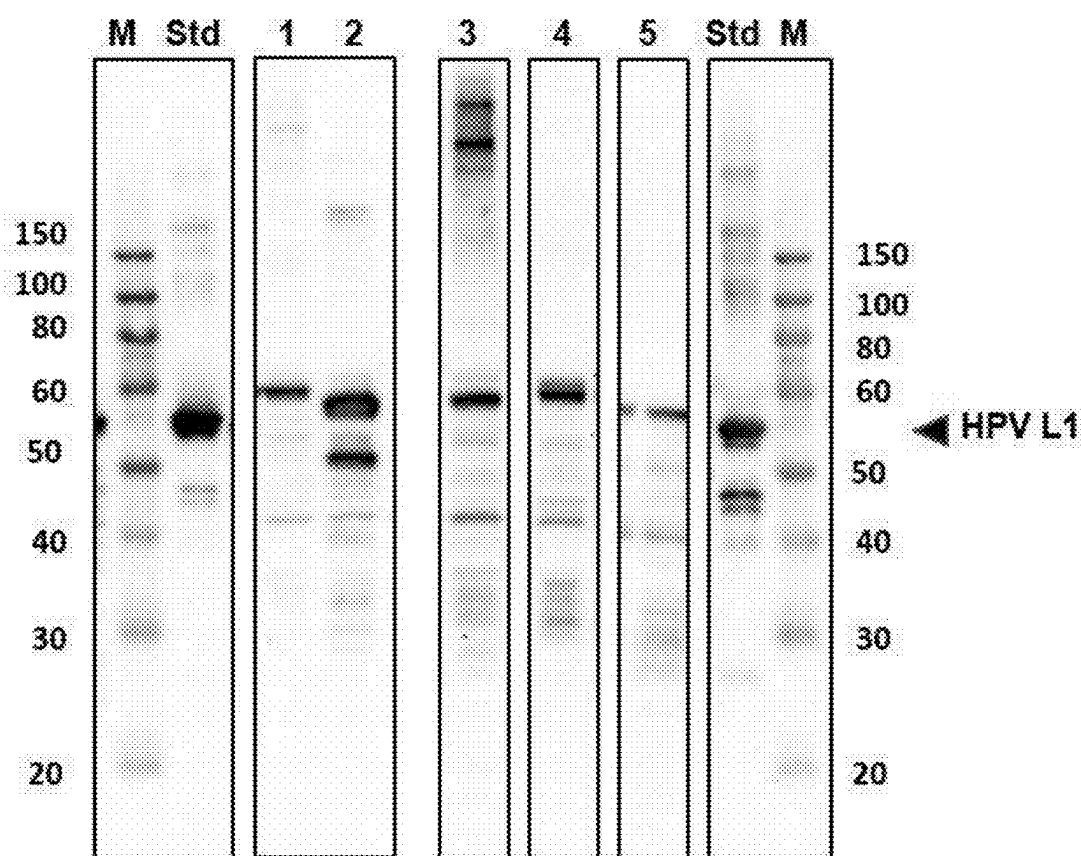
M: Marker

Std: Purified protein standard (HPV 16 L1)

1: *K. phaffii* cell lysate producing HPV 35 L1 original sequence

2: *K. phaffii* cell lysate producing HPV 35 L1 Δ 2-5 AAs @ n-term

Fig. 8



M: Marker

Std: Purified protein standard (HPV 16 L1)

1: *K. phaffii* cell lysate producing HPV 45 L1 original sequence

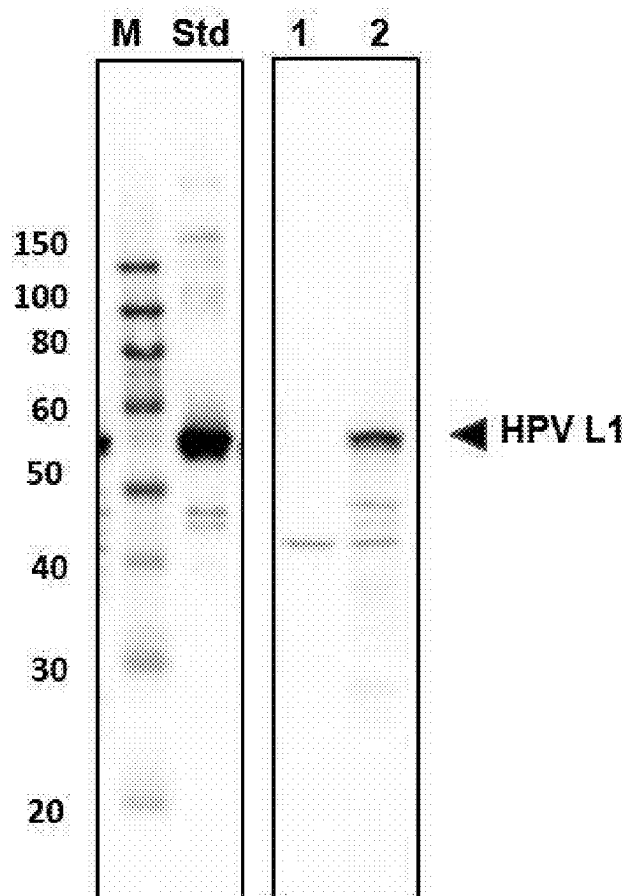
2: *K. phaffii* cell lysate producing HPV 45 L1 Δ 2-31 AAs @ n-term

3: *K. phaffii* cell lysate producing HPV 45 L1 Δ 2-10 AAs @ n-term

4: *K. phaffii* cell lysate producing HPV 45 L1 Δ 2-17 AAs @ n-term

5: *K. phaffii* cell lysate producing HPV 45 L1 Δ 2-22 AAs @ n-term

Fig. 9



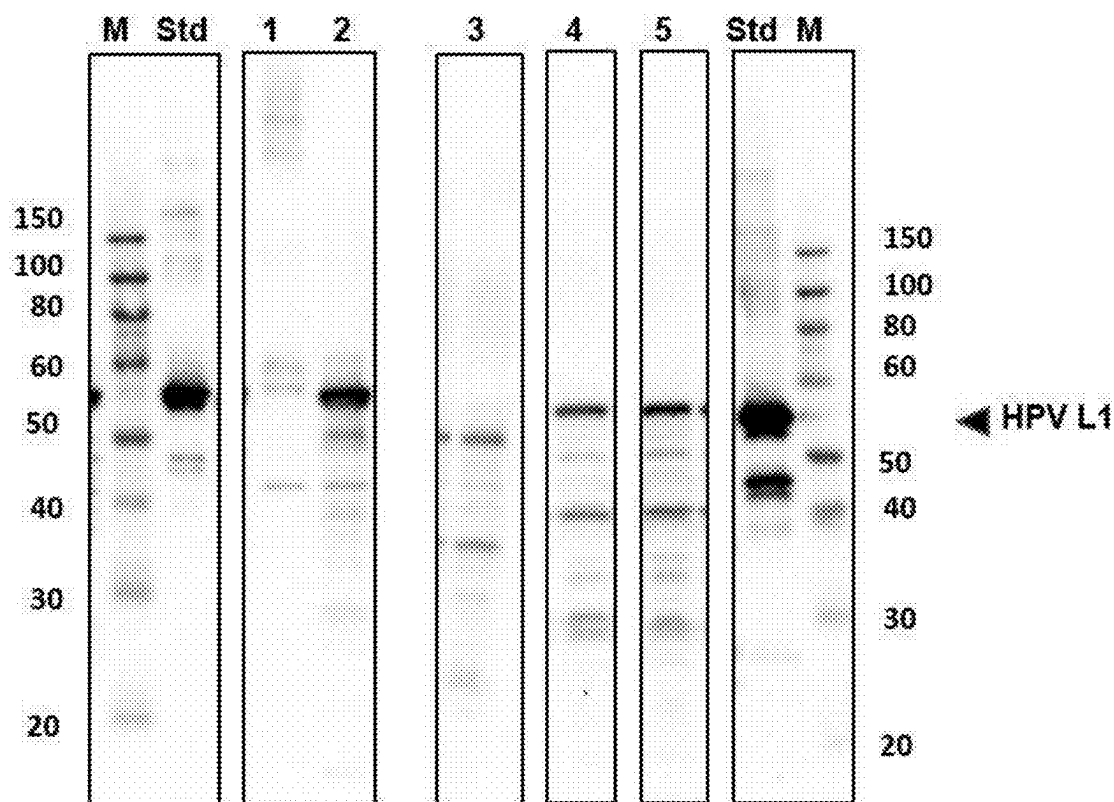
M: Marker

Std: Purified protein standard (HPV 16 L1)

1: *K. phaffii* cell lysate producing HPV 52 L1 original sequence

2: *K. phaffii* cell lysate producing HPV 52 L1 Δ 2-31 AAs @ n-term

Fig. 10



M: Marker

Std: Purified protein standard (HPV 16 L1)

1: *K. phaffii* cell lysate producing HPV 58 L1 original sequence

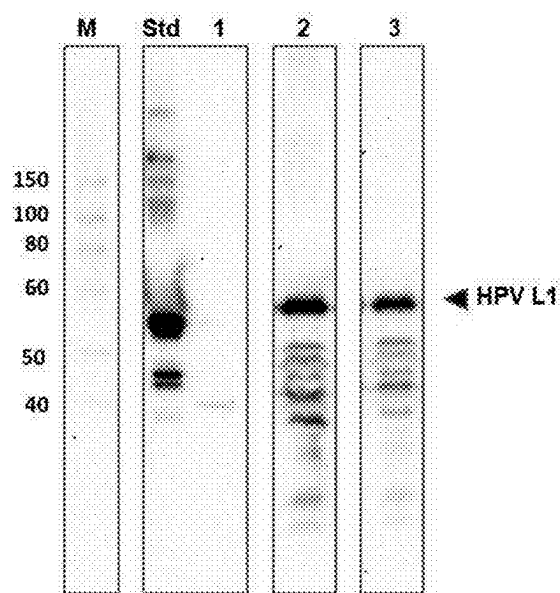
2: *K. phaffii* cell lysate producing HPV 58 L1 Δ 2-31 AAs @ n-term

3: *K. phaffii* cell lysate producing HPV 58 L1 Δ 2-10 AAs @ n-term

4: *K. phaffii* cell lysate producing HPV 58 L1 Δ 2-17 AAs @ n-term

5: *K. phaffii* cell lysate producing HPV 58 L1 Δ 2-22 AAs @ n-term

Fig. 11



M: Marker

Std: Purified protein standard (HPV 16 L1)

1: *K. phaffii* cell lysate producing HPV 16 L1 from control nucleotide sequence

2: *K. phaffii* cell lysate producing HPV 16 L1 from nucleotide sequence codon optimized for *K. phaffii*

3: *K. phaffii* cell lysate producing HPV 16 L1 from sequence codon optimized for *H. sapiens*

Fig. 12

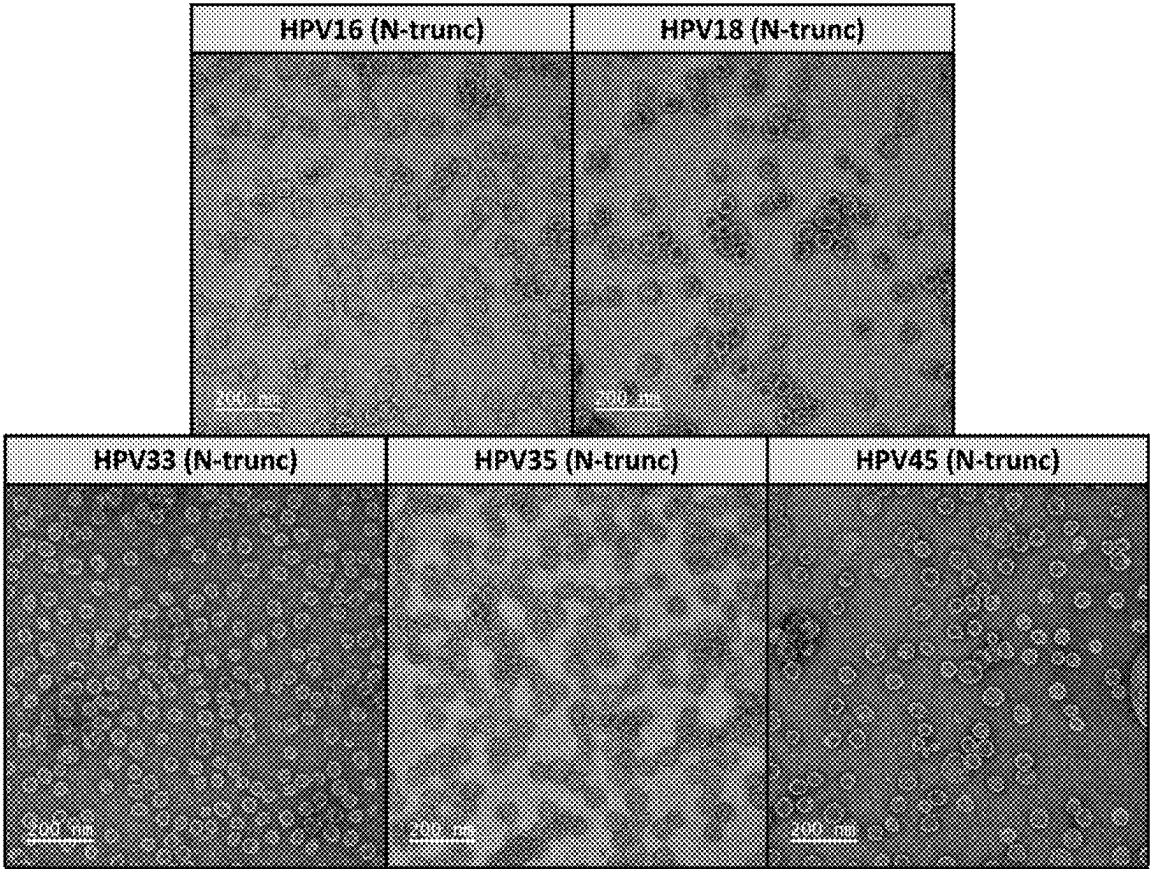


Fig. 13

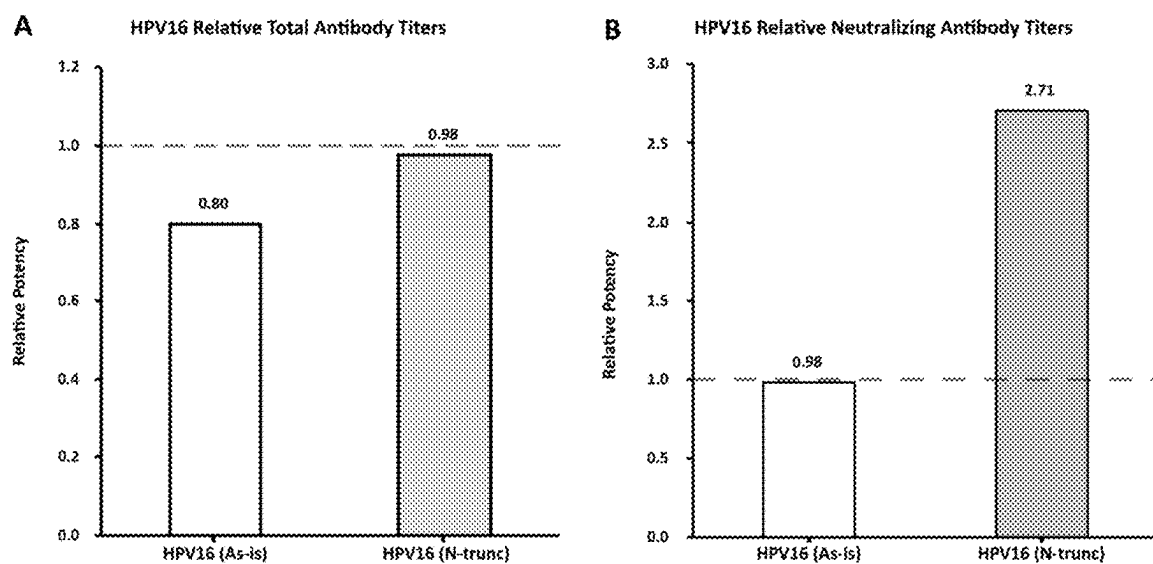


Fig. 14

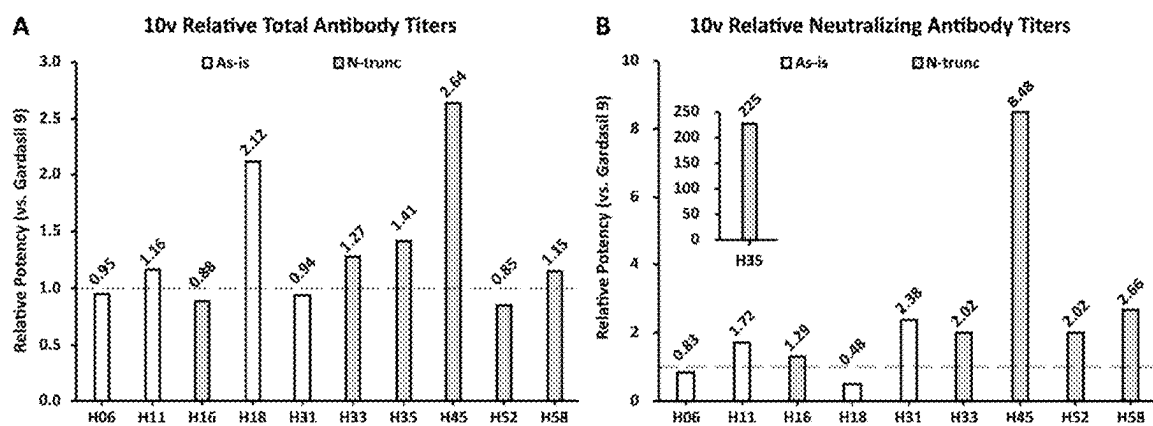


Fig. 15

TRUNCATED RECOMBINANT L1 PROTEIN OF HUMAN PAPILLOMAVIRUS

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims the benefit of and priority to U.S. Provisional Application No. 63/563,554, filed Mar. 11, 2024, the entire contents of which are incorporated herein by reference.

FIELD OF THE DISCLOSURE

[0002] The invention relates to a truncated recombinant L1 protein of Human papillomavirus (HPV), a recombinant polynucleotide encoding the protein, a recombinant vector comprising the polynucleotide, an isolated cell comprising the recombinant vector, a composition comprising the protein, and a method for producing the protein. The disclosure also relates to an HPV virus-like particle (VLP) comprising the protein, an immunogenic composition comprising the protein or the VLP, and a method for inducing an immune response to HPV in an individual.

DESCRIPTION OF THE TEXT FILE SUBMITTED ELECTRONICALLY

[0003] The instant application contains a sequence listing, which has been submitted in XML format via PatentCenter. The XML file named, "OPP20250,645US_Sequence_Listing.xml," was created on Feb. 25, 2025, and is approximately 162,315 bytes in size, the entire contents of which are incorporated herein by reference.

BACKGROUND OF THE DISCLOSURE

[0004] Human papillomavirus (HPV) is a group of more than 200 viruses that can infect the skin and mucous membranes. HPV can cause a variety of health problems, including genital warts, cervical cancer, and other cancers. HPV is spread through skin-to-skin contact, usually during sexual activity. The virus can also be spread through oral sex and other forms of close contact. There is no cure for HPV, but there are treatments available for the symptoms of HPV infection. Because there is no cure for HPV, vaccination is the best way to protect the body from HPV infection and the diseases that HPV can cause. The HPV vaccine is highly effective at preventing HPV infection, and it can also help to prevent the development of cervical cancer.

[0005] Currently, there are over 90 different types of HPV, mainly causing papillary disease in the skin and mucosa of humans. HPV types are divided into three groups depending on their relation with tumorigenesis: (1) group of low or no cancerogenic risk, containing types 6 and 11; (2) group of medium cancerogenic risk, containing types 22, 33, 35, 39, 52, 56, 58, 59 and 68; and (3) group of high cancerogenic risk, containing types 16, 18, 31 and 45.

[0006] Human papillomavirus (HPV) is a non-enveloped, double-stranded DNA virus with a circular genome approximately 7.2-8 kb in length. It contains eight open reading frames (ORFs) that encode for various proteins involved in viral replication, transcription, and transformation. The genome can be divided into three regions:

[0007] (1) Early region (E): Approximately 4.5 kb long, encodes six non-structural proteins (E1, E2, E4-E7) involved in viral replication, transcription, and transformation.

[0008] (2) Late region (L): Approximately 2.5 kb long, encodes the major capsid protein L1 and the minor capsid protein L2, which form the viral particle's outer shell.

[0009] (3) Long control region (LCR): Located between the L region and the E region's initiation point, approximately 800-900 bp long, contains regulatory elements for DNA replication and expression instead of coding for proteins.

[0010] Viral particles are 45-55 nm in diameter, with an icosahedral nucleocapsid composed of L1 and L2 proteins. The nucleocapsid protects the viral DNA within the particle.

[0011] HPV L1 protein, with a molecular weight of 55-60 kDa, is the major capsid protein of the human papillomavirus and the main protein comprising the HPV vaccine. HPV L1 protein expressed in multiple different expression systems can form Virus-like particles (VLPs) which resemble native HPV particles morphologically, without the assistance of the L2 protein. The VLP, consisting of 72 pentamers of the L1 proteins, exhibits icosahedral symmetry. Since the VLPs retain the native epitopes of the viral particles, they are highly immunogenic and can induce the generation of neutralizing antibodies against homologous HPV (Kimbauer, R. F. Booy, et al. 1992 Proc Natl Acad Sci USA 89 (24): 12180-4). Furthermore, the VLPs are safe and have no potential cancerogenic risk as they contain no viral DNA. Therefore, VLP vaccines become the primary candidate for an HPV vaccine.

[0012] The key for development of a vaccine is to efficiently produce VLPs of HPV L1 at large-scale. Currently, the most commonly used expression systems are eukaryotic expression systems including mammalian cells, insect cells and yeast. HPV L1 protein expressed in eukaryotic systems shows little conformational difference from that of the native virus, and can self-assemble into VLPs. These VLPs are typically expressed recombinantly at low levels in the named host organisms, requiring purification via gradient density centrifugation, which is time consuming, low yielding and not relevant for commercial scale manufacturing.

[0013] These manufacturing challenges impact the volumetric productivities of commercial processes, leading to high costs of goods manufactured (COGsM), that are passed on to the patient consumer. The existing price of marketed HPV vaccine products substantially limits their accessibility and uptake with global patients that would benefit from this life-saving product.

[0014] Therefore, there is a continued need for the development of highly manufacturable HPV L1 proteins that retain the ability to form uniform virus-like particles (VLPs) and can thereby induce the production of protective antibodies when these particles are delivered in vivo. This necessitates the advancement of efficient and cost-effective technologies for the production of high quality HPV L1 proteins.

SUMMARY OF THE DISCLOSURE

[0015] Accordingly, the present disclosure provides, inter alia, compositions and methods that find use in more efficient vaccination approaches against HPV.

[0016] An embodiment described herein provides a truncated recombinant human papillomavirus (HPV) L1 protein with at least 2 amino acids deleted from positions 2 to 66, or positions corresponding thereto, of the N-terminus.

[0017] Another embodiment described herein provides a recombinant polynucleotide encoding the truncated recombinant HPV L1 protein.

[0018] Another embodiment described herein provides a recombinant vector comprising the polynucleotide.

[0019] Another embodiment described herein provides an isolated cell comprising the recombinant vector.

[0020] Another embodiment described herein provides a composition comprising the truncated recombinant HPV L1 protein.

[0021] Another embodiment described herein provides a recombinant HPV virus-like particle (VLP) comprising the truncated recombinant HPV L1 protein.

[0022] Another embodiment described herein provides an immunogenic composition comprising the truncated recombinant HPV L1 protein or the VLP comprising the same, and optionally comprising pharmaceutically acceptable carriers, excipients and/or adjuvant.

[0023] Another embodiment described herein provides a method for inducing an immune response to HPV in an individual, comprising administering the immunogenic composition to an individual.

[0024] Another embodiment described herein provides a method for preventing an HPV infection or a disease associated with an HPV infection, comprising administering the immunogenic composition to a subject in need thereof. Another embodiment described herein provides a method for producing the truncated recombinant HPV L1 protein, comprising expressing a polynucleotide encoding the HPV L1 protein in a yeast cell.

BRIEF DESCRIPTION OF THE DRAWINGS

[0025] FIG. 1: Exemplary expression vector.

[0026] FIG. 2: Western Blot demonstrating expression of HPV 6 L1 in *K. phaffii*.

[0027] FIG. 3: Western Blot demonstrating expression of HPV 11 L1 in *K. phaffii* FIG. 4: Western Blot demonstrating expression of HPV 16 L1 in *K. phaffii*.

[0028] FIG. 5: Western Blot demonstrating expression of HPV 18 L1 in *K. phaffii*.

[0029] FIG. 6: Western Blot demonstrating expression of HPV 31 L1 in *K. phaffii*.

[0030] FIG. 7: Western Blot demonstrating expression of HPV 33 L1 in *K. phaffii*.

[0031] FIG. 8: Western Blot demonstrating expression of HPV 35 L1 in *K. phaffii*.

[0032] FIG. 9: Western Blot demonstrating expression of HPV 45 L1 in *K. phaffii*.

[0033] FIG. 10: Western Blot demonstrating expression of HPV 52 L1 in *K. phaffii*.

[0034] FIG. 11: Western Blot demonstrating expression of HPV 58 L1 in *K. phaffii*.

[0035] FIG. 12: Western Blot demonstrating expression of HPV 16 L1 in *K. phaffii* with different codon optimization.

[0036] FIG. 13: TEM observation of HPV virus-like particles (VLPs).

[0037] FIG. 14: A) Relative total antibody titers following administration of HPV 16 L1 VLPs to mice in as-is or N-terminal truncated form, as assessed by ELISA on mouse serum. B) Relative neutralizing antibody titer following administration of HPV 16 L1 VLPs to mice in as-is or N-terminal truncated form, as determined by PBNA on mouse serum.

[0038] FIG. 15: A) Relative total antibody titers following administration of various HPV L1 VLPs to mice in as-is or N-terminal truncated form, as assessed by ELISA on mouse serum. B) Relative neutralizing antibody titer following administration of various HPV L1 VLPs to mice in as-is or N-terminal truncated form, as determined by PBNA on mouse serum.

DETAILED DESCRIPTION OF THE DISCLOSURE

[0039] According to one aspect of the present disclosure, there is provided a truncated recombinant human papillomavirus (HPV) L1 protein with at least 2 amino acids deleted from positions 2 to 66, or positions corresponding thereto, of the N-terminus.

[0040] In one embodiment, the HPV L1 protein may have a deletion of amino acids at positions 2-3, 2-5, 2-7, 2-10, 2-15, 2-17, 2-20, 2-22, 2-31, or 2-66, or positions corresponding thereto, of N-terminus.

[0041] In one embodiment, the HPV may be an HPV serotype selected from HPV serotypes 6, 11, 16, 18, 31, 33, 35, 45, 52, and 58.

[0042] In one embodiment, the HPV6 L1 protein may have a deletion of amino acids at positions 2-3 or 2-5, or positions corresponding thereto, of N-terminus; the HPV11 L1 protein may have a deletion of amino acids at positions 2-3 or positions corresponding thereto, of N-terminus; the HPV16 L1 protein may have a deletion of amino acids at positions 2-5, 2-10, 2-15, or 2-20, or positions corresponding thereto, of N-terminus; the HPV18 L1 protein may have a deletion of amino acids at positions 2-5, 2-7, or 2-66, or positions corresponding thereto, of N-terminus; the HPV31, HPV33 or HPV35 L1 protein may have a deletion of amino acids at positions 2-5, or positions corresponding thereto, of N-terminus; the HPV45 or HPV58 L1 protein may have a deletion of amino acids at positions 2-10, 2-17, 2-22, or 2-31, or positions corresponding thereto, of N-terminus; or the HPV52 L1 protein may have a deletion of amino acids at positions 2-31, or positions corresponding thereto, of N-terminus.

[0043] In one embodiment, the HPV L1 protein may comprise of an amino acid sequence selected from one of SEQ ID NOs: 21-41, or a variant having at least about 90%, or at least about 95%, or at least about 98%, at least about 99% identity to one of SEQ ID NOs: 21-41.

[0044] In one embodiment, the HPV L1 protein may consist of an amino acid sequence selected from one of SEQ ID NOs: 21-41.

[0045] According to another aspect of the present disclosure, there is provided a recombinant polynucleotide encoding the truncated recombinant HPV L1 protein.

[0046] In one embodiment, the polynucleotide may be codon-optimized for expression in a yeast cell.

[0047] In one embodiment, the polynucleotide may comprise of a nucleotide sequence selected from one of SEQ ID NOs: 42-62, or a variant having at least about 60%, at least about 70%, at least about 80%, at least about 90%, or at least about 95%, or at least about 98%, at least about 99% identity to one of SEQ ID NOs: 42-62.

[0048] In one embodiment, the polynucleotide may consist of a nucleotide sequence selected from one of SEQ ID NOs: 42-62.

[0049] According to another aspect of the present disclosure, there is provided a recombinant vector comprising the polynucleotide.

[0050] According to another aspect of the present disclosure, there is provided an isolated cell comprising the recombinant vector.

[0051] According to another aspect of the present disclosure, there is provided a composition comprising the truncated recombinant HPV L1 protein. There also is provided a composition comprising the polynucleotide encoding the truncated recombinant HPV L1 protein, vector comprising the polynucleotide, or the cell comprising the recombinant vector.

[0052] According to another aspect of the present disclosure, there is provided a recombinant HPV virus-like particle (VLP) comprising the truncated recombinant HPV L1 protein.

[0053] According to another aspect of the present disclosure, there is provided an immunogenic composition comprising the truncated recombinant HPV L1 protein or the VLP comprising the same, and optionally comprising pharmaceutically acceptable carriers, excipients and/or adjuvant.

[0054] In one embodiment, the immunogenic composition may be used for inducing an immune response to HPV in an individual.

[0055] In one embodiment, the immunogenic composition may be used for the prevention of HPV infection or a disease associated with HPV infection.

[0056] In one embodiment, the disease associated with HPV infection may be condyloma acuminatum, genital warts, cervical neoplasia, or cervical cancer.

[0057] According to another aspect of the present disclosure, there is provided a method for inducing an immune response to HPV in an individual, comprising administering the immunogenic composition to an individual.

[0058] According to another aspect of the present disclosure, there is provided a method for preventing an HPV infection or a disease associated with an HPV infection, comprising administering the immunogenic composition to a subject in need thereof. In one embodiment, the disease associated with HPV infection is condyloma acuminatum, genital warts, cervical neoplasia, or cervical cancer. In one embodiment, the subject is likely to be exposed or has been recently exposed to HPV. In one embodiment, the subject is a human aged 9 to 45 years. Preferably, the subject is a female aged 9 to 45 years, and more preferably, the subject is a female aged 26 years or younger.

[0059] According to another aspect of the present disclosure, there is provided a method for producing the truncated recombinant HPV L1 protein, comprising expressing a polynucleotide encoding the HPV L1 protein in a yeast cell.

[0060] Hereinafter, the present disclosure will be described in more detail.

[0061] Where the terms “comprise”, “comprises”, “comprised” or “comprising” are used in this specification (including the claims) they are to be interpreted as specifying the presence of the stated features, integers, steps or components, but not precluding the presence of one or more other feature, integer, step, component or group thereof.

[0062] The discussion of documents, acts, materials, devices, articles and the like is included in this specification solely for the purpose of providing a context for the present invention. It is not suggested or represented that any or all of these matters formed part of the prior art base or were

common general knowledge in the field relevant to the present disclosure before the priority date of each claim of this application.

[0063] The disclosure is at least partially based on the discovery that a truncated recombinant HPV L1 protein capable of inducing the generation of neutralization antibodies against HPV can be expressed in a yeast expression system with high expression levels, and the truncated HPV L1 protein can be produced with a high yield.

[0064] Therefore, in one aspect, the disclosure relates to a truncated recombinant HPV L1 protein with at least 2 amino acids deleted from positions 2 to 66, or positions corresponding thereto, of the N-terminus.

[0065] The term “a truncated HPV L1 protein” refers to the protein with one or more amino acids deleted at the N- and/or C-terminal of wild-type HPV L1 protein, wherein the example of the wild-type HPV L1 protein includes, but is not limited to, the full-length L1 proteins such as SEQ ID NOs: 1-10.

[0066] In a preferred embodiment, the truncated HPV L1 protein has a deletion of 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, or 65 amino acids from positions 2 to 66, or positions corresponding thereto, of the N-terminus, as compared with wild type HPV L1 protein.

[0067] In another preferred embodiment, the truncated HPV L1 protein has a deletion of amino acids at positions 2-3, 2-4, 2-5, 2-6, 2-7, 2-8, 2-9, 2-10, 2-11, 2-12, 2-13, 2-14, 2-15, 2-16, 2-17, 2-18, 2-19, 2-20, 2-21, 2-22, 2-23, 2-24, 2-25, 2-26, 2-27, 2-28, 2-29, 2-30, 2-31, 2-32, 2-33, 2-34, 2-35, 2-36, 2-37, 2-38, 2-39, 2-40, 2-41, 2-42, 2-43, 2-44, 2-45, 2-46, 2-47, 2-49, 2-50, 2-51, 2-52, 2-53, 2-54, 2-55, 2-56, 2-57, 2-58, 2-59, 2-60, 2-61, 2-62, 2-63, 2-64, 2-65, or 2-66, or positions corresponding thereto, of N-terminus, as compared with wild type HPV L1 protein. In an exemplary embodiment, the truncated HPV L1 protein may have a deletion of amino acids at positions 2-3, 2-5, 2-7, 2-10, 2-15, 2-17, 2-20, 2-22, 2-31, or 2-66 of N-terminus, as compared with wild type HPV L1 protein.

[0068] In another preferred embodiment, the HPV may be an HPV serotype of low cancerogenic risk, medium cancerogenic risk, or high cancerogenic risk. For example, the HPV may be an HPV serotype such as HPV6, 11, 16, 18, 22, 23, 31, 33, 35, 39, 45, 52, 56, 58, 59, 68, etc., but is not limited thereto. In an exemplary embodiment, the HPV may be an HPV serotype selected from HPV serotypes 6, 11, 16, 18, 31, 33, 35, 45, 52, and 58.

[0069] In another preferred embodiment, the HPV6 L1 protein may have a deletion of amino acids at positions 2-3 or 2-5, or positions corresponding thereto, of N-terminus; the HPV11 L1 protein may have a deletion of amino acids at positions 2-3 or positions corresponding thereto, of N-terminus; the HPV16 L1 protein may have a deletion of amino acids at positions 2-5, 2-10, 2-15, or 2-20, or positions corresponding thereto, of N-terminus; the HPV18 L1 protein may have a deletion of amino acids at positions 2-5, 2-7, or 2-66, or positions corresponding thereto, of N-terminus; the HPV31, HPV33 or HPV35 L1 protein may have a deletion of amino acids at positions 2-5, or positions corresponding thereto, of N-terminus; the HPV45 or HPV58 L1 protein may have a deletion of amino acids at positions 2-10, 2-17, 2-22, or 2-31, or positions corresponding thereto, of N-terminus.

minus; the HPV52 L1 protein may have a deletion of amino acids at positions 2-31, or positions corresponding thereto, of N-terminus.

[0070] In another preferred embodiment, the HPV L1 protein may comprise of an amino acid sequence selected from one of SEQ ID NOs: 21-41, or a variant having at least about 90%, or at least about 95%, or at least about 98%, at least about 99% identity to one of SEQ ID NOs: 21-41.

[0071] In another preferred embodiment, the HPV L1 protein may consist of an amino acid sequence selected from the group consisting of SEQ ID NOs: 21-41.

[0072] In another preferred embodiment, the truncated HPV L1 protein may include a variant in which amino acid sequence is different from the truncated HPV L1 protein according to the disclosure by one or more amino acids, such as conservative amino acid substitutions, or which has an identity of at least about 60%, at least about 80%, at least about 81%, at least about 82%, at least about 83%, at least about 84%, at least about 85%, at least about 86%, at least about 87%, at least about 88%, at least about 89%, at least about 90%, at least about 91%, at least about 92%, at least about 93%, at least about 94%, at least about 95%, at least about 96%, at least about 97%, at least about 98%, or at least about 99% to the truncated HPV L1 protein according to the disclosure and retains the essential characteristics of the truncated protein. The term “essential characteristics” may be one or more of the following characteristics: capable of inducing the generation of neutralization antibodies against HPV; capable of being expressed in a yeast cell; capable of obtaining purified protein with a high yield by the expression.

[0073] The term “identity” refers to the match degree between two polypeptides or between two nucleic acids. When two sequences for comparison have the same base or amino acid monomer subunit at a certain site (e.g., each of two DNA molecules has an adenine at a certain site, or each of two polypeptides has a lysine at a certain site), the two molecules are identical at the site. Generally, the comparison of two sequences and the percent identity between two amino acid sequences can be determined using a known computer program or algorithm in the art.

[0074] The term “conservative substitution” refers to amino acid substitutions which would not negatively affect or change the biological activity of a protein/polypeptide comprising the amino acid sequence. For example, a conservative substitution may be introduced by standard techniques known in the art such as site-directed mutagenesis and PCR-mediated mutagenesis. Conservative amino acid substitutions include substitutions wherein an amino acid residue is substituted with another amino acid residue having a similar side chain, for example, a residue similar to the corresponding amino acid residue biophysically or functionally (such as, having similar size, shape, charges, chemical properties including the capability of forming covalent bond or hydrogen bond, etc.). The families of amino acid residues having similar side chains have been defined in the art. Therefore, a corresponding amino acid residue is preferably substituted with another amino acid residue from the same side-chain family. Methods for identifying amino acid conservative substitutions are well known in the art (for example, Brummell et al., *Biochem.* 32:1180-1187 (1993); Kobayashi et al., *Protein Eng.* 12(10): 879-884 (1999); and Burks et al., *Proc. Natl Acad. Sci USA* 94:412-417 (1997), which are incorporated herein by reference).

[0075] For non-conservative amino acid modifications, a single amino acid variation in a protein sequence can significantly impact the structural stability, functional activity, or immunogenicity of the protein. In particular, substitution, deletion, or insertion of specific amino acids can alter the secondary and tertiary structures of the protein, thereby affecting expression efficiency, binding affinity, and the ability to induce an immune response. For example, in the major structural protein L1 of the human papillomavirus (HPV) antigen, the deletion of N amino acids at the N-terminus, compared to the deletion of N+1 amino acids, may result in substantial differences in protein expression levels or functional properties. Such variations cannot be predicted solely based on sequence differences and must be experimentally validated to assess their effects. Therefore, in the present invention, the impact of specific amino acid deletions on protein expression and function was experimentally evaluated, and an antigen with an optimal sequence was selected.

[0076] In other aspect, the disclosure relates to a recombinant polynucleotide encoding the truncated recombinant HPV L1 protein, a recombinant vector comprising the polynucleotide, an isolated cell comprising the recombinant vector, a composition comprising the protein, and a method for producing the protein in a yeast expression system.

[0077] According to the disclosure, the term “yeast expression system” refers to an expression system comprising or consisting of a yeast (strain) and a vector, wherein the yeast (strain) includes, but is not limited to *Komagataella phaffii* (*Pichia pastoris*).

[0078] It will be appreciated by those skilled in the art that as a result of the degeneracy of the genetic code, each amino acid may be encoded by more than one codon, with the codons for the same amino acid having different frequencies of usage in wild-type genes. For example, yeast cell's preference for codons may result in low translation efficiency and expression levels of recombinant proteins. Thus, it may be advantageous to produce nucleotide sequences encoding proteins possessing a substantially different codon usage. Codons may be selected to tune the rate at which translation of the peptide occurs in a particular host in accordance with the frequency with which particular codons are utilized by the host. Properly tuning the codon usage may lead to improved overall expression.

[0079] In this regard, the present disclosure provides a codon-optimized gene encoding the major capsid protein L1 of human papilloma virus, which is capable, when introduced into a yeast cell, of efficiently expressing the major capsid protein L1 of human papilloma virus which then self-assembles into virus-like particles, in an expression amount that meets the requirements of industrial production.

[0080] When optimizing a heterologous protein for expression in *Komagataella phaffii* (*Pichia pastoris*), it has been reported that the nucleotide sequence may vary depending on the codon optimization strategy employed. Typically, sequence variations range from approximately 10% to 40%, with 20% to 30% being the most commonly reported range in scientific literature (For example, Wang J R et al., *Biomed Res Int.* 2015:248680).

[0081] In a preferred embodiment, the polynucleotide encoding the truncated recombinant HPV L1 protein may be codon-optimized for expression in a yeast cell, preferably *Komagataella phaffii*. For example, the polynucleotide may consist of a nucleotide sequence selected from the group

consisting of SEQ ID NOs: 42-62, 67, and 69-72, but is not limited thereto. For example, the polynucleotide described herein may comprise of a nucleotide sequence selected from one of SEQ ID Nos: 42-62, 67, and 69-72, or a variant having at least about 60%, at least about 70%, at least about 80%, at least about 90%, or at least about 95%, or at least about 98%, at least about 99% identity to one of SEQ ID NOs: 42-62, 67, and 69-72. The nucleotide is suitable to be efficiently expressed in yeast cell, particularly in *Komagataella phaffii* cell.

[0082] The codon-optimized gene provided by the present disclosure has the following advantages: 1) this optimized gene is more suitable for efficiently expressing target protein in yeast host and meets the requirements of industrial production; and 2) low cost, high yield and more uniform and stable quality of products can be achieved with the use of *Pichia* yeast expression system.

[0083] The term “vector” refers to a nucleic acid vehicle which can have a polynucleotide inserted therein. For expression in an appropriate expression system, an L1 nucleic acid encoding a polypeptide is operably linked into an expression vector and introduced into a host cell to enable the expression of the L1 protein by that cell. The gene with the appropriate regulatory regions will be provided in the proper orientation and reading frame to allow for expression. Methods for gene construction are known in the art. For example, in the vector, the polynucleotide may be operably linked to a promoter. The term “operatively linked” refers to a functional linkage between a nucleotide expression control sequence (e.g., a promoter sequence) and another nucleotide sequence. The control sequence may be operably linked to control transcription and/or translation of another nucleotide sequence.

[0084] The vector may be constructed as a vector for typically cloning or expression. In the present disclosure, any suitable vector can be used, especially those used for cloning and expression in yeast cell. More preferably, said expression vector is selected from currently commonly used yeast expression vectors, such as pPICZ, pD912, pPIC6, pGAPZ and pAO815, but is not limited thereto. Such expression vectors are commercially available from diverse vendors.

[0085] Moreover, an isolated cell comprising said gene encoding HPV L1 is also included in the present disclosure. Said cell is preferably a yeast cell, particularly a *Komagataella* yeast cell. The yeast cell strains are commercially available.

[0086] Delivery (introduction) of the polynucleotide or the vector comprising the same into the host cell may be carried out by using a delivering method widely known in the art. With regard to the delivering method, for example a micro-injection method, a calcium phosphate precipitation method, an electroporation method, a liposome-mediated transfection method, and gene bombardment may be used, but is not limited thereto.

[0087] A method of selecting the transformed host cell may be easily carried out using a phenotype expressed by a selection marker according to a method well known in the art. For example, when the selection marker is a specific antibiotic resistance gene, the transformant may be easily selected by culturing the transformant in a medium containing the antibiotic. The transformed host cells may be cultured for a period sufficient for polypeptide expression or secretion, and the polypeptide may be separated and purified

by a method of separating and purifying proteins commonly used, for example, a method of using solubility, such as salting out, solvent precipitation, etc., a method of using a molecular weight difference, such as dialysis, ultrafiltration, gel filtration, sodium dodecyl sulfate-polyacrylamide gel electrophoresis, etc., a method of using a charge, such as ion exchange chromatography or hydroxylapatite chromatography, etc., a method of using specific affinity, such as affinity chromatography, etc., a method of using a hydrophobicity difference, such as reverse-phase high performance liquid chromatography, etc., a method of using an isoelectric point difference, such as isoelectric point electrophoresis, etc.

[0088] In other aspects, the disclosure relates to an HPV virus-like particle (VLP) comprising the protein, an immunogenic composition comprising the protein or the VLP, and a method for inducing an immune response to HPV in an individual.

[0089] Some embodiments concern using the protein produced herein for both prophylactic administration to reduce the risk of infection, for treatment of an infection and for diagnostics of an existing condition. Suitability of the protein produced for use as vaccines or as diagnostic agents can be confirmed by reaction with antibodies or monoclonal antibodies that react or recognize conformational epitopes present on the intact version of virus and based on ability to elicit the production of neutralizing antiserum. Other suitable assays for determining whether neutralizing antibodies are produced are known in the art and contemplated herein.

[0090] In a preferred embodiment, the HPV L1 protein may be used to generate immunogenic compositions against HPV infection. In accordance with these embodiments, these immunogenic compositions can contain the protein described herein in an effective amount to induce formation of neutralizing antibodies when introduced to a subject in need thereof. Immunogenic or vaccine compositions contemplated herein can further contain a pharmaceutically acceptable carrier or excipients or another anti-viral agent or other known agent in the art to complement the compositions.

[0091] An elicited immune response may be either prophylactic, preventing later infection by the specific viral type targeted, or may be therapeutic, reducing the severity of infection or disease. An immune response can include a humoral, e.g., antibody, response to a provided antigen and/or a cell mediated response to a provided antigen in a vaccine. Methods to measure an immune response are known to those skilled in the art. If one or both types of immune response are present, they can protect a subject from developing the condition. In accordance with certain embodiments, ability of a composition disclosed herein to protect from disease refers to the ability of the protein generated herein to treat, ameliorate and/or prevent disease caused by a virus contemplated herein or cross-reactive agent, for example, by eliciting an immune response in the subject. It is to be noted that a subject may be protected by a composition of the present disclosure even without the detection of a humoral or cell-mediated response to the composition. Protection can be measured by methods known to those skilled in the art.

[0092] Immunogenic compositions or vaccines can be administered in therapeutically effective amounts. That is, in amounts sufficient to produce a protective immunological response. In accordance with the embodiment, dosages ranging from about 0.001 mg to about 1 mg protein can be

introduced to a subject in need thereof or to a subject in order to reduce the onset of a condition associated with infection (e.g. HPV infection). Single or multiple dosages can be administered as determined by a health professional.

[0093] The composition may be formulated as either a monovalent or multivalent vaccine. In one embodiment, the composition may comprise the L1 protein from serotypes HPV 6, 11, 18, 16, 31, 33, 35, 45, 52, and 58. For example, the composition may comprise serotypes HPV 6 at 30 µg/human dose, HPV 11 and 18 at 40 µg/human dose, HPV 16 at 60 µg/human dose, and HPV 31, 33, 35, 45, 52, and 58 at 20 µg/human dose of the L1 protein. The preferred human dose is 0.5 mL.

[0094] The term “pharmaceutically acceptable carriers and/or excipients” refers to carriers and/or excipients that are pharmacologically and/or physiologically compatible with subjects and active ingredients, and are well known in the art (for example, Remington’s Pharmaceutical Sciences. Edited by Gennaro A R, 19th ed. Pennsylvania: Mack Publishing Company, 1995), including, but not limited to pH adjusting agents, surfactants, and ionic strength enhancers. For example, pH adjusting agents include, but are not limited to, phosphate buffers; surfactants include, but are not limited to: anion surfactants, cation surfactants, or non-ionic surfactants (for example, Tween-80); and ionic strength enhancers include, but are not limited to, NaCl.

[0095] The term “adjuvant” refers to a substance that is added to the composition to induce, increase, stimulate or strengthen a cellular or humoral immune response to administration of a vaccination described herein. Any adjuvant known in the art that is compatible with compositions disclosed herein is contemplated. Suitable adjuvants include, but are not limited to, other bacterial cell wall components, aluminum-based salts, such as aluminum hydroxide, calcium-based salts, silica, polynucleotides, toxins, such as cholera toxin, toxoids, such as cholera toxoid, serum proteins, other viral coat proteins, other bacterial-derived preparations, block copolymer adjuvants, saponins and their derivatives, and Freund’s adjuvants, such as Freund’s complete adjuvant. More preferably, the adjuvant may be an aluminum-based adjuvant, such as aluminum hydroxide. The adjuvant is included in an amount effective to stimulate the immune system without causing significant adverse effects. Typical concentrations may range from about 0.001% to 10% by weight of the total composition, although other concentrations may be used as appropriate based on factors such as the type of antigen, the type of adjuvant, the desired immune response, administration route, target population. In a preferred embodiment, the adjuvant may be present at a concentration of about 120 µg/mL.

[0096] The composition may be administered parenterally, locally or systemically, including by way of example, oral drops or tablet, intranasal, intravenous, intramuscular, subcutaneous, inhalation, and topical administration. Preferably, it is administered intramuscularly. In one embodiment, the composition was administered intramuscularly in a mouse immunogenicity study. The manner of administration is affected by factors including the natural route of infection. The dosage administered will depend upon factors including the age, health, weight, kind of concurrent treatment, if any, and nature and type of the particular viral, e.g., human papillomavirus. The vaccine or immunogenic composition may be employed in dosage form such as capsules, liquid

solutions, suspensions, or elixirs, for oral administration, or sterile liquid formulations such as solutions or suspensions for parenteral or intranasal use.

[0097] The composition may be administered in a regimen designed to elicit an optimal immune response. The dosing schedule, including the interval between administrations and the total number of doses, may vary depending on factors such as the type of antigen, the presence of an adjuvant, the route of administration, and the target population. Generally, the composition may be administered in one or multiple doses at appropriate intervals to achieve the desired level of immunity. In a preferred embodiment, the administration occurs twice at an interval of two weeks. In a preferred embodiment suitable for administration to humans, the composition may be administered as two doses at months 0 and 6-12, or alternatively as three doses at months 0, 2, and 6. However, alternative dosing schedules may be used as needed based on clinical considerations and immunogenicity data. Any pharmaceutical formulation known in the art for a vaccine is contemplated herein. It is contemplated that formulations can contain other agents of use in vaccination of a subject including, but not limited to other active or inactive ingredients or compositions known to one skilled in the art.

[0098] All contemplated immunogenic or vaccine compositions can be prepared by any method known to one skilled in the art. In certain embodiments, the virus compositions are lyophilized and are mixed with a pharmaceutically acceptable excipient (e.g. water, phosphate buffered saline (PBS), wetting agents etc.). In other embodiments, immunogenic compositions can include stabilizers that are known to reduce degradation of the formulation and prolong shelf-life of the compositions.

Beneficial Effect

[0099] HPV L1 proteins expressed in eukaryotic expression systems retain their native conformation and can self-assemble into VLPs. Nevertheless, low expression levels experienced when using eukaryotes like insect cells limit production at large-scale and result in high costs. The n-truncated HPV L1 protein provided by the present disclosure describes a protein suitable for yeast expression systems with high expression levels. It enables the production of intact VLPs at higher expression levels than those reported for other expression hosts, while increasing production yields and enabling large-scale industrial applications.

EXAMPLES

[0100] Hereinafter, the present disclosure will be described in more detail with reference to the following examples. However, these examples are only for illustrating the present disclosure, and the scope of the present disclosure is not limited by these examples.

Example 1. Construction of Vector and Yeast Strains for Expression of Mutated HPV L1 Proteins

[0101] The coding region for L1 proteins of HPV types 6, 11, 16, 18, 31, 33, 35, 45, 52, and 58 (SEQ ID NOs: 1-10) were codon optimized for *Komagataella phaffii* (*Pichia pastoris*) (SEQ ID NOs: 11-20) and synthesized by Thermo Fisher GeneArt. The DNA fragment was amplified using PCR and cloned into a custom vector using Gibson Assem-

bly. The custom vector contained a bacterial origin of replication, a promoter specific to *Komagataella phaffii*, a terminator specific to *Komagataella phaffii*, and a selectable marker (e.g. antibiotic resistance) (FIG. 1). The circular plasmid containing the HPV L1 coding region and custom vector was transformed into *E. coli* (TOP10 Chemically Competent *E. coli*, Thermo Fisher) for amplification. The amplified plasmid was purified from *E. coli* culture using QIAprep Spin Miniprep Kit (Qiagen) and linearized using PCR. Linearized plasmid was transformed into *Komagataella phaffii* cells (wild type NRRL Y-11430 retrieved from United States Department of Agriculture (USDA) and Agricultural Research Service (ARS)) via electroporation. Stable *Komagataella phaffii* transformants were obtained via homologous recombination and selected for on appropriate antibiotic-resistant solid media.

Example 2. Expression of HPV L1 Proteins in Yeast

[0102] *Komagataella phaffii* strains containing a recombinant HPV L1 gene, such as those constructed in Example 1, were cultivated in 24-well deep well plates (3 mL cultivation volume) at room temperature or in 1 L baffled shake flasks (200 mL cultivation volume) at 25° C. or at 30° C. Strains were cultivated in custom chemically-defined media. Glycerol-containing medium was used for the first 24 hours to build biomass. The medium was then exchanged for a methanol containing medium to promote protein expression. Strains were cultivated in Methanol-containing medium for 24 hours. The yeast cells were then harvested by centrifugation and stored at -80° C. until use. Cell pellets were resuspended in a lysis buffer and lysed with 0.5 mm acid washed Silica glass beads (MilliporeSigma) using a vortexer or a BeadBeater Blender (BioSpec). The lysate was centrifuged at 16,000×g for 30 minutes at 4° C. Cell lysate was sterile filtered using a 0.2 µm filter and stored at -80° C. prior to further purification.

[0103] Samples of soluble cell lysate were analyzed for protein expression using SDS-PAGE and Western blot. Samples for SDS-PAGE were reduced using NuPAGE Sample Reducing Agent (Thermo Fisher), boiled at 75° C. for 7 minutes, and run on 12% Tris-Glycine gels (Thermo Fisher). Samples for Western blot were first run on SDS-PAGE and then transferred to a nitrocellulose membrane using an iBlot 2 Dry Blotting System (Thermo Fisher). The membrane was then blocked in TBST-M (Tris-Buffered Saline (TBS, Thermo Fisher) with 0.5% Tween 20 (Sigma) and 5% Skim Milk (Fisher Scientific)) at room temperature for 1 hour, incubated in a solution of primary antibody (Purified mouse Anti-L1 protein of human papilloma virus, BD Bioscience Cat No. 554171) diluted in TBST-M at 4° C. overnight, washed three times with TBST ((Tris-Buffered Saline (TBS, Thermo Fisher) with 0.5% Tween 20 (Sigma))), incubated with secondary antibody (Goat anti-Mouse IgG (H+L) Cross-Adsorbed Secondary Antibody, HRP, Thermo Fisher Cat No. G-21040) diluted in TBST-M at room temperature for 1 hour, washed another three times with TBST, and finally washed once with TBS (Thermo Fisher) for 10 minutes. Western blots were imaged using electrochemiluminescence.

[0104] L1 proteins of HPV types 6, 11, 16, 18, 31, 33, 35, 45, 52, and 58 were expressed in *Komagataella phaffii* as illustrated (FIGS. 2-11; Lane 1).

Example 3. Purification of Mutated Proteins

[0105] The HPV L1 protein produced by *Komagataella phaffii* was purified by a two-step chromatography method, i.e. HS-HE method, the cell lysate was purified and finally, high-purity virus-like particles were obtained. For the purification step, the supernatant samples were loaded onto a salt-containing buffer (such as NaCl or KCl buffer solution). Then the media was incubated with a commonly used reducing agent and loaded onto the cation exchange chromatography, and then washed using a salt concentration gradient buffer to separate impurities from HPV L1 VLPs protein. Subsequently, a high-concentration salt-containing buffer solution was used to elute the bound HPV L1 VLPs protein, and the eluted preliminarily purified HPV L1 VLPs protein was collected. Thus, the obtained preliminarily purified HPV L1 VLPs protein was once again incubated with a commonly used reducing agent and loaded onto affinity chromatography for fine purification. Similar to the previous step, it was washed using a salt concentration gradient buffer to separate impurities from HPV L1 VLPs protein, and subsequently high concentration salt-containing buffer solution was used to elute the bound HPV L1 VLPs protein, and then the secondly purified HPV L1 VLPs protein was collected. Then tangential flow filtration was applied to the second elution for the final step of the purification and VLP formation of HPV L1 protein. The detailed concentration of salt can vary depending on the serotype of HPV.

First Step Chromatography:

[0106] Resin: POROST™ 50 HS strong cation exchange resin produced by Thermo Fisher was used.

Second Step Chromatography:

[0107] Resin: POROST™ 50 HE Heparin affinity resin produced by Thermo Fisher was used.

Example 4. Study of Codon Optimization in Intracellular Expression

[0108] The coding region for L1 protein of HPV type 16 (SEQ ID NO: 2) was codon optimized for *Komagataella phaffii* (*Pichia pastoris*) (SEQ ID NO: 12) or for *Homo sapiens* (SEQ ID NO: 21) and synthesized by Thermo Fisher GeneArt. Yeast (*K. phaffii*) strains expressing each version of the codon optimized HPV type 16 were constructed as described in Example 1 and expression was evaluated as described in Example 2. Genes that were codon optimized for *K. phaffii* (SEQ ID NO: 12) or *H. sapiens* (SEQ ID NO: 64) showed higher expression than the control that was not codon optimized (SEQ ID NO: 74) (FIG. 12).

Example 5. Evaluating N-Terminal Truncations to Promote HPV L1 Expression

[0109] In order to improve the expression of L1 in *K. phaffii*, a series of deletions were made in the DNA expression plasmids from the 5'-end of the HPV L1 coding regions. Vectors were constructed as in Example 1 for L1 proteins of HPV types 6, 11, 16, 18, 31, 33, 35, 45, 52, and 58 (SEQ ID NOs: 21-41), were codon optimized for *Komagataella phaffii* (*Pichia pastoris*) (SEQ ID NOs: 42-62), and were expressed as in Example 2. Proteins with deletions were expressed as illustrated in FIGS. 2-11; Lane 2+. For all

serotypes demonstrated, expression of protein with deletion maintained or improved expression levels observed on Western blot (FIGS. 2-11).

Example 6. Evaluation of Titer According to Amino Acid Changes

[0110] Cell lysate samples from Examples 2 and 5 were analyzed via ELISA to confirm function (through binding to relevant antibodies) and quantitatively measure protein expression titer. Briefly, capture antibody was diluted in PBS, loaded into 96-well microtiter plates, and incubated overnight at 4° C. The plate was then washed three times with PBS-T (1×PBS with 0.1% Tween 20) and blocked with 1×PBS with 5% skim milk (blocking buffer) for 4 hours at room temperature. Blocking buffer was then removed and cell lysate samples diluted in PBS with 0.5% skim milk and HPV L1 protein standard diluted in 10 mM sodium phosphate, pH 6.5, 330 mM sodium chloride, 0.01% Tween 80 were added to the plate along with 60 mM sodium phosphate and incubated overnight at room temperature with shaking at 100 rpm. The plate was then washed three times with PBS-T. Detection antibody, diluted in PBS with 0.5% skim milk was added and incubated at room temperature for 1 hour and then washed three times with PBS-T. Streptavidin-HRP was added to the plate and incubated at room temperature for 1 hour. The plate was washed 3 times with PBS-T. TMB substrate was added to the plate and incubated at room temperature for 5 minutes. TMB stop solution was added and OD was measured at 450 nm. Samples were compared to the standards to determine protein titer.

[0111] Results showed that expression of N-terminally truncated HPV L1 in yeast was higher than the full length sequence for serotypes 11, 16, 18, 33, 35, 45, 52, and 58 (TABLE 1).

TABLE 1

Expression Titer for HPV L1 in <i>K. phaffii</i> as Determined by ELISA			
Serotype	Amino Acid Deletions	SEQ ID NO:	Estimated Titer (mg/L)
11	None	10	4.51
11	Δ2-3 @n-term	41	20.91
16	None	2	8.5
16	Δ2-5 @n-term	23	21.9
18	None	3	1.9
18	Δ2-5 @n-term	27	4.5
33	None	5	26.8
33	Δ2-5 @n-term	30	49.6
35	None	6	7.5
35	Δ2-5 @n-term	31	15.5
45	None	7	Below LOD*
45	Δ2-31 @n-term	35	35.8
52	None	8	Below LOD
52	Δ2-31 @n-term	36	68.9
58	None	9	Below LOD
58	Δ2-31 @n-term	40	204.2

*LOD: Limit of detection

Example 7. Productivity Improvement at the Bioreactor Level

[0112] *Komagataella phaffii* strains containing a recombinant HPV L1 gene, such as those constructed in Example 1, were inoculated in a Daisy Petal™ perfusion bioreactor at 25° C. or 30° C. Strains were cultivated in custom chemi-

cally-defined media. Glycerol-containing medium was fed for the first 40 hours to build biomass. The medium was then changed to a methanol containing medium to promote protein expression. Strains were cultivated in Methanol-containing medium for at least 53 hours. The yeast cells were then harvested by centrifugation and stored at -80° C. until use. Cell pellets were lysed and Western Blots were performed as described in Example 2. ELISA was performed as described in Example 6.

[0113] Results showed that expression of N-terminally truncated HPV L1 in yeast was higher than that of the full length sequence for serotypes 11, 16, and 18 (TABLE 2). Productivity was measured as space-time yield, the total mass of protein produced per cultivation volume per cultivation day. The space-time yield for each serotype was normalized to the space-time yield achieved with the full length protein.

TABLE 2

Normalized Space-Time Yield for HPV L1 in <i>K. phaffii</i> in Daisy Petal™ perfusion bioreactor as Determined by ELISA			
Serotype	Amino Acid Deletions	SEQ ID NO:	Normalized Space-Time Yield
11	None	10	1.0
11	Δ2-3 @n-term	41	4.2
16	None	2	1.0
16	Δ2-5 @n-term	23	5.3
18	None	3	1.0
18	Δ2-5 @n-term	27	2.6

Example 8. TEM Observation of HPV Virus-Like Particle (VLP)

[0114] The proper formation of HPV virus-like particles is crucial for the functional aspects of the VLP vaccine. Through the observation using Transmission Electron Microscopy (TEM) of the HPV VLP, we can confirm the formation of VLP with a consistent shape and verify whether the formed VLP exhibits an appropriate structure. Transmission Electron Microscopy involves the convergence of electron beams generated by tungsten filaments, field emission sources, etc., through multiple magnetic lenses. These focused beams pass through the specimen, are detected by a fluorescent screen or CCD detector after passing through several magnetic lenses and are then output. Approximately 20-30 nmol concentration of the sample was dropped onto the plasma-treated carbon grid. To enhance the contrast of the specimen during electron microscopy observation, it was stained with 2% uranyl acetate and then dried. Then the Images were observed using Transmission Electron Microscopy at magnifications of 40,000× and 100,000×. The observation of the shape of the HPV VLP L1 particles formed by HPV16 L1 Δ2-5 through the Transmission Electron Microscopy revealed the consistent formation of VLP of HPV L1 protein (FIG. 13).

Example 9. Immunogenicity of HPV Virus-Like Particle in Animals

9.1 Evaluation of Immunoprotective Characteristics of N-Terminal Truncation on HPV L1 in Animals.

[0115] The study compared the immunoprotective characteristics of HPV L1 VLPs made from wildtype HPV16 L1

and an N-terminal truncated HPV16 L1 in animals. The HPV16 Δ2-5 variant was used for the truncated version. The manufactured HPV16 VLPs were adsorbed onto aluminum hydroxide adjuvant at a concentration of 120 μg/mL and diluted 300-fold from the human dose for this animal study. Each experimental group consisted of 7 mice immunized with the diluted antigens. The negative control group of 4 mice received an empty vehicle, while the control group was immunized with an in-house HPV VLP made of wildtype HPV16 L1 expressed in insect cells. Immunizations were administered on Days 0, and 14. Blood samples were collected from the eyes and jugular vein on Day 0, and from the heart on Day 28. Sera were isolated for total antibody and neutralizing antibody titration. Blood samples were incubated at room temperature for 30 minutes for coagulation, then centrifuged for 10 minutes at 2,000 g to separate sera. The supernatants were used for total antibody and neutralizing antibody titration.

TABLE 3

Test substance	Antigen content μg/human dose
Vehicle	0
In-house Control (WT, Insect)	60
HPV16 mono (WT, Yeast)	60
HPV16 mono (N-trunc, Yeast)	60

[0116] To evaluate the immunogenicity of the injected HPV VLPs, VLP based Enzyme-Linked Immunosorbent Assay (VLP ELISA) was used to titrate total antibodies, and Pseudovirus-based Neutralization Assay (PBNA) was used to titrate neutralizing antibodies in the immunized mice sera. For VLP ELISA, purified VLPs were coated on immune plates a day before the assay. After removing unbound VLPs, the plates were incubated with skim milk blocking buffer. Serially diluted mice sera, including negative and positive control groups, were then incubated. Anti-mouse IgG HRP was added for signal development, and signals were quantified by a microplate reader at 450 nm after the TMB substrate/stop solution was added.

[0117] PBNA, which has replaced traditional immunogenicity tests for HPV due to its difficulty to culture in vitro and strong host specificity, uses HPV pseudovirus composed of major capsid proteins L1 and L2 to mimic the immunogenic characteristics of natural HPV VLPs. By incubating the serum of immunized mice with HPV pseudovirus expressing SEAP (Secreted Alkaline Phosphatase) reporter system, the strength of immunogenicity can be measured quantitatively. 293TT cells were inoculated on 96-well plates at a density of 30,000 cells/well and incubated at 37° C. with 5% CO₂. Serially diluted mice sera, including negative and control groups, were mixed with appropriate subtypes of HPV pseudovirus and incubated at 4° C. for 1 hour. The sera and pseudovirus mixtures were added to inoculated 293TT cells and incubated for 4 days. The chemifluorescence of the culture media was measured by a fluorescence plate reader. Dose-response curves were generated for each experimental group based on the signals produced by the serially diluted sera in VLP ELISA and PBNA. The specific dilution factors where the signal reached 50% of the maximum signal of the curves were designated as EC₅₀ for total antibody titration and neutral-

izing antibodies. Relative potency of each group was determined by normalizing each EC₅₀ value to the EC₅₀ of the control group. [0118] While the HPV16 truncated L1 VLP injected mice sera showed nearly identical total antibody titers to the control group (wildtype HPV16 L1 expressed in insect cells), the wildtype HPV16 was 20% lower in total antibody relative potency compared to the control group (FIG. 14A). Although the wildtype HPV16 L1 VLP showed relatively identical neutralizing antibody relative potency to the control group, the truncated variant showed about a 2.71-fold increase in relative potency (FIG. 14B).

9.2 Evaluation of Immunoprotective Characteristics of N-Terminal Truncated HPV L1 VLPs in Multivalent Forms in Animals.

[0119] Another study was conducted to examine the immunoprotective characteristics of truncated HPV L1 VLPs when formulated into a multivalent form. For this study, the following HPV types were selected based on manufacturing feasibility: As-is HPV6, HPV11, HPV18, HPV31 (denoted as H06, H11, H18, and H31 in FIG. 15, respectively), and N-terminal truncated HPV16 Δ2-5, HPV33 Δ2-5, HPV35 Δ2-5, HPV45 Δ2-31, HPV52 Δ2-31, and HPV58 Δ2-31 (denoted as H16, H33, H35, H45, H52, and H58 in FIG. 15, respectively). The manufactured HPV VLPs were adsorbed onto aluminum phosphate adjuvant and formulated into a decavalent suspension with the corresponding concentrations for each antigen listed in Table 4. The decavalent suspension was diluted 300-fold from the human dose for this animal study. Both the control and experimental groups consisted of 7 mice immunized with the diluted antigens. For the control group, Gardasil®9, a commercially available nonavalent product, was diluted 300-fold from the human dose. The negative control group of 3 mice received an empty vehicle. Immunizations and analyses were performed as described previously.

TABLE 4

Test substance	Antigen content μg/human dose (serotype)
Vehicle	0
Control	30 (6), 40 (11, 18), 60 (16), 20 (31, 33, 45, 52, 58)
Gardasil ®9)	
10v HPV VLPs	30 (6), 40 (11, 18), 60 (16), 20 (31, 33, 35, 45, 52, 58)

[0120] The relative total antibody titers appeared to be unaffected by the N-terminal truncation, as both the As-is and N-terminal truncated subtypes showed either nearly identical or superior relative potency compared to the control group (FIG. 15A). However, all N-terminal truncated subtypes exhibited at least a 29% increase in relative neutralizing antibody titers compared to the control group, while some As-is subtypes showed up to a 52% decrease in relative potency (FIG. 15B).

[0121] Based on the above description, it will be understood by those skilled in the art that the present disclosure may be implemented in a different specific form without changing the technical spirit or essential characteristics thereof. In this regard, it should be understood that the above embodiment is not limitative, but illustrative in all aspects. The scope of the disclosure is defined by the appended claims rather than by the description preceding them, and therefore all changes and modifications that fall within

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SEQ	Info.
ID No: type	Info.
40 P	HPV58 Δ2-31 @n-term
41 P	HPV11 Δ2-3 @n-term
N-truncated Nucleotide Sequences codon optimized for <i>K. phaffii</i>	
42 N	HPV6 Δ2-3 @n-term (codon optimized for <i>K. phaffii</i>)
43 N	HPV6 Δ2-5 @n-term (codon optimized for <i>K. phaffii</i>)
44 N	HPV16 Δ2-5 @n-term (codon optimized for <i>K. phaffii</i>)
45 N	HPV16 Δ2-10 @n-term (codon optimized for <i>K. phaffii</i>)
46 N	HPV16 Δ2-15 @n-term (codon optimized for <i>K. phaffii</i>)
47 N	HPV16 Δ2-20 @n-term (codon optimized for <i>K. phaffii</i>)
48 N	HPV18 Δ2-5 @n-term (codon optimized for <i>K. phaffii</i>)
49 N	HPV18 Δ2-7 @n-term (codon optimized for <i>K. phaffii</i>)
50 N	HPV31 Δ2-5 @n-term (codon optimized for <i>K. phaffii</i>)
51 N	HPV33 Δ2-5 @n-term (codon optimized for <i>K. phaffii</i>)
52 N	HPV35 Δ2-5 @n-term (codon optimized for <i>K. phaffii</i>)
53 N	HPV45 Δ2-10 @n-term (codon optimized for <i>K. phaffii</i>)
54 N	HPV45 Δ2-17 @n-term (codon optimized for <i>K. phaffii</i>)
55 N	HPV45 Δ2-22 @n-term (codon optimized for <i>K. phaffii</i>)
56 N	HPV45 Δ2-31 @n-term (codon optimized for <i>K. phaffii</i>)
57 N	HPV52 Δ2-31 @n-term (codon optimized for <i>K. phaffii</i>)
58 N	HPV58 Δ2-10 @n-term (codon optimized for <i>K. phaffii</i>)
59 N	HPV58 Δ2-17 @n-term (codon optimized for <i>K. phaffii</i>)
60 N	HPV58 Δ2-22 @n-term (codon optimized for <i>K. phaffii</i>)
61 N	HPV58 Δ2-31 @n-term (codon optimized for <i>K. phaffii</i>)
62 N	HPV11 Δ2-3 @n-term (codon optimized for <i>K. phaffii</i>)
As-Is Nucleotide Sequences codon optimized for <i>H. sapiens</i>	
63 N	HPV6 (codon optimized for <i>H. sapiens</i>)
64 N	HPV16 (codon optimized for <i>H. sapiens</i>)
65 N	HPV18 (codon optimized for <i>H. sapiens</i>)
66 N	HPV31 (codon optimized for <i>H. sapiens</i>)
67 N	HPV33 Δ2-5 @n-term (codon optimized for <i>H. sapiens</i>)
68 N	HPV35 (codon optimized for <i>H. sapiens</i>)
69 N	HPV35 Δ2-5 @n-term (codon optimized for <i>H. sapiens</i>)
70 N	HPV45 Δ2-31 @n-term (codon optimized for <i>H. sapiens</i>)
71 N	HPV52 Δ2-31 @n-term (codon optimized for <i>H. sapiens</i>)
72 N	HPV58 Δ2-31 @n-term (codon optimized for <i>H. sapiens</i>)
73 N	HPV6
74 N	HPV16
75 N	HPV18 Δ2-66 @n-term
76 N	HPV31
77 N	HPV33 Δ2-5 @n-term
78 N	HPV35
79 N	HPV35 Δ2-5 @n-term
80 N	HPV45 Δ2-31 @n-term
81 N	HPV52 Δ2-31 @n-term
82 N	HPV58 Δ2-31 @n-term

SEQUENCE LISTING

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source                       1..500
                               mol_type = protein
                               organism = synthetic construct

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KVSQGYQYRV KVVLPDPNKF ALPDSSLFDP TQLRLVWACT GLEVGGRGQF
NKGDDVDVNSG SGGNPGQDNR VNVGMDYKQT QLCMVGCAAP LGEHWGKGK
CPPLELITSV IQDGDMDVDTG FGAMNFADLQ TNSKSDVPIDI CGTTCKYPD
RLFFFLRKEQ MFARHFFNRA GEVGEPVPDT LIIKSGNRT SVGSSIIYVN
QLFNPKPYWLQ KAQGHNGNIC WGNQLFVPTV TDTSTNTML CASVTTSST
HVEEYDLQFI FQLCSITLSA EVMAYIHTMN PSVLEDWNFG LSPPPNGTL
ITCQKPTPEK EKPPDPYKNLS FWEVNLKEKF SSELDDQYPLG RKFFLLQSGY
RPAVSKASA PKRKRAKTRK

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SEQ ID NO: 2 moltype = AA length = 505
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 source 1..505
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 2

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LVPKVSGLQY	RVFRIHLDPD	NKFGFPDTSF	YNPDTQRLVW	ACVGVEVGRG	QPLGVGISGH	120
PLLNLDDTE	NASAYAANAG	VDNRECISMD	YKQTQLCLIG	CKPPIGEHWG	KGSPCTNVAV	180
NPGDCPPLEL	INTVIQDGM	VDTGFGAMDF	TTLQANKSEV	PLDICTSICK	YPDYIKMVSE	240
PYGDSLFFYL	RREQMFVRHL	FNRAGAVGEN	VPDDLVIKGS	GSTANLASSN	YFPTPSGSMV	300
TSDAQIFNKP	YWLQRAQGHN	NGICWGNQLF	VTVDVTRST	NMSLCAAIST	SETTYKNTNF	360
KEYLRHGEEY	DLQFIFQLCK	ITLTADVMTY	IHSMNSTILE	DWNFGLQPPP	GGTLEDTYRF	420
VTSQAIACQK	HTPPAPKEDP	LKKYTFWEVN	LKEKFSADLD	QFPLGRKFLL	QAGLKAKPKF	480
TLGKRKATPT	TSSTSTTAKR	KKRKL				505

SEQ ID NO: 3 moltype = AA length = 507
 FEATURE Location/Qualifiers
 source 1..507
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 3

MALWRPSDNT	VYLPPPSVAR	VVNTDDYVTP	TSIFYHAGSS	RLLTVGPNPYF	RVPAGGGNKQ	60
DIPKVSAYQY	RVFRVQLPDP	NKFGLPDTSI	YNPETQRLVW	ACAGVEIGRG	QPLGVGLSGH	120
PFYNKLDDE	SSHAATSNVS	EDVRDNVSVD	YKQTQLCILG	CAPAIGEHWA	KGTACKSRPL	180
SQGDCCPPLEL	KNTVLEDGDM	VDTGYGAMDF	STLQDTKCEV	PLDICQSICK	YPDYLMQMSAD	240
PYGDSMFFCL	RREQLFARHF	WNRAGTMGDT	VPQSLYIKGT	GMRASPGSCV	YSPSPSGSIV	300
TSDSQLFNKP	YWLHKAQGHN	NGVCWHNQLF	VTVDVTRST	NLTICASTQS	PVPQGYDATK	360
FKQYSRHHVEE	YDLQFIFQLC	TITLTADVMS	YIHSMNSSIL	EDWNFGVPPP	PTTSLVDTYR	420
FVQSVAITCQ	KDAAPAENKD	PYDKLKFWNV	DLKEKFSLDL	DQYPLGRKFL	VQAGLRKRKPT	480
IGPRKRSAPS	ATTSSKPAKR	VRVRARK				507

SEQ ID NO: 4 moltype = AA length = 504
 FEATURE Location/Qualifiers
 source 1..504
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 4

MSLWRPSEAT	VYLPVPVPSK	VVSTDEYVTR	TNIYYHAGSA	RLLTVGHPYY	SIPKSDNPCK	60
IVVPKVSGLQ	YRVFRVRLPD	PNKFGFPDTS	FYNPETQRLV	WACVGLEVGR	GQPLGVGISG	120
HPLLNKFDDT	ENSNRYAGGP	GTDNRECISM	DYKQTQLCLL	GCKPPIGEHW	GKGSPCSNNA	180
ITPGDCPPLE	LKNSVIQDGD	MVDTGFGAMD	FTALQDTKSN	VPLDICNSIC	KYPDYLKMAV	240
EPYGDTLFFY	LRREQMFVRH	FFNRSGTVGE	SVPTDLYIKG	SGSTATLANS	TYFPTPSGSM	300
VTSDAQIFNK	PYWMQRAQGH	NNGICWGNQL	FVTVDVTRTS	TNMSVCAAIA	NSDTTFKSSN	360
FKEYLRHGEE	FDLQFIFQLC	KITLSADIMT	YIHSMNPAIL	EDWNFGLTTP	PSGSLEDTYR	420
FVTSQAITCQ	KTAPQKPKED	PFKDYVFWEV	NLKEKFSADL	DQFPLGRKFL	LQAGYRARPK	480
FKAGKRSAPS	ASTTTTAKRK	KTKK				504

SEQ ID NO: 5 moltype = AA length = 499
 FEATURE Location/Qualifiers
 source 1..499
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 5

MSVWRPSEAT	VYLPVPVPSK	VVSTDEYVSR	TSIYYAGSS	RL LAVGHPYF	SIKNPTNAKK	60
LLVPKVSGLQ	YRVFRVRLPD	PNKFGFPDTS	FYNPDTQRLV	WACVGLEIGR	GQPLGVGISG	120
HPLLNKFDDT	ETGNKYPGQP	GADNRECLSM	DYKQTQLCLL	GCKPPTGEHW	GKGVACTIONA	180
PANDCPPLEL	INTIIEDGDM	VDTGFGCMTD	KTLQANKSDV	PIDICGSTCK	YPDYLMKMTSE	240
PYGDSLFFFL	RREQMFVRHF	FNRAGTLGEA	VPDDLVIKGS	GTTASIQSSA	FFPTPSGSMV	300
TSESQLFNKP	YWLQRAQGHN	NGICWGNQVF	VTVDVTRST	NMTLCTQVTS	DSTYKNENFK	360
EYIRHVEEYD	LQFVFQLCKV	TLTAEVMTYI	HAMNPDILED	WQFGLTPPPS	ASLQDTYRFV	420
TSQAITCQKT	VPPKEKEDPL	GKYTFWEVDL	KEKFSADLDQ	FPLGRKFLLQ	AGLKAKPKLK	480
RAAPTSTRTS	SAKRKKVKK					499

SEQ ID NO: 6 moltype = AA length = 502
 FEATURE Location/Qualifiers
 source 1..502
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 6

MSLWRSNEAT	VYLPVSVSK	VVSTDEYVTR	TNIYYHAGSS	RL LAVGHPYY	AIKKQDSNKI	60
AVPKVSGLQY	RVFRVKLPDP	NKFGFPDTSF	YDPASQRLVW	ACTGVEVGRG	QPLGVGISGH	120
PLLNLDDTE	NSNKYVNGSG	TDNRECISMD	YKQTQLCLIG	CRPPIGEHWG	KGTPCNANQV	180
KAGECPPLEL	LNTVLQDGM	VDTGFGAMDF	TTLQANKSDV	PLDICSSICK	YPDYLMKMVSE	240
PYGDMLEFFYL	RREQMFVRHL	FNRAGTVGET	VPADLYIKGT	TGTLPTSTSYF	PTPSGSMVTS	300
DAQIFNKPYW	LQRAQGHNNG	ICWSNQLFVT	VVDVTRSTNM	SVCSAVSTSD	STYKNDNFKE	360

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YLRHGEEDL	QFIFQLCKIT	LTADVMTYIH	SMNPSILEDW	NFGLTPPPSG	TLEDTRYVT	420
SQAVTCQKPS	APKPKDDPLK	NYTFWEVDLK	EKFSADLDQF	PLGRKFLLQA	GLKARPNFRL	480
GKRAAPASTS	KKSSTKRRKV	KS				502

SEQ ID NO: 7 moltype = AA length = 539
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 source 1..539
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 7

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YHAGSSRLLT	VGNPYFRVVP	NGAGNKQAVP	KVSAYQYRVF	RVALPDPNKF	GLPDSTIYNP	120
ETQRLVWACV	GMEIGRGQPL	GIGLSGHPFY	NKLDDTESAH	AATAVITQDV	RDNVSVQYKQ	180
TQLCILGCVF	AIGEHWAKGT	LCKPAQLQPG	DCPPLELKNT	IIEDGDMVDT	GYGAMDFSTL	240
QDTKCEVPLD	ICQICKYKPD	YLQMSADPYG	DSMFFCLRRE	QLFARHFVNR	AGVMGDTVPT	300
DLVIKGTSAV	MRETPGSCVY	SPSPSGSIIT	SDSQLFNKPY	WLHKAQGHNN	GICWHNQLFV	360
TVVDTRSTN	LTLCASTQNP	VPSTYDPTKF	KQYSRHVEEY	DLQFIFQLCT	ITLTAEVMSY	420
IHSMNSSILE	NWNFGVPPPP	TTSLVDTYRF	VQSVAVTCQK	DTTPPEKQDP	YDKLKFWTV	480
LKEKFSDDL	QYPLGRKFLV	QAGLRRRPTI	GPRKRPAAST	STASTASRA	KRVIRIRSKK	539

SEQ ID NO: 8 moltype = AA length = 529
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 mol_type = protein
 organism = synthetic construct

SEQUENCE: 8

MVQILFYILV	IFYVAVGVNV	FHIFLQMSVW	RPSEATVYLP	PVPVSKVVST	DEYVSRTSIY	60
YYAGSSRLLT	VGHYPYFSIKN	TSSGNKKVL	VPKVSGLQYR	VFRIKLDPDN	KFGFPDTSFY	120
NPETQRLVWA	CTGLEIGRGQ	PLGVGISGHP	LLNKFDDTET	SNKYAGKPGI	DNRECLSMYD	180
KQTQLCILGC	KPPIGEHWGK	GTPCNNSGN	PGDCPPLQLI	NSVIQDGMV	DTGFGCMDFN	240
TLQASKSDVP	IDICSSVCKY	PDYLMASEP	YGDSLFFFLR	REQMFVRHFF	NRAGTLGDPV	300
PGDLIYQGSN	SGNTATVQSS	AFPTTPSGSM	VTSESQLPNK	PYWLQRAQGH	NNGICWGNQL	360
FVTVDTTTRS	TNMTLCAEVK	KESTYKNENF	KEYLRHGEEF	DLQFIFQLCK	ITLTADVMTY	420
IHKMDATILE	DWQFGLTPPP	SASLEDTYRF	VTSTAITCQK	NTPPKGKEDP	LKDYMFWEVD	480
LKEKFSADLD	QFPLGRKFL	QAGLQARPKL	KRPASSAPRT	STKKKKVKR		529

SEQ ID NO: 9 moltype = AA length = 524
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 mol_type = protein
 organism = synthetic construct

SEQUENCE: 9

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YYAGSSRLLA	VGNPYFSIKS	PNNNKKVLVP	KVSGQLQYRVF	RVRLPDPNKF	GFPDTSFYNP	120
DTQRLVWACV	GLEIGRGQPL	GVGVSGHPYL	NKFDDTETSN	RYPAPQGSND	RECLSMYDKQ	180
TQLCLIGCKP	PTGEHWGKGV	ACNNNAATD	CPPLELFNSI	IEDGDMVDTG	FGCMDFTGLQ	240
ANKSDVPIDI	CNSTCKYPDY	LKMASEPYGD	SLFFFLRREQ	MFVRHFFNRA	GKLGEAVPDD	300
LYIKGSGNTA	VIQSAFFPT	PSGSIVTSES	QLFNKPYWLQ	RAQGHNNGIC	WGNQLFVTVV	360
DTTRSTNMTL	CTEVTKEGTY	KNDNFKEYVR	HVEEYDLQFV	FQLCKITLTA	EIMTYIHTMD	420
SNILEDWQFG	LTPPPASLQ	DTYRFVTSQA	ITCQKTAPPK	EKEDPLNKYT	FWEVNLKEKF	480
SADLDQFPLG	RKFLQSGLK	AKPRLKRSAP	TTRAPSTKRR	KVKK		524

SEQ ID NO: 10 moltype = AA length = 501
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 mol_type = protein
 organism = synthetic construct

SEQUENCE: 10

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KVSGYQYRVF	KVVLDPDNKF	ALPDSSLFDP	TTQRLVWACT	GLEVGRGQPL	GVGVSGHPLL	120
NKYDDVENSG	GYGNGPGQDN	RNVVGMQYKQ	TQLCMVGCAP	PLGEHWGKGT	QCSNTSVQNG	180
DCPPLELITS	VIQDGMVDT	GFGAMNFADL	QTNKSDVPLD	ICGTVCCKYPD	YLQMAADPYG	240
DRLFYFLRKE	QMFARHFVNR	AGTVGEPVDP	DLVVKGGNNR	SSVASSIYVH	TPSGSLVSSE	300
AQLFNKPYWL	QKAQGHNNGI	CWGNHLFVTV	VDTRSTNMT	LCASVSKSAT	YTNSDYKEYM	360
RHVEEFDLQF	IFQLCSITLS	AEVMAYIHTM	NPSVLEDWNF	GLSPPPNGTL	EDTRYVQSQ	420
AITCQKPTPE	KEKQDPYKDM	SFWEVNLKEK	FSSELDQFPL	GRKFLLQSGY	RGRTSARTGI	480
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 mol_type = other DNA
 organism = synthetic construct

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gcttttgccag	actcttcctt	gttcgaccca	actactcaga	gattgggttg	ggcttgtagc	300
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aaacaagtacg	acgacgttga	aaactccggt	tctgggtggtg	accaggtca	agacaacaga	420
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tgtgcctcgg	ttactacctc	ctccacttac	actaactccg	actacaaga	gtacatgaga	1080
cacgtcgaag	agtacgacct	gcagttcatt	ttccagctgt	gttccatcac	tttgtccgct	1140
gaggttatgg	cctacatcca	cactatgaac	ccatccggtt	tggaggactg	gaacttcggt	1200
ttgtctccac	cacctaaccg	tactttggag	gatacctaca	gatacgttca	gtcccagggt	1260
atcacttgcc	aaaagccaac	tccagagaaa	gagaagccag	atccttaca	gaacctgtcc	1320
ttctgggaag	tcaacctgaa	agaaaagttc	tccagcgcag	tggatcagta	cccatggggt	1380
agaaagttct	tgtgcagtc	cggttacaga	ggtagatcct	ccattagaac	cgggtgttaag	1440
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tag						1503

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 source 1..1518
 mol_type = other DNA
 organism = synthetic construct

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agattgctgg	ctgtttggtca	tccatacttc	ccaatcaaga	agccaaacaa	caacaagatc	180
ctgggtgccaa	aggtttccgg	tctgcagtac	agagttttca	gaatccactt	gccagatcca	240
aaacaagttcg	gtttcccgaga	cacatccttc	tacaaccagc	acactcagag	attggttttg	300
gcctgtgttg	gtgttgaaag	tggtagaggt	caaccacttg	gtgttggtat	ttctgggtcac	360
ccattgtctga	acaagttgga	cgatactgaa	aacgcttccg	cttacgctgc	taacgctggg	420
gttgataaca	gagaatgcat	ctccatggac	tacaagcaga	cccagttgtg	tttgatcggg	480
tgcaggccac	catattggtga	acattggggg	aagggttctc	catgtacca	cgttgctggt	540
aaaccaggtg	attgtccacc	attggagctg	atcaacactg	ttattcagga	cggtgacatg	600
gtcgacactg	gttttggtgc	tatggacttc	actaccttgc	aggctaacaa	gtccgaaggt	660
ccattggaca	tctgtacctc	catctgcaag	taccagact	acatcaagat	gggtgtctgag	720
ccatacgggtg	actccctggt	cttctacttg	agaagagaac	agatgttcgt	caggcacctg	780
ttcaacagag	ctggtgctgt	tggtgaaaac	gttccagacg	acctgtacat	caaaggttct	840
ggttctactg	ctaaccctgc	ctcctctaac	tactttccaa	ctccatccgg	ttccatgggt	900
acttccgacg	ctcagatttt	caacaagcca	tactgggtgc	aaagagccca	gggtcacaac	960
aacggtatct	gttggggtaa	ccagttgttc	gttactgttg	tcgacactac	cagatccacc	1020
aaacatgtctt	tgtgtgtctg	tcttccact	tccgagacta	cctacaagaa	cactaacttc	1080
aaagaataacc	tgaggcacgg	tgaagagtac	gacttgcagt	tcattttcca	gctgtgcaag	1140
atcactttga	cgcgcgacgt	tatgacttac	atccactcca	tgaactccac	tatcttggag	1200
gattggaaact	ttgggtctgc	accaccacca	ggtggttact	tgaagatata	ctacagattc	1260
gttacttccc	aggtatctgc	ttgtcagaag	cacactccac	cagctccaaa	agaggatcca	1320
ttgaagaagt	acactttctg	ggaagtcaac	ctgaaagaga	agttctccgc	tgacttggat	1380
cagttcccac	tgggtagaaa	gttcttgggt	caggctgggt	tgaaggccaa	gccaaagttt	1440
accttgggta	agagaaaggg	tactccaact	acttctccca	cctctactac	cgttaagagg	1500
aagaagagaa	agttgtag					1518

SEQ ID NO: 13 moltype = DNA length = 1524
 FEATURE Location/Qualifiers
 source 1..1524
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 13

atggccctgt	ggcgtccatc	cgacaacact	gtttatttgc	caccaccatc	cgttgccaga	60
gttgttaaca	ctgacgacta	cgttactccc	acctccattt	tctaccacgc	tggttcttct	120
agattgctga	ccgttggtta	cccatacttc	agagttccag	ctggtgggtg	taacaagcag	180
gacattccaa	aggtttccgc	ctaccagtac	agagtgttca	gagtccaatt	gccagatcca	240
aaacaagttcg	gtttgccaga	cacctccatc	tacaaccagc	agactcaaa	attggttttg	300
gcttgtgctg	gtgtcgagat	tggttagaggt	caaccacttg	gtgttggttt	gtctggtcac	360
ccattctaca	acaagttgga	gcagactgaa	tcttcccacg	ctgctacttc	taacgtttcc	420
gaggatgtca	gagacaacgt	gtccgttgac	tacaagcaga	cccagttgtg	tatcttgggt	480
tgtgtccag	ctattgtgga	acattgggct	aagggaactg	cctgtaagtc	cagaccattg	540
tctcaagggt	attgcccacc	attggagttg	aagaacaccg	ttttggagga	cggtgacatg	600
gttgatactg	gttacggtgc	tatggacttc	tccaccttgc	aagacactaa	gtgtgaggtc	660
ccattggaca	tctgtcagtc	catctgtaag	taccagact	acttgcagat	gtccgctgat	720

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ccatacgggtg actctatggt cttctgctg agaagagagc agttgttcgc tagacacttc 780
tggaacacagag ctggtactat ggggtgacact gttccacagt ccttgtagat caaaggtact 840
ggtagtgagag cttccccagg ttctctgtgt tactctccat ctccatccgg ttccatcggt 900
acttccgatt ctcagctggt caacaagcca tactgggtgc acaaggctca gggtcataac 960
aacgggtgttt gttggcacaa ccagctgttc gttactgttg ttgacactac cagatccacc 1020
aacttgacta tctgtgtctt cactcaatcc ccagttccag gtcaatacga cgctactaag 1080
ttcaagcagt actccagaca cgctgaagag tacgacttgc agttcatctt ccagttgtgc 1140
actatcactt tgaccgcga cgttatgttc tacatccact ctatgaacct ctccatcttg 1200
gaggactgga actttgtgtt tctctctctt ccaactactt ccttggttga cactacaga 1260
ttcggtcagt ccgtcgctat cacttgtcaa aaggatgctg ctctgctga gaacaaggac 1320
ccatacagata agctgaagtt ctggaacgtc gacctgaaag agaagttctc cttggacttg 1380
gatcagtagc cactgggtag aaagtctctg gtccaagccg gtttgagaag aaagccaact 1440
atcggtccaa gaaagagatc tgctccttcc gctactactt cttccaagcc agctaagaga 1500
gttagagtca gagccagaaa gttag 1524

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SEQ ID NO: 14      moltype = DNA length = 1515
FEATURE            Location/Qualifiers
source              1..1515
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 14
atgtcttgtt ggcgtccatc tgaggctacc gtttacttgc caccagttcc agtttccaag 60
gttgtttcca ctgacgagta cgtccaccagg accaacatct actaccatgc tggttctgct 120
agattgctga ctggttggtca ccttacttac tccattccaa agtctgacaa cccaagaag 180
atcgctgctc caaagggttc cggactgcag tacagagtgt tcagagtcag attgccagat 240
ccaaacaagt tcggtttccc cgacacatcc ttctacaacc cagagactca aagattgggt 300
tggtcctgtg ttggtttgga ggttggtaga ggtcaaccac ttggtgttgg tatctcgggt 360
catcaactgc tgaacaagtt tgacgacact gagaactcca acagatacgc tgggtgtcca 420
ggtactgaca acagagaatg tatctccatg gactacaagc agaccagtt gtgtttgttg 480
ggttgcaagc cactatttgg tgaacattgg ggtaagggtt ctccatgttc caacaacgct 540
attaccocag gtgactgacc accattggag ttgaagaact ccgttattca ggacgggtgac 600
atggttgaca ctggtttcgg tgctatggat ttcaactgct tgcaagacac caagtccaac 660
gttccattgg acatctgtaa ctccatctgc aagtaccag actacttgaa gatggttgc 720
gagcctacag gtgacacatt gttctctac ctgagaagag aacagatggt cgtcagacac 780
ttcttcaaca gatccggtac ttgttggtgag tccgttccaa ctgacctgta catcaaaagt 840
tctggttcca ctgctacttt ggcacactct acttacttcc caactccatc cgttccatg 900
gttacttccg acgctcagat ttccaacaag ccatactgga tgcaaaagag ccagggtcac 960
aacaacggta tttgttgggg taaccagtgg ttgcttacgg ttgtcgacac taccagatcc 1020
accaacatgt ctggtttgtg tgctatcgct aactccgaca ccactttcaa gtccccaac 1080
ttcaagaagt acctgagaca cgttgaagag ttgcacctgc agttcatatt ccagctgtgc 1140
aagatcactt tgcgcgcga catcatgacc tacatccact ctatgaacct agccatcttg 1200
gaggattgga actcgggttt gactactcca ccatctggat ccttgaggga cactacaga 1260
ttcggttact ttcaggcact cactgttcaa aagactgctc cacaaaagcc aaaagaggac 1320
ccattcaagg actacgtgtt ctgggaagtc aacttgaag agaagttctc cgtgacctg 1380
gatcagttcc cattgggtag aaagtctctg ttgcaggctg gttcacagag cagaccaaaag 1440
tttaaggctg gtaagagatc tgctccttcc gcttctacta ctacccagc caagagaaaag 1500
aaaaactaaga gtaa 1515

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SEQ ID NO: 15      moltype = DNA length = 1500
FEATURE            Location/Qualifiers
source              1..1500
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 15
atgtctgttt ggaggccatc tgaggctacc gtttacttgc caccagttcc agtttccaag 60
gttgtttcca ctgacgagta cgtttccaga acctccatct actactacgc tggttcttcc 120
agattgctgg ctggttggtca cccatacttc tccatcaaga acccaactaa cgccaagaag 180
ttgctgtgct caaagggttc tggattgcag tacagagtgt tcagagtcag attgccagat 240
ccaaacaagt tcggtttccc cgacacatcc ttctacaacc cagacactca aagattgggt 300
tggtcctgtg ttggtttgga gatttggtaga ggtcaaccac ttggtgttgg tatctctggt 360
caccctttgc tgaacaagtt tgacgacact gagactggta acaagtaacc aggtcaacct 420
ggtgctgaca acagagaatg tctgtctatg gactacaagc agaccagtt gtgtttgttg 480
ggttgtaagc caccactgg tgaacattgg ggtaagggtg ttgctgttac taatcgtgct 540
ccagtaacg actgtccacc attggagtgg attaacacca tcatcgagga cggtagacatg 600
gttgatactg gttcgggtt catggacttc aagaccttgc aagctaacaa gtccgacgtg 660
ccaattgaca tctgtgttcc cacttgttaag taccocgact acttgaagat gacctctgag 720
ccatacgggt actccctgtt cttttctctg cgtcgtgaac agatgttcgt cagacacttc 780
ttcaacagag ccggtacttt ggggtgaagct gttccagacg acctgtacat caaaggttct 840
ggtactaccg cttccattca gtctctgct ttcttccaa ctccatccgg ttctatgggt 900
acttctgagt cccagctgtt caacaagcca tactgggtgc aaagagccca gggtcacaac 960
aacggtagct tgtgggttaa ccagggttttc gttactgttg ttgacactac cagatccacc 1020
aacatgactt ggtgtactca ggttacttcc gactccacct acaagaacga gaacttcaaa 1080
gagtacatca ggcacgtcga agagtacgac ttgcagttcg ttttccagtt gtgcaagggtc 1140
actttgaccg ctgaggttat gacttacatc cagctatga accccgacat ttggaggat 1200
tggaattcgg gtttgactcc accaccatct gcttctctgc aagacaccta cagattcgtt 1260
acttcccagg ccatcacctg tcaaaagacc gttccaccaa aagagaaaag ggaccattg 1320

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-continued

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ggcaagtaca ctttctggga agttgacctg aaagagaagt tctccgctga cttggatcag 1380
ttcccactgg gtagaaagt cttgttgacg gctgggttga aggccaaagg aaagttgaaa 1440
agagctgctc caacctccac tagaacctct tccgctaaaa gaaagaaggt caagaagtaa 1500

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SEQ ID NO: 16      moltype = DNA length = 1509
FEATURE           Location/Qualifiers
source            1..1509
                  mol_type = other DNA
                  organism = synthetic construct

```

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SEQUENCE: 16
atgtcccttt ggagatccaa cgaggctacc gtttacttgc caccagtttc cgtttccaag 60
gttgtttcca ctgacgagta cgtcaccagg accaacatct actaccatgc tggttcttcc 120
agattgctgg ctgttggtca tcttactac gccattaaga agcaggactc caacaagatc 180
gccgttccaa aggtttctgg tctgcagta agagtgttc gagtcaagt gccagatcca 240
aacaagttcg gtttcccga cacttcttct taccacctg cttctcaaag attggtttgg 300
gcttgtaacc gtgttgaggt tggtagaggt caaccacttg gtgttggtat ttctggtcac 360
ccattgctga acaagttgga cgacactgag aactctaaca agtacgtcg taactccggt 420
actgacaaca gagaatgtat ctccatggac tacaagcaga cccagttgtg tttgatcggt 480
tgcagaccac ctattggtga acattggggg aagggtactc catgtaacgc caaccagggt 540
aaggctgggt aatgtccacc attggagttg ttgaacactg tcttgacgga cggtgacatg 600
gttgatactg gtttcggtgc tatggacttc actaccttgc aagctaacaa gtccgacgtt 660
ccattggaca tctgttccct catctgtaag taccagact acctgaagat ggtgtctgaa 720
ccatacggcg acatgttgtt cttctacctg agaagagaa aaatgttcgt caggcacctg 780
ttcaacagag ctggtactgt tggtagaaact gttccagccg acttgatcat caagggtact 840
actggtactt tgcctccac ctttacttcc ccaactccat ctggttccat ggttacttcc 900
gacgtcaga ttctcaacaa gccatactgg ttgcaaaagag cccagggtca caacaacggt 960
atctgttgtt ctaaccagtt gttcgttacc gttgtgcaga ctaccagatc caccacatcg 1020
tctgtttgtt ccgctgtttc tacctccgac tccacttaca agaacgacaa cttcaaaaga 1080
taacctgagac acggtgaaga gtaacgacct cagttcattt tccagctgtg caagatcact 1140
ttgaccgccc acgttatgac ctacatccac tctatgaacc catccatctt ggaggactgg 1200
aaactcgggt tgaactccac accatctgga actttggagg acacctacag atacgtcact 1260
tcccaggctg ttacttgtca aaagccatct gctccaaagc caaaggacga cccattgaag 1320
aactacactt tctgggaagt cgacctgaaa gaaaagttct ccgctgactt ggatcagttc 1380
ccactgggta gaaagttctt gttgcaggct ggtttgaagg ccagaccaaa cttcagattg 1440
ggtaagagag ctgctccagc ttccacttct aagaagtcct ccaccaagag aagaaagggtc 1500
aagtcttaa 1509

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SEQ ID NO: 17      moltype = DNA length = 1620
FEATURE           Location/Qualifiers
source            1..1620
                  mol_type = other DNA
                  organism = synthetic construct

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SEQUENCE: 17
atggcccaca acattatcta cggtcacggc atcatcatct ttctgaagaa cgtcaacgtg 60
ttcccactct tcttgagat ggctttgtgg cgtccatctg actccactgt ttatttgcca 120
ccaccatccg ttgcccaggt tgtttctact gacgactacg ttccaggac ctccatttcc 180
taccacgctg gttcttccag attgctgact gttggttaacc catacttcag agtcgttcca 240
aacggtgctg gtaacaagca agctgttcca aaggtttccg cctaccagta cagagtttcc 300
agagttgctt tgccagatcc aaacaagttc ggtttgccag actctaccat ctacaaccca 360
gagactcaga tagtggtttg ggctgtgtt ggtatggaaa tcggtagagg tcagccactt 420
ggtattggtt tgtctgttca cccattctac aacaagttgg acgacactga atctgtctac 480
gctgtactcg ctgttatcac tcaagacgtc agagacaacg ttccgtcga ctacaagcag 540
accagttgtg gtatcttggg ttgtgttcca gctattggtg agcattgggc taagggtact 600
ttgtgtaagc cagctcaatt gcagccaggt gattgtccac cattggagtt gaagaacacc 660
atcattgagg acggtgacat ggttgacact ggttacggtg ctatggactt ctctacctg 720
caggacacta agtgtgaggt cccattggac atttgtcagt ccatctgtaa gtaccccgac 780
tacttgcaaa tgtccgctga tccatcaggt gactctatgt tcttctgctt gagaagagag 840
cagttgttcc ctagacactt ctggaacaga gctgggtgta tgggtgacac tgttccaacc 900
gacctgtaca ttaaggaac ttccgctaac atgagagaaa cccaggttc ctgtgtttac 960
tctccatctc catccggttc catcatcact tccgattctc agctgttcaa caagccatac 1020
tgggtgcaca aggtctaggg tcacaacaac ggtatctgtt ggcacaacca gctgttcggt 1080
actgttgttg aactaccag atccaccaac ttgactttgt gtgcctccac tcaaaaccca 1140
gttccatcta cttacgacc caccaagttc aagcaatact ccagacacgt tgaagagtag 1200
gacctgcagt tcatcttcca gttgtgcact atcactttga ccgcccaggt tatgtcctac 1260
atccactcta tgactctctc catcttggag aactggaact tgggtgttcc tctctctcca 1320
actacttctc tgggtgatac ctacagattc gttcagtcct ttgcccgtac ttgtcagaag 1380
gacactactc caccagagaa gcaagaccca taccgacagt tgaagttctg gaccgtcgac 1440
ctgaaagaga agttctctc tgacttggat caataccac tgggtagaaa gttcttggtc 1500
caggctgggt tgagaagaag gccaaactatc ggtccaagaa aaagaccagc tgccttctact 1560
tctactgctt ccactgtctc tagaccagcc aagagagtta gaatcagatc caagaagtaa 1620

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SEQ ID NO: 18      moltype = DNA length = 1590
FEATURE           Location/Qualifiers
source            1..1590
                  mol_type = other DNA
                  organism = synthetic construct

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SEQUENCE: 18

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atggtccaga tctgttctca catcctgggc atctttctact acgttgccgg tgtcaacgtg 60
ttccacatct tcttgcaaat gtcggtttgg aggccatccg aggcctactgt ttatttgcca 120
ccagttccag tctccaaggt tgtttctact gacgagtagc tttccaggac ctccatctac 180
tactatgctg gttcttccag attgctgacc gttggtcacc catacttctc tatcaagaac 240
acctcttccg gtaacggtaa gaaggtcttg gtcccaagg tttccggatt gcagtagaca 300
gtgttcagaa tcaagctgcc agatccaaac aagttcgggt tcccagacac atccttctac 360
aaccagagag ctccagagatt ggtttgggct tgtactgggt tggagatcgg tagaggtcaa 420
ccacttggtg ttggtatttc tggtcaccct ctgctgaaca agtttgacga cactgagact 480
tccaacaagt acgctggtta gccaggtatc gacaacagag aatgcctgtc tatggactac 540
aagcagaccc agttgtgtat cttgggttgc aagccaccta ttggtgaaca ttggggttaag 600
ggtactccat gcaacaacaa ctctggtaac cctggtgatt gtccaccatt gcagttgatt 660
aactccgtca ttcaggacgg tgacatggtc gatactgggt tcggttggtat ggacttcaac 720
accttgcaag cttccaaatt cgacgtgcca attgacattt gttcctcgt ctgtaagtac 780
cccgactact tgcagatggc ttttgaacca tacggtgact cctgttctt tttcttgctg 840
cgtgaacaga tgttcgtcag acacttcttc aacagagccg gtactttggg tgatccagtt 900
cctggtgact ttgtacattca aggttccaac tccggttaaca ccgctactgt tcaactcttc 960
gctttcttcc caactccatc cgtttctatg gttacttctg agtcccagct gttcaacaag 1020
ccatactggt tgcaagagc ccagggtcac aacaacggta tctgttgggg taaccagttg 1080
ttcgttaccg ttgtcgcacac taccagatcc accaacaatga ctttgtgtgc cgaggtcaag 1140
aaagagtcca cctacaagaa cgagaacttc aaagaatacc tgagacacgg tgaagagttc 1200
gacctgcagt tcattttcca gctgtgcaag atcactttga ccgcccagct tatgacctac 1260
atccacaaga tggacgctac cactttggag gactggcaat tccggttgac tctctctct 1320
tctgcttctc tggagatcac ctacagattc gttacttcca ccgccaatcac ctgtcagaag 1380
aacactccac caaagggtta agaggacca ctgaaggatt acatgttctg ggaagtcgac 1440
ctgaaagaga gaaactctgc tgacttggat cagttccac tgggtagaaa gttcttggtg 1500
caagctggtc tgcaggccag accaaagttg aaaagacctg cttcttccgc tccaagaact 1560
tccaccaaga agaagaaggt taagagataa 1590
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SEQ ID NO: 19

moltype = DNA length = 1575

FEATURE

Location/Qualifiers

source

1..1575

mol_type = other DNA

organism = synthetic construct

SEQUENCE: 19

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atggtcttga tcttgtgttg caccttggcc atcttgttct gtgttgctga cgtcaacgtg 60
ttccacatct tcttgcaaat gtcggtttgg aggccatccg aggcctactgt ttatttgcca 120
ccagttccag tctccaaggt tgtttctact gacgagtagc tttccaggac ctccatctac 180
tactacgctg gttcttctag attgctggcc gttggttaacc catacttctc cattaaagtc 240
ccaaacaaca caaagaaggt gctggttcca aaggtttccg gattgcagta cagagtgttc 300
agagtcagat tgcagatcc aaacaagttc ggtttcccg acacatcctt ctacaaccca 360
gacactcaaa gattggtttg ggcctgtgtt ggtttggaga ttggtagagg tcaaccactt 420
ggtgttggtg tttctgggtc ccatacctg aacaagtttg acgacactga gacttccaac 480
agataccag ctcaaccagg ttcgacacac agagaatgtc tgtctatgga ctacaagcag 540
acctagttgt gctcgatcgg ttgtaagcca ccaactggtg aacattgggg taagggtgtt 600
gcctgttaaca acaacgctgc tgcactgat tgtccacat tggagctgtt caactccatt 660
attgaggacg gtgacatggt cgacactggt ttcggttgta tggatttcgg tactctgcag 720
gctaacaagt ccgacgttcc aatcgacatc tghtaactcca cctgtaagta ccagactac 780
ctgaagatgg cttctgagcc atacggtgat tccctgttct ttttcttgcg tctgtgacag 840
atgttcgtca gacacttctt caacagagcc ggtaagttgg gtgaagctgt tccagacgac 900
ctgtacatca aaggttctcg taacacccgt gttattcagt cctctgtttt cttcccaact 960
ccatccggtt ctatcggttc tcttgagtcc cagttgttca acaagccata ctggttgcaa 1020
agagcccagg gtcacaacaa cggtatattt tggggttaacc agctgttcgt tactgttgtt 1080
gacactacca gatccaccaa catgaccttg tgtaccgagg ttaccaaaaga gggaacctac 1140
aagaacgaca acttcaaaaga atacgtcagg cagctcgaag agtacgactt gcaattcgtt 1200
ttccagctgt gcaagatcac tttgaccgct gagatcatga cctacatcca cactatggac 1260
tccaacatct tggaggactg gcaattcgggt ttgactccac caccatctgc ttctctgcaa 1320
gacacctaca gattcgtcac ttcccaggct atcacctgtc aaaagactgc tccaccaaaa 1380
gagaaagagg acccattgaa caagtacact ttctgggaag tcaacctgaa agagaagttc 1440
tccgctgact tggatcagtt cccactgggt agaaagttct tgttgacgtc cggtttgaag 1500
gccaagccaa gattgaagag atccgctcca actactagag ccccatccac taagagggaag 1560
aaggttaaga agtaa 1575
```

SEQ ID NO: 20

moltype = DNA length = 1506

FEATURE

Location/Qualifiers

source

1..1506

mol_type = other DNA

organism = synthetic construct

SEQUENCE: 20

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atgtggcgct cctccgactc cactgtttat gttccaccac caaatccagt ctccaaggtt 60
gttgctactg acgcctactg caagaggacc aatattttct accacgcctc ctcatccaga 120
ttgctggctg ttggtcatcc ttactactcc atcaagaagg tcaacaagac cgtcgttcca 180
aaggtttccg gttaccagta cagagtgttc aaggtcgttt tgccagatcc aaacaagttc 240
gctttgctg actcttctt gttcgacca actactcaga gattggtttg ggcttgtacc 300
ggtttggaag ttggtagagg tcaaccactt ggtgttggtg tttctggtea cccattgctg 360
aacaagtacg acgacgttga aaactccggt ggttacggtg gtaaccagg tcaagataac 420
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agagtcaacg tcggtatgga ctacaagcag acccagttgt gtatgggttg ttgtgtcca 480
ccattgggtg aacattgggg taagggtact cagtggtcca acatttcgt tcagaacggt 540
gattgtccac ctttggagtt gatcacctcc gttattcaag acggtgacat ggttgacact 600
ggtttcggtg ctatgaactt cgtgacttg cagactaaca agtccgacgt tcctttggac 660
atctgtggta ctgtctgtaa gtaccagac tacttgcaaa tggctgctga tccatacgg 720
gacagactgt tctctacact gagaaaagaa cagatgttcg ccagacactt cttcaacaga 780
gctggtagtg ttggtgaacc agttccagac gacttggttg tcaaaaggtg taacaacaga 840
tcctccgttg cttcctctat ctacgttcac actccatccg gttccttggt ttcttctgaa 900
gcccagttgt tcaacaagcc atactggttg caaaaggccc agggtcacaa caacggtatt 960
tgttggggtt accacttggt cgttaccgtt gtcgacacta ccagatccac caatatgact 1020
ttgtgtgctt ccgtttccaa gtcgctact tacactaact ccgactacaa agagtacatg 1080
agacacgtcg aagagttcga cctgcagttc attttccagc tgtgttccat tactttgtcc 1140
gcccaggtta tggcctacat ccacactatg aacctatccg ttttggagga ctggaacttc 1200
ggtttgtctc caccacctaa cggtagtttg gaggatacct acagatacgt tcagtcaccg 1260
gctatcactt gccaaaagcc aactccagag aaagagaagc aggaccata caaggacatg 1320
tctttctggg aagtcacact gaaagaaaag ttctcctcag agctggatca gttccactg 1380
ggtagaaagt tcttggtgca gtccggttac agaggtagaa cttccgctag aactggtatc 1440
aagaggcctg ctgtttctaa gccatctact gcccacaaaga gaaagcgtac caagaccaag 1500
aaatag 1506

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SEQ ID NO: 21      moltype = AA length = 498
FEATURE           Location/Qualifiers
source            1..498
                  mol_type = protein
                  organism = synthetic construct

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SEQUENCE: 21
MPSDSTVYVP PPNPVSKVVA TDAYVTRTNI FYHASSSRLL AVGHPYFSIK RANKTVVPKV 60
SGYQYRVFKV VLPDPNKFAL PDSSLFDPPT QRLVWACTGL EVGRGQPLGV GVGHPFLNKL 120
YDDVENSMSG GNPQGDNRVN VGMDYKQTQL CMVGCAPPLG EHWGKGKQCT NTPVQAGDCP 180
PLELITSVIQ DGMVDVTGFG AMNFADLQTN KSDVPIDICG TTCKYPDYLO MAADPYGDR 240
FFFLRKEQMF ARHFFNRAGE VGEFVPDTLI IKSGSNRTSV GSSIVVNTPS GSVLSSEAQL 300
FNKPYWLQKA QGHNNIGICW NQLFVTVDVT TRSTNMTLCA SVTTSSTYT N SDYKEYMRHV 360
EEYDLQFIFQ LCSITLSAEV MAYIHTMNPS VLEDWNFGLS PPPNGTLEDY YRYVQSQAIT 420
CQKPTPEKEK PDPYKNLSFW EVNLKEKFS ELDQYPLGRK PLLQSGYRGR SSIRTGVKRP 480
AVSKASAAPK RKRAKTKR 498

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SEQ ID NO: 22      moltype = AA length = 496
FEATURE           Location/Qualifiers
source            1..496
                  mol_type = protein
                  organism = synthetic construct

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SEQUENCE: 22
MDSTVYVPPP NPVSKVATD AYVTRTNIFY HASSSRLLAV GHPYFSIKRA NKTVPKVS 60
YQYRVFKVVL PDPNKFALPD SSLFDPPTQR LVWACTGLEV GRGQPLGVGV SGHPFLNKYD 120
DVENSMSGGN PGQDNRVNVG MDYKQTQLCM VGCAPPLGEH WGKGKQCTNT PVQAGDCPPL 180
ELITSVIQDG DMVDVTGFGAM NFADLQTNKS DVPIDICGTT CKYPDYLOMA ADPYGDR 240
FLRKEQMFAR HFFNRAGEVG EPVPDTLIIK GSGNRTSVGS SIYVNTPSGS LVSSEAQLFN 300
KPYWLQKAGG HNNGICWGNQ LFTVTVDVTR STNMTLCASV TTSSTYTNSD YKEYMRHVEE 360
YDLQFIFQLC SITLSAEVMA YIHTMNPSVL EDWNFGLSPP PNGTLEDYR YVQSQAITCQ 420
KPTPEKEKPD PYKNLSFEV NLKEKFSSEL DQYPLGRKFL LQSGYRGRSS IRTGVKRP 480
SKASAAPKRK RAKTKR 496

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SEQ ID NO: 23      moltype = AA length = 501
FEATURE           Location/Qualifiers
source            1..501
                  mol_type = protein
                  organism = synthetic construct

```

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SEQUENCE: 23
MPSEATVYLP PVPVSKVST DEYVARTNIY YHAGTSRLLA VGHYPFPIKK PNNKILVPK 60
VSGLQYRVFR IHLDPNKFPG FPDTSFYNDP TQRLVWACVG VEVGRGQPLG VGISGHPLLN 120
KLDDTENASA YAANAGVDNR ECISMDYKQT QLCLIGCKPP IGEHWGKGSP CTNVAVNPGD 180
CPPLELINTV IQDGMVDVTG FGAMDFTTLQ ANKSEVPLDI CTSICKYPDY IKMVSEPYGD 240
SLEFFYLRRQ MFVRLHFNRA GAVGENVPDD LYIKGSGSTA NLASSNYFPT PSGSMVTS 300
QIFNKPYWLQ RAQGHNNIGI WGNQLFVTVV DTTRSTNMSL CAAISTSETT YKNTNFKEYL 360
RHGEYDLDQF IFQLCKITLT ADVMTYIISM NSTILEDWNF GLQPPPGGTG EDTYRFVTSQ 420
AIACQKHTPP APKEDPLKKY TFWEVNLKEK FSADLDQFPL GRKFLQAGL KAKPKFTL 480
RKATPTTSST STAKRKKRK L 501

```

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SEQ ID NO: 24      moltype = AA length = 496
FEATURE           Location/Qualifiers
source            1..496
                  mol_type = protein
                  organism = synthetic construct

```

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SEQUENCE: 24
MVLPPVPVPS KVVSTDEYVA RTNIYYHAGT SRLLAGVHPY FPIKKPNNNK ILVPKVSGLQ 60
YRVFRIHLPD PNKFGFPDTS FYNPDQRLV WACVGVEVGR GQPLGVGISG HPLLNLDDT 120

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ENASAYAANA	GVDNRECISM	DYKQTQLCLI	GCKPPIGEHW	GKGSPECTNVA	VNPGDCPPLE	180
LINTVIQDGD	MVDTGFGAMD	FTTLQANKSE	VPLDICTSIC	KYPDYIKMVS	EPYGDSLFFY	240
LRREQMFVRH	LFNRAGAVGE	NVPDDLIIKG	SGSTANLASS	NYFPTPSGSM	VTSDAQIFNK	300
PYWLQRAQGH	NNGICWGNQL	FVTVVDTRRS	TNMSLCAAIS	TSETTYKNTN	FKEYLRHGEE	360
YDLQFIFQLC	KITLTADVMT	YIHSMNSTIL	EDWNFGLQPP	PGGTLEDTYR	FVTSQAIACQ	420
KHTPPAPKED	PLKKYTFWEV	NLKEKFSADL	DQFPLGRKFL	LQAGLKAKPK	FTLGKRKATP	480
TTSSTSTTAK	RKKRKL					496

SEQ ID NO: 25 moltype = AA length = 491
 FEATURE Location/Qualifiers
 source 1..491
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 25

MVPVSKVVST	DEYVARTNIY	YHAGTSRLLA	VGHYPYPIKK	PNNNKILVPK	VSGLQYRVFR	60
IHLDPDNKFG	FPDTSFYNDP	TQRLVWACVG	VEVGRGQPLG	VGISGHPLL	KLDDTENASA	120
YAANAGVDNR	ECISMDYKQT	QLCLIGCKPP	IGEHWGKGS	CTNVAVNPGD	CPPLELINTV	180
IQDGMVDVTG	FGAMDFTTLQ	ANKSEVPLDI	CTSICKYPDY	IKMVSEPYGD	SLFFYLRRREQ	240
MFVRLHFNRA	GAVGENVPDD	LYIKGSGSTA	NLASSNYFPT	PSGSMVTSDA	QIFNKPYWLQ	300
RAQGHNNGIC	WGNQLFVTVV	DTTRSTNMSL	CAAISTSETT	YKNTNFKEYL	RHGEEYDLQF	360
IFQLCKITLT	ADVMTYIISM	NSTILEDWNF	GLQPPPGGTL	EDTYRFVTSQ	AIACQKHTTP	420
APKEDPLKKY	TFWEVNLKEK	FSADLDQFPL	GRKFLLQAGL	KAKPKFTLGK	RKATPTTSST	480
STTAKRKKRK	L					491

SEQ ID NO: 26 moltype = AA length = 486
 FEATURE Location/Qualifiers
 source 1..486
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 26

MVVSTDEYVA	RTNIYYHAGT	SRLLAUGHYP	FPIKKPNNNK	ILVPKVSGLG	YRVFRIHLDP	60
PNKFGFPDTS	FYNPDTRQLV	WACVGVEVGR	GQPLGVGISG	HPLLNLDDT	ENASAYAANA	120
GVDNRECISM	DYKQTQLCLI	GCKPPIGEHW	GKGSPECTNVA	VNPGDCPPLE	LINTVIQDGD	180
MVDTGFGAMD	FTTLQANKSE	VPLDICTSIC	KYPDYIKMVS	EPYGDSLFFY	LRREQMFVRH	240
LFNRAGAVGE	NVPDDLIIKG	SGSTANLASS	NYFPTPSGSM	VTSDAQIFNK	PYWLQRAQGH	300
NNGICWGNQL	FVTVVDTRRS	TNMSLCAAIS	TSETTYKNTN	FKEYLRHGEE	YDLQFIFQLC	360
KITLTADVMT	YIHSMNSTIL	EDWNFGLQPP	PGGTLEDTYR	FVTSQAIACQ	KHTPPAPKED	420
PLKKYTFWEV	NLKEKFSADL	DQFPLGRKFL	LQAGLKAKPK	FTLGKRKATP	TTSSTSTTAK	480
RKKRKL						486

SEQ ID NO: 27 moltype = AA length = 503
 FEATURE Location/Qualifiers
 source 1..503
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 27

MPSDNTVYLP	PPSVARVVNT	DDYVTPTSIF	YHAGSSRLLT	VGNPYFRVPA	GGGNKQDIPK	60
VSAYQYRVFR	VQLPDPNFKG	LPDTSIYNPE	TQRLVWACAG	VEIGRGQPLG	VGLSGHPFYN	120
KLDDTESSHA	ATSNVSEDRV	DNVSDYKQT	QLCILGCAPA	IGEHWAKGTA	CKSRPLSQGD	180
CPPLELKNTV	LEDGDMVDTG	YGAMDFSTLQ	DTKCEVPLDI	CQSICKYPDY	LQMSADPYGD	240
SMFFCLRRQF	LFARHFWNRA	GTMGDTVPQS	LYIKGTGMRA	SPGSCVYSPS	PSGSIVTSDS	300
QLFNKPYWLH	KAQGHNNGVC	WHNQLFVTVV	DTTRSTNLTI	CASTQSPVPG	QYDATKFKQY	360
SRHVEEYDLQ	FIFQLCTITL	TADVMSYIHS	MNSSILEDWN	FGVPPPTTTS	LVDTYRFVQS	420
VAITCQKDA	PAENKDPYDK	LKFWNVDLKE	KFSLDLQYQP	LGRKFLVQAG	LRRKPTIGPR	480
KRSAPSATTS	SKPAKRVVR	ARK				503

SEQ ID NO: 28 moltype = AA length = 501
 FEATURE Location/Qualifiers
 source 1..501
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 28

MDNTVYLPPP	SVARVVNTDD	YVTPTSIFYH	AGSSRLLTVG	NPYFRVPAGG	GNKQDIPKVS	60
AYQYRVFRVQ	LPDPNFKGLP	DTSIYNPETQ	RLVWACAGVE	IGRGQPLGVG	LSGHFPYKNL	120
DDTESSHAAT	SNVSEVDNRD	VSDYKQTQL	CILGCAPAIG	EHWAKGTACK	SRPLSQGDGP	180
PLELKNTVLE	DGDMVDVTGY	AMDFSTLQDT	KCEVPLDICQ	SICKYPDYLQ	MSADPYGDSM	240
FFCLRRQQLF	ARHFWNRAGT	MGDTVPQSLY	IKGTGMRAS	GSCVYSPSPS	GSIVTSDSQL	300
FNKPYWLHKA	QGHNNGVCWH	NQLFVTVVDT	TRSTNLITCA	STQSPVPQGY	DATKFKQYSR	360
HVEEYDLQFI	FQLCTITLTA	DVMSYIHSMN	SSILEDWNFG	VPPPTTTSVL	DYRFVQSVAS	420
ITCQKDAAPA	ENKDPYDKLK	FWNVDLKEKF	SLDLQYPLG	RKFLVQAGLR	RKPTIGPRKR	480
SAPSATTSK	PAKRVVRAR	K				501

SEQ ID NO: 29 moltype = AA length = 500
 FEATURE Location/Qualifiers
 source 1..500
 mol_type = protein

-continued

organism = synthetic construct

SEQUENCE: 29

MPSEATVYLP	PVPVSKVST	DEYVTRTNIY	YHAGSARLLT	VGHPPYSIPK	SDNPKKIVVP	60
KVSGLQYRVF	RVRLPDPNKF	GFPDTSFYNP	ETQRLVWACV	GLEVGRGQPL	GVGISGHPLL	120
NKFDDTENS	RYAGGPGTDN	RECISMDYKQ	TQLCLLGCKP	PIGEHWGKGS	PCSNNAITPG	180
DCPPLLELKN	VIQDGMVD	GFGAMDFTAL	QDTKSNVPLD	ICNSICKYPD	YMKMVAEPYG	240
DTLFFYLRR	QMFVRHFFNR	SGTVGESVPT	DLYIKSGGST	ATLANSTYFP	TPSGSMVTS	300
AQIFNKPYWM	QRAQGHNNGI	CWGNQLFVTV	VDTRSTNMS	VCAAIAANS	TFKSSNFKEY	360
LRHGEEFDLQ	FIFQLCKITL	SADIMTYIHS	MNPAILEDWN	FGLTTPPSGS	LEDTRYFVTS	420
QAITCQKTAP	QKPKEDPFK	YVFEVNLKE	KFSADLDQFP	LGRKFLLQAG	YRARPKFKAG	480
KRSAPSASTT	TPAKRKKTKK					500

SEQ ID NO: 30 moltype = AA length = 495

FEATURE Location/Qualifiers

source 1..495

 mol_type = protein

 organism = synthetic construct

SEQUENCE: 30

MPSEATVYLP	PVPVSKVST	DEYVSRTSIY	YYAGSSRLLA	VGHPPYFSIKN	PTNAKKLLVP	60
KVSGLQYRVF	RVRLPDPNKF	GFPDTSFYNP	DTQRLVWACV	GLEIGRGQPL	GVGISGHPLL	120
NKFDDTETGN	KYPGQPGADN	RECLSMYKQ	TQLCLLGCKP	PTGEHWGKGV	ACTNAAPAND	180
CPPLLELINTI	IEDGDMVDTG	FGCMDFKTLQ	ANKSDVPIDI	CGSTCKYPDY	LKMTSEPYGD	240
SLFFFLRREQ	MFVRHFFNRA	GTLGEAVPDD	LYIKSGGTTA	SIQSSAFFPT	PSGSMVTSES	300
QLFNKPYWLQ	RAQGHNNGIC	WGNQVFVTVV	DTTRSTNMTL	CTQVTSDESTY	KNENFKEYIR	360
HVEEYDLQFV	FQLCKVTLTA	EVMTYIHAMN	PDILEDWQFG	LTPPPSASLQ	DTYRFVTSQA	420
ITCQKTVPPK	EKEDPLGKYT	FWEVDLKEKF	SADLDQFFLG	RKFLQAGLKG	AKPKLKRAAP	480
TSTRTSSAKR	KKVKK					495

SEQ ID NO: 31 moltype = AA length = 498

FEATURE Location/Qualifiers

source 1..498

 mol_type = protein

 organism = synthetic construct

SEQUENCE: 31

MSNEATVYLP	PVSVSKVST	DEYVTRTNIY	YHAGSSRLLA	VGHPPYAIKK	QDSNKIAVPK	60
VSGLQYRVFR	VKLDPDNKFG	FPDTSFYDPA	SQRLVWACTG	VEVGRGQPLG	VGISGHPLL	120
KLDDTENSNNK	YVNGSGTDNR	ECISMDYKQT	QLCLIGCREP	IGEHWGKGTG	CNANQVKAGE	180
CPPLLELNTV	LQDGMVDTG	FGAMDFTTLQ	ANKSDVPLDI	CSSICKYPDY	LKMVSEPYGD	240
MLFFYLRRQ	MFVRHLFNRA	GTVGETVPAD	LYIKGTTGTL	PSTSYFPTPS	GSMVTSDAQI	300
FNKPYWLQRA	QGHNNGICWS	NQLFVTVVDT	TRSTNMSVCS	AVSTSDSTYK	NDNFKEYLRH	360
GEEYDLQFIF	QLCKITLTAD	VMTYIHSNMP	SILEDWNFGL	TPPPSGTLED	TYRYVTSQAV	420
TCQKPSAPKP	KDDPLKNYTF	WEVDLKEKFS	ADLDQFPPLGR	KFLQAGLKA	RPNFRLGKRA	480
APASTSKKSS	TKRRKVKS					498

SEQ ID NO: 32 moltype = AA length = 530

FEATURE Location/Qualifiers

source 1..530

 mol_type = protein

 organism = synthetic construct

SEQUENCE: 32

MIIFLKNVN	VPFIQLMAL	WRPSDSTVYL	PPPSVARVVS	TDDYVSRTSI	FYHAGSSRL	60
TVGNPYFRV	PNGAGNKQAV	PKVSAYQYRV	FRVALPDPNK	FGLPDSTIYN	PETQRLVWAC	120
VGMEIGRGQP	LGIGLSGHPF	YNKLDDTESA	HAATAVITQD	VRDNVSVDYK	QTQLCILGCV	180
PAIGEHWAKG	TLCKPAQLQP	GDCPPLLELKN	TIIEDGDMVD	TGYGAMDFST	LQDTKCEVPL	240
DICQSIKCKP	DYQLQMSADPY	GDSMFFCLRR	EQLFARHFWN	RAGVMGDTVP	TDLYIKGTS	300
NMRETPGSCV	YSPSPSGSII	TSDSQLFNKP	YWLHKAQGHN	NGICWHNQLF	VTVDTTTRST	360
NLTLCASTQN	PVPSTYDPTK	FKQYSRHVEE	YDLQFIFQLC	TITLTAEVMS	YIHSNMSSIL	420
ENWNFGVPPP	PTTSLVDTYR	FVQSVAVTCQ	KDTPPEKQD	PYDKLKFWTV	DLKEKFSDDL	480
DQYPLGRKFL	VQAGLRRRPT	IGPRKRPAAS	TSTASTASRP	AKRVIRIRSK		530

SEQ ID NO: 33 moltype = AA length = 523

FEATURE Location/Qualifiers

source 1..523

 mol_type = protein

 organism = synthetic construct

SEQUENCE: 33

MVNVPFIFLQ	MALWRPSDST	VYLPPPSVAR	VVSTDYVSR	TSIFYHAGSS	RLLTVGNPYP	60
RVPNGAGNKG	QAVPKVSAYQ	YRVFRVALPD	PNKFGLPDST	IYNPETQRLV	WACVGMIEGR	120
GQPLGIGLSG	HPFYNKLDDT	ESAAHAATAVI	TQDVRDNVSV	DYKQTQLCIL	GCVPAIGEHW	180
AKGTLCKPAQ	LQPGDCPPL	LKNTIIEDGD	MVDTYGAMG	FSTLQDTKCE	VPLDICQSTIC	240
KYPDYLQMSA	DPYDGMFFC	LRREQLFARH	FWNRAGVMGD	TVPTDLYIKG	TSANMRETPG	300
SCVYSPSPSG	SIITSDSQLF	NKPYWLHKAQ	GHNNGICWHN	QLFVTVVDTT	RSTNLTLCAS	360
TQNPVPSTYD	PTKFKQYSRH	VEEYDLQFIF	QLCTITLTAE	VMSYIHSNMS	SILENWNFGV	420
PPPTTSLVD	TYRFVQSVAV	TCQKDTTPE	KQDPYDKLKF	WTVDLKEKFS	SDLDQYPLGR	480
KFLVQAGLRR	RPTIGPRKRP	AASTTASTA	SRPAKRVRIR	SKK		523

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SEQ ID NO: 34 moltype = AA length = 518
FEATURE Location/Qualifiers
source 1..518
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 34

MIFLQMALWR	PSDSTVYLPP	PSVARVVSTD	DYVSRTSIFY	HAGSSRLLT	GNPYFRVVPN	60
GAGNKQAVPK	VSAYQYRVFR	VALPDPNPKF	LPDSTIYNPE	TQRLVWACVG	MEIGRGQPLG	120
IGLSGHPFYN	KLDDTESAHA	ATAVITQDVR	DNVSVDYKQT	QLCILGCVPA	IGEHWAKGTL	180
CKPAQLQPGD	CPPLELKNTI	IEDGDMVDTG	YGAMDFSTLQ	DTKCEVPLDI	CQSICKYPDY	240
LQMSADPYGD	SMFFCLRREQ	LFAHFHWNRA	GVMGDTVPTD	LYIKGTSANM	RETPGSCVYS	300
PSPSGSIITS	DSQLFNKPYW	LHKAQGHNNG	ICWHNQLFVT	VVDTRSTNL	TLCASTQNPV	360
PSTYDPTKFK	QYSRHVEEYD	LQFIFQLCTI	TLTAEVMSYI	HSMNSSILEN	WNFGVPPPT	420
TSLVDITYRFV	QSVAVTCQKD	TTPPEKQDPY	DKLKFWTVDL	KEKFSSDLQ	YPLGRKFLVQ	480
AGLRRRPTIG	PRKRPAASTS	TASTASRP	AK	RVRIRSKK		518

SEQ ID NO: 35 moltype = AA length = 509
FEATURE Location/Qualifiers
source 1..509
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 35

MPSDSTVYLP	PPSVARVVST	DDYVSRTSIF	YHAGSSRLLT	VGNPYFRVVP	NGAGNKQAVP	60
KVSAYQYRVF	RVALPDPNKF	GLPDSIYNP	ETQRLVWACV	GMEIGRGQPL	GIGLSGHPFY	120
NKLDDTESAH	AATAVITQDV	RDNVSVDYKQ	TQLCILGCVPA	AIGEHWAKGT	LCKPAQLQPG	180
DCPPLELKNT	IIEDGDMVDT	GYGAMDFSTL	QDTKCEVPLD	ICQSICKYPD	YLQMSADPYG	240
DSMFFCLRRE	QLFARHFWRN	AGVMGDTVPT	DLYIKGTSAN	MRETPGSCVY	SPSPSGSIIT	300
SDSQLFNKPY	WLHKAQGHNN	GICWHNQLFV	TVVDTRSTN	LTLCASTQNP	VPSTYDPTKF	360
KQYSRHVEEY	DLQFIFQLCT	ITLTAEVMSY	IHSMNSSILE	NWNFGVPPPP	TSLVDITYRF	420
VQSVAVTCQK	DTTPPEKQDP	YDKLKFWTV	LKEKFSSDLQ	QYPLGRKFLV	QAGLRRRPTI	480
GPRKRPAAST	STASTASRPA	KRVRIRSKK				509

SEQ ID NO: 36 moltype = AA length = 499
FEATURE Location/Qualifiers
source 1..499
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 36

MPSEATVYLP	PVPVSKVYST	DEYVSRTSIY	YYAGSSRLLT	VGHYPFYSIKN	TSSGNGKKVL	60
VPKVSGLQYR	VFRIKLDPDN	KFGFPDTSFY	NPETQRLVWA	CTGLEIGRGQ	PLGVGISGHP	120
LLNKFDDTET	SNKYAGKPGI	DNRECLSMY	KQTQLCILGC	KPPIGEHWGK	GTPCINNNSGN	180
PGDCPPLQLI	NSVIQDGMV	DTGFGCMDFN	TLQASKSDVP	IDICSSVCKY	PDYLMASEP	240
YGDSLFFFLR	REQMFVRHFF	NRAGTLGDPV	PGDLYIQGSN	SGNTATVQSS	AFFPTPSGSM	300
VTSESQLFNK	PYWLQRAQGH	NNGICWGNQL	FVTVVDTTTS	TNMTLCAEVK	KESTYKNEF	360
KEYLRHGEF	DLQFIFQLCK	ITLTADVMTY	IHKMDATILE	DWQFGLTPPP	SASLEDITYRF	420
VTSTAITCQK	NTPPKGKEDP	LKDYMFWEVD	LKEKFSADLD	QFPLGRKFL	QAGLQARPKL	480
KRPASSAPRT	STKKKKVKR					499

SEQ ID NO: 37 moltype = AA length = 515
FEATURE Location/Qualifiers
source 1..515
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 37

MILFCVADV	N	VFIHFLQMSV	WRPSEATVYL	PPVPVSKVVS	TDEYVSRTSI	YYYAGSSRL	60
AVGNPYFSIK	SPNNNKVLV	PKVSGLQYRV	FRVRLPDPNK	FGFPDTSFY	N	PDTQRLVWAC	120
VGLEIGRGQP	LGVGVSHPY	LNKFDDTETS	NRYPAPGSD	NRECLSMYK	Q	TQLCLIGCK	180
PPTGEHWGKG	VACNNAAAT	DCPPLELFNS	IIEDGDMVDT	GFGCMDFTL	Q	ANKSDVPID	240
ICNSTCKYPD	YLMASEPYG	DSLFFFLRRE	QMFVRHFFNR	AGKLGEAVPD	D	LYIKGSGNT	300
AVIQSSAFFP	TPSGSIVTSE	SQLFNKPYWL	QRAQGHNNGI	CWGNQLFVT	V	DTRSTNMT	360
LCTEVTKEGT	YKNDNFKEYV	RHVEEYDLQF	VFQLCKITLT	AEIMTYIHTM	D	SNILEDWQF	420
GLTPPPSASL	QDTYRFVTSQ	AITCQKTAPP	KEKEDPLNKY	TFWEVNLKEK	F	SADLDQFPL	480
GRKFLQSG	L	KAKPRLKRS	A	PTTRAPSTKR	K	VKK	515

SEQ ID NO: 38 moltype = AA length = 508
FEATURE Location/Qualifiers
source 1..508
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 38

MVNVFHIFLQ	MSVWRPSEAT	VYLPVPVSK	VVSTDEYVSR	TSIIYYAGSS	RLAVGNPYF	60	
SIKSPNNKK	VLVPKVSGLQ	YRVFRVRLPD	PNKFGFPDTS	FYNPDTQRLV	WACVGLIGR	120	
GQPLGVGVSG	HPYLNKDDT	ETSNRYPAQ	GSDNRECLSM	DYKQTQLCLI	GCKPPTGEHW	180	
KGKVACNNNA	AATDCPPEL	FNSIIEDGDM	VDTGFGCMD	GTQANKSDV	PIDICNSTCK	240	
YPDYLMASE	PYGDSLFFFL	RREQMFVRHF	FNRAKLG	EA	VPDDLYIKGS	GNTAVIQSS	300
FPPTPSGSIV	TSESQLFNKP	YWLQRAQGH	NGICWGNQLF	VTVDVDTTRST	NMTLCTEVT	K	360

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EGTYKNDNFK	EYVRHVEEYD	LQPVFQLCKI	TLTAEIMTYI	HTMDSNILED	WQFGLTPPPS	420
ASLQDTYRFV	TSQAITCQKT	APPKEKEDPL	NKYTFWEVNL	KEKFSADLDQ	FPLGRKFLLQ	480
SGLKAKPRLK	RSAPTTRAPS	TKRKKVKK				508

SEQ ID NO: 39 moltype = AA length = 503
 FEATURE Location/Qualifiers
 source 1..503
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 39

MIFLQMSVWR	PSEATVYLPP	VPVSKVVSTD	EYVSRTSIYY	YAGSSRLLA	GNPYFSIKSP	60
NNNKKVLVVK	VSGLQYRVFR	VRLPDPNFKF	FPDTSFYNDP	TQRLVWACVG	LEIGRGQPLG	120
VGVSGHPYLN	KFDDTETSNR	YPAQPGSDNR	ECLSMYKQT	QLCLIGCKPP	TGEHWGKGVA	180
CNNNAATDC	PPLELFNSII	EDGDMVDTGF	GCMDFGTLQA	NKSDVPIDIC	NSTCKYPDYL	240
KMASEPYGDS	LFFFLRREQM	FVRHFFNRAG	KLGEAVPDDL	YIKGSGNTAV	IQSSAFFPTP	300
SGSIVTSEQS	LFNKPYWLQR	AQGHNNGICW	GNQLFVTVD	TTRSTNMTLC	TEVTKEGTYK	360
NDNFKEYVRH	VEEYDLQFVF	QLCKITLTAE	IMTYIHTMDS	NILEDWQFGL	TPPPSASLQD	420
TYRFVTSQAI	TCQKTAPPKE	KEDPLNKYTF	WEVNLKEKFS	ADLDQFPLGR	KFLLQSGLKA	480
KPRLKRSAPT	TRAPSTKRKK	VKK				503

SEQ ID NO: 40 moltype = AA length = 494
 FEATURE Location/Qualifiers
 source 1..494
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 40

MPSEATVYLP	PVPVSKVVST	DEYVSRTSIY	YYAGSSRLLA	VGNPYFSIKS	PNNNKKVLVP	60
KVSGLQYRVF	RVRLPDPNKF	GFPDTSFYNP	DTQRLVWACV	GLEIGRGQPL	GVGVSGHPYL	120
NKFDDTETS	RYPAQPGSDN	RECLSMYKQ	TQLCLIGCKP	PTGEHWGKGV	ACNNNAATD	180
CPPLELFNSI	IEDGDMVDTG	FGCMDFGTLQ	ANKSDVPIDI	CNSTCKYPDY	LKMASEPYGD	240
SLFFFLRREQ	MFVRHFFNRA	KLGEAVPDD	LYIKGSGNTA	VIQSSAFFPT	PSGSIVTSES	300
QLFNKPYWLQ	RAQGHNNGIC	WGNQLFVTVD	DTRSTNMTL	CTEVTKEGTY	KNDNFKEYVR	360
HVEEYDLQVF	FQLCKITLTA	EIMTYIHTMD	SNILEDWQFG	LTPPPSASLQ	DTYRFVTSQA	420
ITCQKTAPPK	EKEDPLNKYT	FWEVNLKEKF	SADLDQFPLG	RKFLQSGGLK	AKPRLKRSAP	480
TTRAPSTKRK	KVKK					494

SEQ ID NO: 41 moltype = AA length = 499
 FEATURE Location/Qualifiers
 source 1..499
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 41

MPSDSTVYVP	PPNPVSKVVA	TDAYVKRTNI	FYHASSRLL	AVGHPYYSIK	KVNKTVPVKV	60
SGYQYRVFKV	VLPDPNKFAL	PDSSLFDPTT	QRLVWACTGL	EVGRGQPLGV	GVSGHPLLNK	120
YDDVENSNGY	GGNPGQDNVR	NVGMDYKQTQ	LCMVGCAPLL	GEHWGKGTCQ	SNTSVQNGDC	180
PPLELITSVI	QDGMVDTG	GAMNFADLQT	NKSDVPLDIC	GTVCYKPYDYL	QMAADPYGDR	240
LFFYLRLKEQM	FARHFFNRAG	TVGEPVPDDL	LVKGGNNRSS	VASSIYVHTP	SGSLVSSEAQ	300
LFNKPYWLQK	AQGHNNGICW	GNHLFVTVD	TTRSTNMTLC	ASVSKSATYT	NSDYKEYMRH	360
VEEFDLQFIF	QLCSITLSAE	VMAYIHTMNP	SVLEDWNEFL	SPPPNGTLED	TYRYVQSQAI	420
TCQKPTPEKE	KQDPYKDMSF	WEVNLKEKFS	SELDQFPLGR	KFLQSGYRG	RTSARTGIKR	480
PAVSKPSTAP	KRKRTKTKK					499

SEQ ID NO: 42 moltype = DNA length = 1497
 FEATURE Location/Qualifiers
 source 1..1497
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 42

atgcctccg	actccaccgt	ttacgttcca	ccaccaaatac	cagtttccaa	ggttggtgct	60
actgacgct	acgtcaccag	aactaatatc	ttctaccacg	cctcctcatc	cagattgctt	120
gctggttgct	accataactt	ctccatcaag	agagccaaca	agacgctcgt	tccaaagggt	180
tctggttacc	agtacagagt	gttcaagggtc	gtcttgccag	atccaaacaa	gttcgctttg	240
ccagactctt	cctgttgcga	cccaactact	cagagattgg	tttgggcttg	taccggtttg	300
gaagttggta	gaggtcaacc	acttggtggt	ggtggttctg	gacacccatt	cttgaacaag	360
tacgacgacg	ttgaaaactc	cgttctcggt	ggtaaccacg	gtcaagacaa	cagagttaac	420
gtcgggatgg	actacaagca	gacccagttg	tgatatggtg	gttgtgtccc	accattgggt	480
gaacattggg	gtaagggtaa	gcagtgatcc	aacactccag	ttcaagctgg	tgattgtcca	540
ccttgaggag	tgatcaactt	cgttatttcg	gacggtgaca	tggttgacac	tggtttcggt	600
gctatgaact	tcgctgactt	gcagactaac	aagtcgcagc	ttcctatcga	catctgtggt	660
actacctgta	agtaccacga	ctacttgcaa	atggctgctg	atccatacgg	tgacagactg	720
ttcttcttcc	tcagaaaaga	acagatgttc	gccagacact	tcttcaacag	agctgggtgaa	780
gttggtgaac	cagttccaga	caccttgatc	atcaaagggt	ccggttaacag	aaactccggt	840
ggttcctcta	tctacgttaa	cactccatcc	ggttctttgg	tgtcctctga	agctcagttg	900
ttcaacaagc	catactgggt	gcaaaaggcc	cagggtcaca	acaacgggat	ctgttggggg	960
aaccagttgt	tcgttaactg	tgctgacact	accagatcca	ccaatatgac	tttgtgtgcc	1020
tccgttacta	cctcctccac	ttacactaac	tccgactaca	aagagtacat	gagacacgtc	1080

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gaagagtacg acctgcagtt cattttccag ctgtgttcca tcaactttgtc cgctgaggtt 1140
atggcctaca tccacactat gaacccatcc gttttggagg actggaactt cggtttgtct 1200
ccaccaccta acggtacttt ggaggatacc tacagatacg ttcagtccea ggctatcact 1260
tgccaaaagc caactccaga gaaagagaag ccagatcctt acaagaacct gtccttcttg 1320
gaagtcaccc tgaagaaaaa gttctccagc gagctggatc agtaccattt gggtagaaaag 1380
ttcttgctgc agtccggtta cagaggtaga tcttccatta gaaccggtgt taagaggcct 1440
gctgtttcta aggtctctgc tgtccaaaag aggaagagag ctaagaccaa gagataa 1497

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SEQ ID NO: 43      moltype = DNA length = 1491
FEATURE           Location/Qualifiers
source            1..1491
                  mol_type = other DNA
                  organism = synthetic construct

```

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SEQUENCE: 43
atggactcca ccgtctacgt cccaccacca aatccagttt ctaagggtgt tgctaccgac 60
gcctacgtca ccagaactaa tatcttctac cagcctcct catccagatt gcttgcgttt 120
ggtcacccat acttctccat caagagagcc aacaagaccg tcgttccaaa ggtttctggt 180
taccagtaca gagtggtcaa ggtcgtcttg ccagatccaa acaagttcgc ttgcccagac 240
tcttccttgt tcgacccaac tactcagaga ttggtttggg cttgtaccgg ttggaagtt 300
ggtagaggtc aaccacttgg tgttggtgtt tctggacacc cattcttgaa caagtacgac 360
gacgttgaaa actccggttc tgggtgtaac ccaggtcaag acaacagagt taacgtcgg 420
atggactaca agcagaccca gttgtgtatg gttggttgtg ctccaccatt gggtagaat 480
tggggttaag gtaagcagtg taccacactt ccagttcaag ctggtgattg tccaccttg 540
gagttgatca cttccgttat tcaggacggt gacatggttg acactggttt cggtgctatg 600
aaactcgctg acttcagacc taacaagtcc gacgttccta tcgacatctg tggtaactac 660
tgtaagtacc cagactactt gcaaatggct gctgacccat acggtgacag actgtctctc 720
ttcctgagaa aagaacagat gttcgcagca cactcttcca acagagctgg tgaagttggt 780
gaaccagttc cagacacact gatcatcaaa ggttccggta acagaacctc cgttggttcc 840
tctatctacg ttaacactcc atccggttct ttggtgtcct ctgaagctca gttgttcaac 900
aagccatact ggttgcaaaa gggccagggt cacaacaacg gtatctgttg gggtaaccag 960
ttgttctgta ctgttctcga cactaccaga tccaccaata tgactttgtg tgctccggt 1020
actacctctc ccacttacac taactccgac tacaagagt acatgagaca cgtcgaagag 1080
tacgacctgc agtctctttt ccagctgtgt tccatcactt tgcctcgtag gggtatggcc 1140
tacatccaca ctatgaaccc atccgttttg gaggactgga acttcggttt gctccacca 1200
cctaaccggt ctttggagga tacctacaga tacgttcagt cccaggctat cacttgccaa 1260
aagccaactc cagagaagaa gaagccagat ccttacaaga acctgtcctt ctgggaagtc 1320
aaactgaaag aaaagtcttc cagcgagctg gatcagtagc cattgggtag aaagtctctg 1380
ctgcagtcgg gttacagagg tagatcctcc attagaaccg gtgttaagag gcctgctgtt 1440
tccaaagctt ctgctgctcc aaagaggaag agagctaaga ccaagagata a 1491

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SEQ ID NO: 44      moltype = DNA length = 1506
FEATURE           Location/Qualifiers
source            1..1506
                  mol_type = other DNA
                  organism = synthetic construct

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SEQUENCE: 44
atgccatctg aggtaccggt ttacttgcca ccagttccag tttccaaggt tgtttccact 60
gacgagtacg tcgccagaac taacatctac taccacgctg gtacttccag attgctggct 120
gttgggtcatc catacttccc aatcaagaag ccaacaacaa acaagatcct ggtgccaaag 180
gtttccggtc tgcagtacag agttttcaga atccacttgc cagatccaaa caagttcgg 240
ttccagagca catcttcta caaccagac actcagagat tggtttgggc ctgtgttggt 300
gttgaagtgc gtagaggtta accacttggt gttggtatct ctggtaaccc attgctgaac 360
aagttggacg atactgaaaa cgttcccgct tacgtgcta acgtggtgt tgataacaga 420
gaatgcactc ccatggacta caagcagacc cagttgtgtt tgatcggttg caagccacct 480
attggtgaac attggggtta gggttctcca tgtaccaacg ttgctgttaa cccaggtgat 540
tgtccaccat tggagctgat caacactgtt attcaggacg gtgacatggt cgacactggt 600
tttgggtgta tggacttcac taccttgtag gctaacaagt ccgaagtccc attggacatc 660
tgtacctcca tctgcaagta ccagactac atcaagatgg tgtctgagcc atacggtgac 720
tccctgttct tctacttgag aagagaacag atgttcgtca ggcacctgtt caacagagct 780
gggtgctgtt gtgaaaaagt tccagacgac ctgtacatca aaggttcttg ttctactgct 840
aaactggcct cctctaacta ctttccaact ccatcgggtt ccatgggttac ttccgacgt 900
cagattttta acaagccata ctgggtgcaa agagcccagg gtcacaacaa cggtatctgt 960
tggggttaac agttgtctgt tactgttgtc gacactacca gatccaccaa catgtctttg 1020
tgtgtgcta tctccacttc cgagactacc tacaagaaca ctaacttcaa agaataacct 1080
aggcacggtg aagagtacga cttgcagttc attttccagc tgtgcaagat cactttgacc 1140
gccgacgtta tgacttacat ccactccatg aactccacta tcttggagga ctggaacttt 1200
ggtctgcaac caccaccagg tggtagcttg gaagatacct acagattcgt tacttcccag 1260
gctatcgctt gtcagaagca cactccacca gctccaaaag aggatccatt gaagaagtag 1320
actttctggg aagtcaacct gaaagagaag ttctccgctg acttggatca gttccactg 1380
ggtagaaaag tctgtgtgca ggtgtgtttg aaggccaagc caaagttcac ttgggtaga 1440
agaaaggcta ctccaactac ttctccacc tctactaccg ctaagaggaa gaagagaaag 1500
ttgtaa 1506

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SEQ ID NO: 45      moltype = DNA length = 1491
FEATURE           Location/Qualifiers
source            1..1491

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-continued

mol_type = other DNA
organism = synthetic construct

SEQUENCE: 45

atggtctact	tgccaccagt	tccagtttcc	aaggttggtt	ccactgacga	gtacgtcgcc	60
agaactaaca	tctactacca	cgttggtact	tccagattgc	tggctgttgg	tcateccatac	120
ttcccaatca	agaagccaaa	caacaacaag	atcctggtgc	caaaggtttc	cggtctgcag	180
tacagagttt	tcagaatcca	cttgccagat	ccaaacaagt	tcggtttccc	agacacatcc	240
ttctacaacc	cagacactca	gagattgggt	tgggcctgtg	ttggtgttga	agttggtaga	300
ggtaaccac	ttggtgttgg	tatttctggt	cacccattgc	tgaacaagtt	ggacgatact	360
gaaaacgctt	ccgcttacgc	tgctaacgct	ggtgttgata	acagagaatg	catctccatg	420
gactacaagc	agaccagtt	gtgtttgatc	ggttgcaagc	cacctattgg	tgaacattgg	480
ggtaagggtt	ctccatgtac	caacgttgct	gttaacccag	gtgattgtcc	accattggag	540
ctgatcaaca	ctgttattca	ggacggtgac	atggtcgaca	ctggttttgg	tgctatggac	600
ttcactacct	tgagggttaa	caagtccgaa	gttccattgg	acatctgtac	ctccatctgc	660
aagtaccacg	actacatcaa	gatggtgtct	gagccatacg	gtgactccct	gttctctctac	720
ttgagaagag	aacagatggt	cgtaaggcac	ctgttcaaca	gagctgggtc	tgttggtgaa	780
aacgtttccg	acgacctgta	catcaaaagt	tctggttcta	ctgctaacct	ggcctcctct	840
aactacttct	caactccatc	cggttccatg	gttacttccg	acgctcagat	tttcaacaag	900
ctactactgt	tgcaagagac	ccagggtcac	aacaacggta	tctgttgggg	taaccagttg	960
ttcgtaactg	ttgtcgacac	taccagatcc	accaaactgt	ctttgtgtgc	tgctatctcc	1020
acttccgaga	ctactacaaa	gaacactaac	ttcaagaat	acctgaggca	cggtgaagag	1080
tacgacttgc	agttcatttt	ccagctgtgc	aagatcactt	tgaccgccga	cgttatgact	1140
tacatccact	ccatgaactc	cactatcttg	gaggactgga	actttggtct	gcaaccacca	1200
ccagggtgta	ctttggaaga	tactacacaga	ttcgttactt	cccaggctat	cgcttgtcag	1260
aagcacactc	caccagcttc	aaaagaggat	ccattgaaga	agtacacttt	ctgggaagtc	1320
aaactgaaag	agaagttctc	cgtgactctg	gatcagttcc	cactgggttag	aaagttcttg	1380
ttgcaggctg	gtttgaaggc	caagccaaag	ttcacttttg	gtaagagaaa	ggctactcca	1440
actacttctc	ccacctctac	taccgctaag	aggaagaaga	gaaagttgta	a	1491

SEQ ID NO: 46 moltype = DNA length = 1476
FEATURE Location/Qualifiers
source 1..1476
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 46

atggtttccag	tctccaaggt	tgtttccacc	gacgaatacg	ttgccaggac	caacatctac	60
taccacgctg	gtacttctag	attgctggct	gttggtcatc	catacttccc	aatcaagaag	120
ccaaacaaca	acaagatcct	ggtgccaaaag	gtttccggtc	tgacgtacag	agttttcaga	180
atccacttgc	cagatccaaa	caagttcggg	ttcccagaca	catccttcta	caaccagac	240
actcaagagat	tgggtttggg	ctgtgttggg	gttgaagttg	gtagagggtc	accacttggg	300
gttggtattt	ctggtcaccc	attgctgaac	aagttggacg	atactgaaaa	cgcttccgct	360
tacgctgcta	acgctggtgt	tgataacaga	gaatgcactc	ccatggacta	caagcagacc	420
cagttgtggt	tcgtcgggtg	caagccacct	attggtgaac	attggggtaa	gggttctcca	480
tgtaccaacg	ttgctgttaa	cccagggtgat	tgtccaccat	tggagctgat	caacactggt	540
attcaggacg	gtgacatggt	cgacactggt	tttggtgcta	tggacttcac	taccttgacg	600
gctaacaagt	cgaagtttcc	attggacatc	tgtacctcca	tctgcaagta	cccagactac	660
atcaagatgg	tgtctgagcc	atacgggtgac	tcctctgtct	tctacttgag	aagagaacag	720
atgttctgta	ggcacctggt	caacagagct	ggtgctgttg	gtgaaaaact	tcacagacgac	780
ctgtacatca	aaggtttctg	ttctactgct	aacctggcct	cctctaacta	ctttccaact	840
ccatccggtt	ccatggtttac	ttccgacgct	cagattttca	acaagccata	ctgggtgcaa	900
agagcccagg	gtcacacaac	cggatatctg	tggggtaaac	agttgttctg	tactgttgtc	960
gacactacca	gatccacca	catgtctttg	tgtgctgcta	tctccacttc	cgagactacc	1020
tacaagaaca	ctaacttcaa	agaataacct	aggcacgggt	aagagtacga	cttgacgttc	1080
attttccagc	tgtgcaagat	cactttgacc	gccgacgtta	tgacttacat	ccactccatg	1140
aaactccacta	tcttggaagga	ttggaaactt	ggtctgcaac	caccaccagg	tggtactttg	1200
gaagatacct	acagattcgt	tacttcccag	gctatcgctt	gtcagaagca	cactccacca	1260
gctccaaaag	aggatccatt	gaagaagtac	actttctggg	aagtcacact	gaaagagaag	1320
ttctccgctg	acttggatca	gttcccactg	ggtagaaagt	tcttgttgca	ggctgggttg	1380
aaggccaagc	caaagttcac	tttgggtaag	agaaaggcta	ctccaactac	ttctccacc	1440
tctactaccg	ctaagaggaa	gaagagaaag	ttgtaa			1476

SEQ ID NO: 47 moltype = DNA length = 1461
FEATURE Location/Qualifiers
source 1..1461
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 47

atggttggtt	ccaccgacga	gtacgttgcc	agaactaaca	tctactacca	cgctggtact	60
tcagatttgc	tggctgttgg	tcateccatac	ttcccaatca	agaagccaaa	caacaacaag	120
atcctggtgc	caaaggtttc	cggtctgcag	tacagagttt	tcagaatcca	cttgccagat	180
ccaaacaagt	tcggtttccc	agacacatcc	ttctacaacc	cagacactca	gagattgggt	240
tgggcctgtg	ttggtgttga	agttggtaga	ggtcaaccac	ttggtgttgg	tatttctggt	300
caccatttgc	tgaacaagtt	ggacgatact	gaaaacgctt	ccgcttacgc	tgctaacgct	360
ggtgttgata	acagagaatg	catctccatg	gactacaagc	agacccagtt	gtgtttgatc	420
ggttgcaagc	cacctattgg	tgaacattgg	ggtaagggtt	ctccatgtac	caacgttgct	480
gttaaccacg	gtgattgtcc	accattggag	ctgatcaaca	ctgttattca	ggacggtgac	540

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atgggtcgaca ctgggttttgg tgcctatggac ttcactacct tgcagggttaa caagtccgaa 600
gttccattgg acatctgttac ctccatctgc aagtaaccag actacatcaa gatgggtgct 660
gagccatacgt gtgactccct gttcttctac ttgagaagag aacagatggt cgtcaggcac 720
ctgttcaaca gagctgggtgc tgggtgtgaa aacgttccag acgacctgta catcaaagg 780
tctgggttcta ctgctaacct ggccctctct aactactttc caactccatc cggttccatg 840
gttacttccg acgctcagat ttccaacaag ccatactggt tgcaaaagag ccagggtcac 900
aacaacggta tctgttgggg taaccagttg ttcgttactg ttgtcgacac taccagatcc 960
accaacatgt ctttgtgtgc tgctatctcc acttccgaga ctacctacaa gaacactaac 1020
ttcaagaat acctgaggca cgggtgaagag tacgacttgc agttcathtt ccagctgtgc 1080
aagatcactt tgaccgcccga cgttatgact tacatccact ccatgaactc cactatcttg 1140
gaggactgga acttttggtc gcaaccacca ccagggtgta ctttgggaaga tacctacaga 1200
ttcgttactt cccagggtat cgttgtcag aagcacactc caccagctcc aaaagaggat 1260
ccattgaaga agtacacttt ctgggaagtc aacctgaaa agaagttctc cgtgacttg 1320
gatcagttcc cactgggtag aaagttcttg ttgagggtg gtttgaagg caagcctaa 1380
ttcactttgg gtaagagaaa ggctactcca actacttctc ccactctac taccgctaa 1440
aggaagaaga gaaagtgtga a 1461

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SEQ ID NO: 48      moltype = DNA length = 1512
FEATURE           Location/Qualifiers
source            1..1512
                  mol_type = other DNA
                  organism = synthetic construct

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SEQUENCE: 48
atgccctccg acaacaccgt ctacttgcca ccaccatctg ttgccagagt tgttaacact 60
gacgactacg ttaactccac ctccattttc taccagctg gttcttctag attgctgacc 120
gttggttaac catacttcag agttccagct ggtgggtgta acaagcagga catccaaaag 180
gtttccgctc accagtagag agtggtcaga gtccaattgc cagatccaaa caagtccggt 240
ttgccagaca cctccatcta caaccagag actcaaatg ttggttgggc ttgtgctggt 300
gtcgagattg gtagaggta accacttggt gttggtttgt ctgggtcacc attctacaac 360
aagttggacg acactgaatc ttcccacgct gctacttcta acgtttccga ggatgtcaga 420
gacaacgtgt ccgttgacta caagcagacc cagttgtgta tcttgggttg tgctccagct 480
attggtgaac attgggctaa gggaaactgc tgaagtcca gaccattgct tcaagggtgat 540
tgcccaccat tggagttgaa gaacactggt ttggaggacg gtgacatggt tgacactggt 600
taagggtgta tggacttctc taccttgcag gacactaagt gtgaggtccc attggacatt 660
tgtcagttca tctgtaagta ccccagctac ttgcaaatgt ccgctgatcc atacgggtgac 720
tctatgttct tctgcctgag aagagagcag ttgttcgcta gacacttctg gaacagagct 780
ggtagctatg gtgacactgt tcacagctcc ttgtacatca aaggtactgg tatgagagct 840
tcccagggtt cctgtgttga ctctccatct ccatccggtt ccatcggtac ttccgattct 900
cagctgttca acaagcaca ctgggtgcac aaggctcagg gtcaatacaa cgggtgtttgt 960
tggcacaacc agctgttctg tactgttgtt gacactacca gatccaccaa cttgactatc 1020
tgtgttccca ctcaatcccc agttccaggt caatacagac ctactaagtt caagcagtag 1080
tcacagacag tcgaagagta cgaactgcag ttcactctcc agttgtgcac tatcactttg 1140
accgcccagc ttatgtccca catccactct atgaactcct ccatcttggg ggactgggac 1200
tttgggtgtc ctctctctcc aactacttcc ttggttgata cctacagatt cgttcagttc 1260
gtcgctatca cttgtcaaaa ggaatgctgt cctgctgaga acaaggaccc atacgataag 1320
ctgaagttct ggaacgtcga cctgaaaagag aagttctcct tggacttggg tcagtaccca 1380
ctgggttaga agttcttggg ccaagccggt ttgagaagaa agccaactat cgggtccaaga 1440
aagagatctg ctcttccgcg tactacttct tccaagccag ctaagagagt tagagtcaga 1500
gccagaaagt aa 1512

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SEQ ID NO: 49      moltype = DNA length = 1506
FEATURE           Location/Qualifiers
source            1..1506
                  mol_type = other DNA
                  organism = synthetic construct

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SEQUENCE: 49
atggacaaca ccgtctacct gccaccacca tctgttgcta gaggttgtaa caccgacgac 60
tacgttactc ccacctccat ttctaccac gctggttctt ctgattgctg gaccgttggt 120
aaccataact tcagagtttc agctgggtgt ggtaacaagc aggacattcc aaaggtttcc 180
gcctaccagt acagagtggt cagagtccaa ttgccagatc caaacaagtt cggtttgcca 240
gacacctcca tctacaaccc agagactcaa agattgggtt gggcttgtgc ttgtgtcgag 300
attggttaga gtcaaccact tgggtgttgt ttgtctggtc accacttcta caacaagttg 360
gacgacactg aatcttccca cgtgctact tctaactgtt ccgaggatgt cagagacaac 420
gtgtccgttg actacaagca gaccagttg tgtatcttgg gttgtgctcc agctattggt 480
gaacattggg ctaagggaac tgctgtgaag tccagacat tgctcgaagg tgattgccca 540
ccattggagt tgaagaacac tgttttgagg gacggtgaca tggttgacac tggttacggt 600
gctatggact tctctacctt gcaggacact aagtgtaggg tcccattgga catttgtcag 660
tcactctgta agtaccocga ctacttgcaa atgtccgctg atccatacgg tgactctatg 720
ttcttctgct tgagaagaga gcagttgttc gctagacact tctggaacag agctggtact 780
atgggtgaca ctgttccaca gtctctgtac atcaaaagta ctggtatgag agcttcccca 840
ggttcctgtg ttactctccc atctccatcc gggtccatcg ttacttccga ttctcagctg 900
ttcaacaagc catactggtt gcacaaggct cagggtcata acaacgggtt ttgttggcgc 960
aaccagctgt tcgttactgt tgttgacact accagatcca ccaacttgac tatctgtgct 1020
tccactcaat ccccagttcc aggtcaatac gacgctacta agttcaagca gtactccaga 1080
cacgtcgaag agtacgactt gcagttcatc ttccagttgt gcactatcac tttgaccgcc 1140
gacgttatgt cctacatcca ctctatgaac tcctccatct tggaggactg gaactttggt 1200

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-continued

gttcctcctc	ctccaactac	ttccttggtt	gatacctaca	gattcgttca	gtccgtcgct	1260
atcacttgtc	aaaaggatgc	tgctcctgct	gagaacaagg	accatacga	taagctgaag	1320
ttctggaacg	tcgacctgaa	agagaagttc	tccttggact	tggatcagta	cccactgggt	1380
agaaagttct	tgggtccaa	cggtttgaga	agaaagccaa	ctatcggtcc	aagaaagaga	1440
ttctctcctt	cgctacttac	ttcttccaag	ccagctaaga	gagttagagt	cagagccaga	1500
aagtaa						1506

SEQ ID NO: 50 moltype = DNA length = 1503
 FEATURE Location/Qualifiers
 source 1..1503
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 50

atgccatctg	aggctaccgt	ttacttgcca	ccagttccag	tttccaaggt	tgtttccact	60
gacgagtacg	tcaccaggac	caacatctac	taccatgctg	gttctgctag	attgctgact	120
gttggtcacc	cttactactc	cattccaaag	tctgacaacc	ccaagaagat	cgtcgttcca	180
aaggtttccg	gactgcagta	cagagtgttc	agagtcagat	tgccagatcc	aaacaagttc	240
ggtttccccc	acacatcctt	ctacaaccca	gagactcaaa	gattgggttg	ggcctgtggt	300
ggtttgagg	ttggtagagg	tcaaccactt	ggtgttggtg	tttccggtca	tccactgctg	360
aacaagtttg	acgacactga	gaactccaac	agatacgctg	gtggtccagg	tactgacaac	420
agagaatgta	tctccatgga	ctacaagcag	accagttgt	gtttgttggg	ttgcaagcca	480
cctattgggtg	aacattgggg	taagggttct	ccatgttcca	acaacgctat	taccacaggt	540
gattgtccac	ctatggagtt	gaagaactcc	gttattcagg	acggtgacat	ggttgacact	600
ggtttcgggtg	ctatggattt	cactgccttg	caagacacca	agtccaacgt	tccattggac	660
atctgttaact	ccatctgcac	gtaccagac	tacttgaaga	tggttgcgtg	gccatacggg	720
gacaccttgt	tcttctacct	gagaagagaa	cagatgttcg	tcagacactt	cttcaacaga	780
tcoggtactg	ttggtgagtc	cgttccaact	gacctgtaca	tcaaaggttc	tggttccact	840
gctactttgg	ccaactctac	ttacttccca	actccatccg	gttccatggt	tacttccgac	900
gctcagattt	tcaacaagcc	atactggatg	caaagagccc	agggtcacaa	caacgggtatt	960
tggtggggta	accagttggt	cgttaccgtt	gtcgacacta	ccagatccac	caacatgtct	1020
gtttgtgctg	ctatcgctaa	ctccgacacc	actttcaagt	cctccaactt	caaagaatac	1080
ctgagacacg	gtgaagagt	cgacctgcag	ttcattttcc	agctgtgcaa	gatcactttg	1140
tcogccgaca	tcatgacctc	catccactct	atgaaccagg	ccatcttgga	ggattggaac	1200
ttcggtttga	ctactccacc	attggatcc	ttggaggaca	cctacagatt	cgttacttct	1260
caggccatca	cctgtcaaaa	gactgtctca	caaaagccaa	aagaggaccc	attcaaggac	1320
tacgtgttct	gggaagtcac	cttgaaagag	aagttctccg	ctgacctgga	tcagtcccca	1380
ttgggtagaa	agttcttggg	gcaggctggt	tacagagcca	gaccaaaagt	taagggtggt	1440
aagagatctg	ctccttccgc	ttctactact	accccagcca	agagaaagaa	aaactaagaag	1500
ttaa						1503

SEQ ID NO: 51 moltype = DNA length = 1488
 FEATURE Location/Qualifiers
 source 1..1488
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 51

atgccatctg	aggctaccgt	ttacttgcca	ccagttccag	tttccaaggt	tgtttccact	60
gacgagtacg	tttccagaac	ctccatctac	tactacgctg	gttcttccag	attgctggct	120
gttggtcacc	catactcttc	catcaagaac	ccaactaacg	ccaagaagtt	gctggtccca	180
aaggtttctg	gattgcagta	cagagtgttc	agagtcagat	tgccagatcc	aaacaagttc	240
ggtttccccc	acacatcctt	ctacaaccca	gacactcaaa	gattgggttg	ggcctgtggt	300
ggtttgagg	ttggtagagg	tcaaccactt	ggtgttggtg	tctctggtca	cccttctgctg	360
aacaagtttg	acgacactga	gactggtaac	aagtaccagg	gtcaacctgg	tgctgacaac	420
agagaatgtc	tgtctatgga	ctacaagcag	accagttgt	gtttgttggg	ttgtaagcca	480
ccaactgggtg	aacattgggg	taagggtggt	gcttgtaact	atgctgctcc	agctaacgac	540
tgtccaccat	tggtgtgat	taacaccatc	atcgaggacg	gtgacatggt	tgatactggt	600
ttcggttgca	tggtacttca	gaccttgcaa	gctaacaagt	ccgacgtgcc	aattgacatc	660
tggtgttcca	cttgtaagta	cccgcactac	ttgaagatga	cctctgagcc	atacgggtgac	720
ttcctgttct	ttttcttgcg	tcgtgaacag	atgttcgtca	gacacttctt	caacagagcc	780
ggtagctttg	gtgaagctgt	tcagacgac	ctgtacatca	aaggttcttg	tactaccgct	840
tcatttcagt	cctctgtggt	cttcccaact	ccatccggtt	ctatgggtac	ttctgagttc	900
cagctgttca	acaagccata	ctgggtgcaa	agagcccagg	gtcacacaaa	cggtatctgt	960
tggtgttaac	aggttttctg	tactgttgct	gacactacca	gatccaccaa	catgactttg	1020
tgtactcagg	ttacttccga	ctccacctac	aagaacgaga	acttcaaaag	gtacatcagg	1080
cacgtcgaag	agtacacgtg	gcagttcggt	ttccagttgt	gcaagggtcc	tttgaccgct	1140
gagggttatga	cttacatcca	cgctatgaac	cccgcacttt	tggtgagattg	gcaattcggt	1200
ttgactccac	caccatctgc	ttctctgcaa	gacacctaca	gattcgttac	ttcccaggcc	1260
atcacctgtc	aaaagaccgt	tcacacaaa	gagaaagagg	accattggg	caagtacact	1320
ttctgggaag	ttgacctgaa	agagaagttc	tcgctgact	tggatcagtt	cccactgggt	1380
agaaagttct	tggtgcaggc	tggtttgaag	gccaaagcaa	agttgaaaag	agctgctcca	1440
acctccacta	gaacctcttc	cgctaaaaga	aagaagggtca	agaagtaa		1488

SEQ ID NO: 52 moltype = DNA length = 1497
 FEATURE Location/Qualifiers
 source 1..1497
 mol_type = other DNA

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organism = synthetic construct
SEQUENCE: 52
atgtccaacg aggtaccgt ttacttgcca ccagtttccg tttccaaggt tgtttccact 60
gacgagtagc tcaccaggac caacatctac taccatgctg gttcttccag attgctggct 120
gttgggtcatc cttactacgc cattaaagaag caggactcca acaagatcgc cgttccaaag 180
gtttcttggtc tgcagtacag agtggttcaga gtcaagttgc cagatccaaa caagttcgggt 240
ttccccgaca cttctttcta cgaacctgct tctcaaagat tgggttgggc ttgtaccgggt 300
gttagaggttg gtagaggtca accacttggt gttggtatgt ctggtcaccg attgctgaac 360
aagttggagc acactgagaa ctctaacaag tacgtcggta actccggtag tgacaacaga 420
gaatgtatct ccatggacta caagcagacc cagttgtggt tgatcggttg cagaccacct 480
attggtgaac attggggtaa gggtagctca tgtaacgcca accaggttaa ggtcggtaga 540
tgtccaccat tggagttggt gaacactgtc ttgcaggacg gtgacatggt tgatactggt 600
ttcgggtgcta tggacttctc taccttgcaa gctaacaagt ccgacgttcc attggacatc 660
tgttctctca tctgtaagta ccagactac ctgaagatgg tgtctgaacc atacggcgac 720
atgttggttct tctacctgag aagagaacaa atgttcgtca ggcacctgtt caacagagct 780
ggtagctgttg gtgaaactgt tccagccgac ttgtacatca agggtagctac tggtagtttg 840
ccctccacct cttacttccc aactccatct ggttccatgg ttacttccga cgtcagatc 900
ttcaacaagc catactgggt gcaaaagacc cagggtcaca acaacggtag ctgttgggtct 960
aaccagttgt tcgttaccgt tgcgcacct accagatcca ccaacatgct tgtttgttcc 1020
gctgttttcta cctccgactc cacttacaag aacgacaact tcaaaagata cctgagacac 1080
ggtagaaggt acgacctgca gttcattttc cagctgtgca agatcacttt gaccgcccac 1140
gttatgacct acatccacca tatgaaccca tccatcttgg aggaactggaa cttcggtttg 1200
actccaccac catctggaac tttagaggac acctacagat acgtcacttc ccaggctggt 1260
acttgtaaaa agccatctgc tccaaagcca aaggacgacc cattgaagaa ctacactttc 1320
tgggaagtgc acctgaaaga aaagttctcc gctgacttgg atcagttccc actgggtaga 1380
aagttcttgt tgcaggttgg ttggaaggcc agaccaaact tcagattggg taagagagct 1440
gtccagctt ccacttctaa gaagtcctcc accaagagaa gaaaggtcaa gtcttaa 1497

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SEQ ID NO: 53      moltype = DNA length = 1593
FEATURE            Location/Qualifiers
source              1..1593
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 53
atgattatca tcttctgaa gaacgtcaac gtgttcccaa tcttcttgca gatggctttg 60
tggcggtccat ctgactccac tgtttatttg ccaccaccat ccgttgccag agttgtttct 120
actgacgact acgtttccag gacctccatt ttctaccacg ctggttcttc cagattgctg 180
actgttggtg acccatactc cagagtcggt ccaaacgggtg ctggttaaca gcaagctggt 240
ccaaaggttt ccgcctacca gtacagagtt ttcagagttg ctttgccaga tccaaacaag 300
ttcggtttgc cagactctac catctacaac ccagagactc agagattggt ttgggcctgt 360
gttggtagtg aaatcggtag aggtcagcca cttggtattg gttgtctggt tccccattc 420
tacaacaagt tggacgacac tgaatctgct cacgtgctca ctgctgttat cactcaagac 480
gtcagagaca acgtttccgt gcactacaag cagaccaggt tgtgtatctt gggttgtggt 540
ccagctattg gtgagcattg ggctaagggt actttgtgta agccagctca attgcagcca 600
ggtagattgc caccattgga gttgaagaac accatcattg aggcaggtga catggttgac 660
actggttagc gtgctatgga cttctctacc ttgcaggaca ctaagtgtga ggtccattg 720
gacatttgct agtccatctg taagtacccc gactacttgc aaatgtccgc tgatccatac 780
ggtagctcta tgttctctcg cctgagaaga gagcagttgt tcgctagaca cttctggaac 840
agagctgggt ttatgggtga cactgttcca accgacctgt acattaaggg aacttccggt 900
aacatgagag aaaccccagg ttctctgtgt tactctccat ctccatccgg tccatccatc 960
acttccgatt ctacgtctgt caacaagcca tactggttgc acaaggctca gggtcacaa 1020
aacggtagct gttggcacaa ccagctgttc gttactgttg ttgacactac cagatccacc 1080
aacttgactt gttgtgcctc cactcaaaaac ccagttccat ctacttacga ccccaccaag 1140
ttcaagcaat actccagaca cgttgaagag tacgacctgc agtccatctt ccagttgtgc 1200
actatcactt tgaccgccga ggttatgtcc tacatccact ctatgaactc ctccatcttg 1260
gagaactgga acttcggtgt tcctctctct ccaactactt ccttggttga tactacaga 1320
ttcgttcagt ccgttgccgt tacttgtcag aaggacacta ctccaccaga gaagcaagac 1380
ccatacgaca agttgaagtt ctggaccgtc gacctgaaag agaagttctc ctctgacttg 1440
gatcaatacc cactgggtag aaagttcttg gtccaggctg gtttgagaag aaggccaact 1500
atcggtccaa gaaaaagacc agctgcttct acttctactg cttccactgc ttctagacca 1560
gccagagag ttagaatcag atccaagaag taa 1593

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SEQ ID NO: 54      moltype = DNA length = 1572
FEATURE            Location/Qualifiers
source              1..1572
                    mol_type = other DNA
                    organism = synthetic construct

```

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SEQUENCE: 54
atggtcaacg tgttcccaat cttcttgtag atggctttgt ggcgtccatc tgactccact 60
gtttatttgc caccaccatc cgttgccaga gttgtttcta ctgacgacta cgtttccagg 120
acctccattt tctccacgcg tggttcttcc agattgctga ctggttgtaa cccatacttc 180
agagtcgttc caaacgggtg tggttaacaag caagctgttc caaaggtttc cgcctaccag 240
tacagagttt tcagagttgc ttgcccagat ccaaacaagt tcggtttgcc agactctacc 300
atctacaacc cagagactca gagattggtt tgggcctgtg ttggtatgga aatcggtaga 360
ggtagccca ctaggtattg ttgtctggt caccattct acaacaagtt ggacgacact 420
gaatctgtc acgctgtctac tgctgttata actcaagacg tcagagacaa cgtttccgtc 480

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gactacaagc agaccaggt gtgtatcttg ggttggttc cagctattgg tgagcattgg 540
gctaagggtta ctttgtgtta gccagctcaa ttgcagccag gtgattgtcc accattggag 600
ttgaagaaca ccatcattga ggacgggtgac atggttgaca ctggttacgg tgctatggac 660
ttctctacct tgcaggacac taagtgtgag gtcccatgg acatttgtca gtccatctgt 720
aagtaccocg actacttgca aatgtccgct gatccatacg gtgactctat gttctctgc 780
ctgagaagag agcagttggt cgttagacac ttctggaaca gagctgggtg tatgggtgac 840
actgttccaa ccgacctgta cattaaggga acttccgcta acatgagaga aaccccaggt 900
tccgtgtgtt actctccatc tccatccggt tccatcatca ctcccgattc tcagctgttc 960
aacaagccat actggttgca caaggctcag ggtcacaaca acggtatctg ttggcacaac 1020
cagctgttcg ttactgttgt tgacactacc agatccacca acttgacttt gtgtgcctcc 1080
actcaaaacc cagttccatc tacttacgac cccaccaagt tcaagcaata ctccagacac 1140
gttgaagagt acgacctgca gtccatcttc cagttgtgca ctatcacttt gaccgccgag 1200
gttatgtcct acatccactc tatgaactcc tccatcttgg agaactggaa ctccggtgtt 1260
cctcctctcc caactatttc cttggttgat acctacagat tcggttcagtc cgttgccggt 1320
acttgtcaga aggacactac tccaccagag aagcaagacc catacgacaa gttgaagttc 1380
tggaccgtcg accgaaaga gaagttctcc tctgacttgg atcaataccc actgggtaga 1440
aagttcttgg tccagggtgg ttgagaaga aggccaaacta tcggtccaag aaaaagacca 1500
gctgcttcta cttctactgc ttccactgct tctagaccag ccaagagagt tagaatcaga 1560
tccaagaagt aa 1572

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SEQ ID NO: 55      moltype = DNA length = 1557
FEATURE           Location/Qualifiers
source            1..1557
                  mol_type = other DNA
                  organism = synthetic construct

```

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SEQUENCE: 55
atgatcttcc tgcagatggc tttgtggcgt ccatctgact ctactgttta cttgccacca 60
ccatccgttg ccagagtgtg ttctactgac gactacgttt ccaggacctc cattttctac 120
cacgctggtt cctccagatt gctgactgtt ggtaaccatc acttcagagt cgttccaaac 180
gggtgctgga acaagcaagc tgttccaaag gtttccgcct accagtacag agtttccaga 240
gttgctttgc cagatccaaa caagttcggg ttgctgactc ccaccatcta caaccagag 300
actcaagatg tggtttgggc ctgtgtcggg atggaaattg gtagaggtea gccacttggg 360
atcggtttgt ctggtcacc atctctacaac aagtgggacg acactgaatc tgctcacgct 420
gctactgctg ttatcactca agacgtcaga gacaacgttt ccgtcacta caagcagacc 480
cagttgtgta tcttgggttg tgttccagct attggtgagc attgggctaa gggtaacttg 540
tgtaagccag ctcaattgca gccaggtgat tgtccaccat tggagttgaa gaacaccatc 600
attgaggacg gtgactgttg tgacactggg tacggtgcta tggacttctc taccttgca 660
gacactaagt gtgaggtccc attggacatt tgtcagttca tctgtaagta ccccagactc 720
tgcaaatgtc ccgctgac atacggtgac tctatgttct tctgcctgag aagagagcag 780
ttgttcgcta gacacttctg gaacagagct ggtgttatgg gtgacactgt tccaaccgac 840
ctgtacatta agggaaactc cgctaacatg agagaaaccc caggttctctg tgtttactct 900
ccatctccat ccggttccat catcacttcc gattctcagc tgttcaacaa gccactactg 960
ttgcacaagg tcaagggtca caacaacggt atctgttggc acaaccagct gttcgttact 1020
gttgttgaca ctaccagatc caccaacttg actttgtgtg cctccactca aaaccaggt 1080
ccatctactc acgacccac caagttcaag caatactcca gacactgtga agagtacgac 1140
ctgcagttca tctccagtt gtgcactatc actttgaccg ccgaggttat gtccatcac 1200
cactctatga actcctccat ctgggagaa tggaaactcg gtgttctcc tctccaaact 1260
acttctctgg ttgatactca agatctcgtt cagtcctgtg ccgttacttg tcagaaggac 1320
actactccac cagagaagca agaccatac gacaagttga agttctggac cgtcgaacct 1380
aaagagaagt tctctctgca cttggatcaa taccactggt gtagaaagt cttggtccag 1440
gctggtttga gaagaaggcc aactatcggt ccaagaaaaa gaccagctgc ttctacttct 1500
actgcttcca ctgcttctag accagccaag agagttagaa tcagatccaa gaagtaa 1557

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SEQ ID NO: 56      moltype = DNA length = 1530
FEATURE           Location/Qualifiers
source            1..1530
                  mol_type = other DNA
                  organism = synthetic construct

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SEQUENCE: 56
atgccatctg actccacggt ttatttgcca ccaccatccg ttgctagagt tgtttctact 60
gacgactacg tgtccagaac ctccatttcc taccacgctg gttcttccag attgctgact 120
gttggtaacc catacttcag agtcgttcca aacggtgctg gtaacaagca agctgttcca 180
aaggtttccg cctaccagta cagagtttcc agagttgctt tgccagatcc aaacaagttc 240
ggtttgccag actctaccat ctacaaccca gagactcaga gattggtttg ggcctgtgtt 300
ggtatggaaa tcggtagagg tcagccactt ggtattggtt tgtctgggtc cccattctac 360
aacaagttgg acgacactga atctgctcac gctgctactg ctgttatcac tcaagacgtc 420
agagacaacg tttccgtcga ctacaagcag acccagttgt gtatcttggg ttgtgttcca 480
gctattggtg agcattgggc taagggtact ttgtgtaagc cagctcaatt gcagccaggt 540
gattgtccac cattggaggt gaagaacacc atcattgagg acggtgacat ggttgacact 600
ggttacggtg ctatggactt ctctaccttg caggacacta agttgaggtt cccattggac 660
atttgtcagt ccactctgaa gatacccgac tacttgcaaa tgtccgctga tccatcagg 720
gactctatgt tcttctgctc gagaagagag cagttgttcg ctgacactct ctggaacaga 780
gctggtgtta tgggtgacac tgttccaacc gacctgtaca ttaaggggaa ttccgctaac 840
atgagagaaa ccccaggttc ctgtgtttac tctccatctc catccgggtc catcatcact 900
tccgattctc agctgttcaa caagccatac tgggtgcaca aggtcagggg tcacaacaac 960
ggtatctggt ggcaacaaca gctgttcgtt actgttgttg acactaccag atccaccaac 1020

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ttgactttgt	gtgcctccac	tcaaaaccca	gttccatcta	cttacgaccc	caccaagttc	1080
aagcaatact	ccagacacgt	tgaagagtag	gacctgcagt	tcattctcca	gttgtgcact	1140
atcactttga	ccgcccaggt	tatgtcctac	atccactcta	tgaactcctc	catctggag	1200
aatcggaact	tcggtgttcc	tcctcctcca	actacttctc	tggttgatac	ctacagattc	1260
gttcagtcgg	ttgcccgttac	ttgtcagaag	gacactactc	caccagagaa	gcaagaccca	1320
tacgacaagt	tgaagtctcg	gaccgtcgac	ctgaaagaga	agttctcctc	tgacttggat	1380
caatacccac	tgggtagaaa	gttcttggtc	caggctgggt	tgagaagaag	gccaactatc	1440
gggtcaagaa	aaagaccagc	tgcttctact	tctactgctt	ccactgcttc	tagaccagcc	1500
aagagagtta	gaatcagatc	caagaagtaa				1530

SEQ ID NO: 57 moltype = DNA length = 1500
 FEATURE Location/Qualifiers
 source 1..1500
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 57

atgccatctg	aggctacogt	ttacttgcca	ccagttccag	tttccaaggt	tgtttccact	60
gacgagtagc	tttccagaac	ctccatctac	tactacgctg	gttcttccag	attgtcgact	120
gttggtccac	catacttctc	tatcaagaac	acctcttccg	gtaacggtaa	gaaggctctg	180
gtcccaaaag	tttccggatt	gcagtagaca	gtgttcagaa	tcaagctgcc	agatccaaac	240
aagttccggt	tccagacac	atccttctac	aaccagaga	ctcagagatt	ggtttgggct	300
tgtaactggt	tggagatcgg	tagaggtaaa	ccacttggtg	ttggtatttc	tggtcaccct	360
ctgctgaaca	agtttgacga	actgagact	tccaacaagt	acgctggtaa	gccaggtatc	420
gacaacagag	aatgctctgc	tatggactac	aagcagaccc	agttgtgtat	cttgggttgc	480
aagccaccta	ttggtgaaca	ttggggtaag	ggtactccat	gcaacaacaa	ctctggtaac	540
cctggtgatt	gtccaccatt	gcagttgatt	aactccgcca	ttcaggacgg	tgacatgggt	600
gatactgggt	tcggttggtat	ggacttcaac	accttgcaag	cttccaagtc	cgacgtgcca	660
attgacattt	gttcctccgt	ctgtaagtac	cccgactact	tgcaaatggc	ttctgagcca	720
tacggtgact	ccctgttctt	ttcttgcgt	cgtgaacaga	tggtcgtcag	acacttcttc	780
aacagagccg	gtactttggg	tgatccagtt	cctggtgact	tgatcatcca	aggttccaac	840
tcggtaaca	ccgtaactgt	tcaatcttcc	gcttcttccc	caactccatc	cggttctatg	900
gttactttct	agttccagct	gttcaacaag	ccatactggt	tgcaaaagag	ccagggtcac	960
aacaacggta	tctgttgggg	taaccagttg	ttcgttactg	tcgttgacac	taccagatcc	1020
accacataga	cttgtgtgct	cgaggtcaag	aaagagtcca	cctacaagaa	cgagaacttc	1080
aaagaatacc	tgagacacgg	tgaagagttc	gacctgcagt	tcattttcca	gctgtgcaag	1140
atcactttga	ccgcccagct	tatgacctac	atccacaaga	tggaacgtac	cattttggag	1200
gactggcaat	tcggtttgac	tcctcctcct	tctgcttctc	tggaagatac	ctacagattc	1260
gttactttcca	ccgccactac	ctgtcagaag	aacactccac	caaaaggtaa	agaggaccca	1320
ctgaaggatt	acatgttctg	ggaagtgcac	ctgaaagaga	agttctctgc	tgacttggat	1380
caagttccac	tgggttagaaa	gttcttgggt	caagctgggc	tcagggccac	accaagtttg	1440
aaaagacctg	ctcttctcgc	tccaagaact	tccacaaga	agaagaaggt	taagagataa	1500

SEQ ID NO: 58 moltype = DNA length = 1548
 FEATURE Location/Qualifiers
 source 1..1548
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 58

atgatcttgt	tctgtgtcgc	cgacgtcaac	gtgttccaca	ttttcttgca	aatgtccgtc	60
tggtgtccat	ccgaagctac	tggtttattg	ccaccagttc	cagttctcca	gggtgtttct	120
actgacaggt	acgtttccag	gacctccatc	tactactacg	ctggttcttc	tagattgtcg	180
gcccgttgga	accataactt	ctccattaag	tcccacaaca	acaacaagaa	gggtgctggtc	240
ccaaagggtt	ccggattgca	gtacagagtg	ttcagagtca	gattgcccaga	tccaaacaag	300
ttcggtttcc	ccgacacatc	cttctacaac	ccagacactc	aaagattggg	ttgggcctgt	360
gttggtttgg	agattggtag	aggtaacca	cttggtgttg	gtgtttcttg	tcaccatac	420
ctgaacaagt	ttgacgacac	tgagacttcc	aacagatacc	cagctcaacc	aggttccgac	480
aacagagaa	gtctgtctat	ggactacaag	cagacccagt	tggtgtctga	cggttgtaag	540
ccaccaactg	gtgaacattg	gggtaagggt	gttgctgtga	acaacaacgc	tgctgctact	600
gattgtccac	cattggagct	gttcaactcc	attattgagg	acgggtgacat	gggtcgacact	660
ggtttcgggt	gtatggattt	cggtactctg	caggctaaca	agtcagcagt	tccaatcgac	720
atctgttaact	ccactgttaa	gtaccagac	tactgaaga	tggttccoga	gccatacgg	780
gattccttgt	tctttttctc	gcgtcgtgaa	cagatgttct	tcagacactt	cttcaacaga	840
gcccgttaagt	tgggtgaagc	tggtccagac	gacctgtaca	tcaaagggtc	tggttaacacc	900
gctgttattc	agtcctctgc	ttcttctcca	actccatccg	gttctatcgt	tacttctgag	960
tcacagttgt	tcacaagacc	atactgggtg	caaagagccc	agggtcacaa	caacggattt	1020
tggtggggtta	accagctggt	cgttactggt	gttgacacta	ccagatccac	caacatgacc	1080
ttgtgtaccg	aggttaccaa	agagggaacc	tacaagaacg	acaacttcaa	agaatacgtc	1140
aggcacgtcg	aagagtacga	cttgcaattc	gttttccagc	tggtgcaagat	cactttgacc	1200
gctgagatca	tgacctacat	ccacactatg	gactccaaca	tcttgaggga	ctggcaattc	1260
ggtttgactc	caccaccatc	tgcttctctg	caagacacct	acagattcgt	cacttcccag	1320
gctatcacct	gtcaaaagac	tgctccacca	aaagagaaag	aggaccattt	gaacaagatc	1380
actttctggg	aagtcacact	gaaagagaag	ttctccgctg	acttggatca	gttcccactg	1440
ggtagaaagt	tctgttggtc	gtccggtttg	aaggccaagc	caagattgaa	gagatccgct	1500
ccaactacta	gagccccatc	cactaagagg	aagaagggtta	agaagtaa		1548

SEQ ID NO: 59 moltype = DNA length = 1527

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FEATURE                      Location/Qualifiers
source                        1..1527
                               mol_type = other DNA
                               organism = synthetic construct

SEQUENCE: 59
atgggtcaacg tgttccacat cttctgtcag atgtctgttt ggaggccatc cgaggctact 60
gtttacttgc caccagttcc agtttccaag gttgtttcca ctgacgagta cgtttccaga 120
acctccatct actactacgc tggttcttcc agattgctgg ctggttgtaa cccatacttc 180
tccatcaagt ccccaaacaa caacaagaag gtgctgggtcc caaagggttc cggattgcag 240
tacagagtgt tcagagtcag attgccagat ccaacaagat tcggtttccc cgacacatcc 300
ttctacaacc cagacactca aagattgggt tgggcctgtg ttggtttgga gattggtaga 360
ggtaaccacac ttggtgttgg tgtttctggt caccatatac tgaacaagtt tgacgacact 420
gagacttcca acagataccc agctcaacca ggttccgaca acagagaatg tctgtctatg 480
gactacaagc agacccagtt gtgtctgata ggttgaagc caccaactgg tgaacattgg 540
ggtaaggggtg ttgcctgtaa caacaacgct gctgctactg attgtccacc attggagctg 600
ttcaactcca ttattgagga cggtagacatg gtccgacactg gtttcggttg tatggatttc 660
ggtagcttgc aggtcaacaa gtccgacggt ccaatcgaca tctgtaactc cactgttaag 720
taccagactc acctgaagat ggcttctgag ccatacgggt attccctgtt ctttttcttg 780
cgtcgtgaac agatgttctg cagacacttc ttcaacagag cggtaagttt ggggtgaagct 840
gttccagacg acctgtacat caaaggttct ggtaaacacg ctgttattca gtcctctgct 900
ttcttcccaa ctccatccgg ttctatcggt acttctgagt cccagttgtt caacaagcca 960
tactgggtgc aaagagccca gggtcacaac aacggtattt gttggggtta ccagctgttc 1020
gttactgttg ttgacatac cagatccacc aacatgacct tgtgtacga ggttaccaaa 1080
gagggaaacct acaagaacga caacttcaaa gaatacgtca ggcacgtcga agagtacgac 1140
ttgcaattcg ttttccagct gtgcaagatc actttgaccg ctgagatcat gacctacatc 1200
cacactatgg actccaacat cttggaggac tggcaattcg gtttgactcc accaccatct 1260
gttctctctg aagacaccta cagattcgct acttcccagg ctatcacctg tcaaaagact 1320
gtccacccaa aagagaaaaga ggaccattg aacaagtaca ctttctggga agtcaacctg 1380
aaagagaagt tctccgctga cttggatcag ttcccaactg gtagaaagtt cttgtgtcag 1440
tccggtttga aggccaaagc aagattgaag agatccgctc caactactag agcccatc 1500
actaagagga agaaggttaa gaagtaa 1527

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SEQ ID NO: 60                moltype = DNA length = 1512
FEATURE                      Location/Qualifiers
source                        1..1512
                               mol_type = other DNA
                               organism = synthetic construct

SEQUENCE: 60
atgatcttcc tgcagatgtc tgtttggagg ccatctgagg ctactgttta cttgccacca 60
gttccagttt ccaaggttgt ttccactgac gagtacgttt ccagaacctc catctactac 120
tacgtctggtt cttccagatt gctggctgtt ggtaaccatc acttctccat caagtcccca 180
aacaacaaca agaagggtgt ggtcccaaaag gtttccggat tgcagtagac agtgttcaga 240
gtcagattgc cagatccaaa caagttcgtt ttcccgcaca catccttcta caaccagac 300
actcaaagat tggtttgggc ctgtgttgtt ttggagattg gtagagggtc accacttggt 360
gttgggtgtt ctggtcacc atacctgaac aagtttgacg acactgagac ttccaacaga 420
taccagctc aaccaggttc cgacaacaga gaatgtctgt ctatggaacta caagcagacc 480
cagttgtgtc tgatcggttg taagccacca actggtgaac attggggtta ggggtgttgc 540
tgtaacaaca acgctgctgt tactgattgt ccaccattgg agctgttcaa ctccattatt 600
gaggacgggtg acatggctga cactggtttc ggttgtatgg atttcggtac tctgcaggct 660
aacaagtcgc acgttccaat cgacatctgt aactccacct gtaagtacc accactacgt 720
aagatggctt ctgagccata cggtagttcc ctgttctttt tcttgctgctg tgaacagatg 780
ttcgttcagc acttcttcaa cagagccggt aagttgggtg aagctgttcc agacgacctg 840
tacatcaaa gttctgttaa caccgctgtt attcagctct ctgctttctt cccaactcca 900
tccggttcta tcgttacttc tgagtcccag ttgttcaaca agccatactg gttgcaaaaga 960
gcccaagggtc acaacaacgg tatattgttg ggtaaccagc tgttcgttac tgttgttgac 1020
actaccagat ccaccaacat gaccttgtgt accgaggtta ccaaagaggg aacctacaag 1080
aacgacaact tcaagaata cgtcaggcac gtccaagagt acgacttgca attcgttttc 1140
cagctgtgca agatcaactt gaccgctgag atcatgacct acatccacac tatggactcc 1200
aacatcttgg aggactggca attcggtttg actccaccac catctgcttc tctgcaagac 1260
acctacagat tcgtcaacttc ccaggctatc acctgtcaaa agactgctcc accaaaagag 1320
aaagaggacc cattgaacaa gtacactttt tgggaagtca acctgaaaga gaagttctcc 1380
gctgacttgg atcagttccc actgggtaga aagttcttgt tgcagtcagg ttgaaggcc 1440
aagccaagat tgaagagatc cgctccaact actagagccc catccactaa gaggaagaag 1500
gttaagaagt aa 1512

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SEQ ID NO: 61                moltype = DNA length = 1485
FEATURE                      Location/Qualifiers
source                        1..1485
                               mol_type = other DNA
                               organism = synthetic construct

```

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SEQUENCE: 61
atgccatctg aggtacogt ttacttgcca ccagttccag tttccaaggt tgtttccact 60
gacgagtacg tttccagAAC ctccatctac tactacgctg gttcttccag attgctggct 120
gttggttaac catacttctc catcaagtcc ccaaacaaca acaagaaggt gctgggtcca 180
aaggtttccg gattgcagta cagagtgttc agagtccagt tgcagatcc aaacaagttc 240
ggtttccccc acacatcctt ctacaacca gacactcaaa gattggtttg ggcctgtgtt 300

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ggtttgagga ttggttagagg tcaaccactt ggtgttggtg tttctggtea cccatacctg 360
aacaagtttg acgacactga gacttccaac agatacccag ctcaaccagg ttccgacaac 420
agagaatgtc tgtctatgga ctacaagcag acccagttgt gtctgatcgg ttgtaagcca 480
ccaactgggtg aacattggggg taagggtggt gcctgtaaca acaacgctgc tgctactgat 540
tgtccaccaa tggagctggt caactccatt attgaggacg gtgacatggt cgacactggg 600
ttcggttgta tggatttcgg tactctgcag gctaacaagt ccgacgttcc aatcgacatc 660
tgtaactcca cctgtaagta cccagactac ttgaagatgg cttccgagcc atacggtgat 720
tccttgttct ttttctcgcg tcgtgaacag atgttcgtca gacacttctt caacagagcc 780
ggtaagttgg gtgaagctgt tccagacgac ctgtacatca aaggttctgg taacaccgct 840
gttattcagt cctctgcttt ctcccaact ccacccggtt ctatcggtac ttctgagtc 900
cagttgttca acaagccata ctggttgcaa agagcccagg gtcacaacaa cggtatattgt 960
tggggtaacc agctgttcgt tactgttgtt gacactacca gatccaccaa catgacctg 1020
tgtaccgagg ttaccaaaga ggaacactac aagaacgaca actcaaaaga atacgtcagg 1080
cacgtcgaag agtgaactgt gcaatcggt ttccagctgt gcaagatcaa ttgacccgct 1140
gagatcatga cctacatcca cactatggac tccaacatct tggaggactg gcaatcgggt 1200
ttgactccac caccatctgc ttctctgcaa gacacctaca gattcgtcac ttcccaggct 1260
atcacctgtc aaaaactgct tcacacaaaa gagaaagagg acccattgaa caagtacact 1320
ttctgggaag tcaacctgaa agagaagttc tccgctgact tggatcagtt cccactgggt 1380
agaaagttct tgttgacgtc cggtttgaag gccaaagcaa gatgaagag atccgctcca 1440
actactagag ccccatccac taagagggaag aaggtaaga agtaa 1485

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SEQ ID NO: 62      moltype = DNA length = 1497
FEATURE           Location/Qualifiers
source            1..1497
                  mol_type = other DNA
                  organism = synthetic construct

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SEQUENCE: 62
atgcatctg actccaccgt ttatgtccca ccaccaaatc cagtttccaa ggtgtgtgct 60
actgacgct acgtcaagag gaccaatatt ttctaccacg cctcctcacc cagattgctg 120
gctgttggtc atccttacta ctccatcaag aaggtcaaca agaccgtcgt tccaaagggt 180
tccggttacc agtacagagt gtccaagggt gttttgccag atccaaacaa gtccgctttg 240
cctgactctt ccttggtcga cccaactact cagagattgg tttgggcttg taccggtttg 300
gaagttggta gaggtcaacc acttggtggt ggtgtttctg gtcacccatt gctgaacaag 360
taacgacgag ttgaaaactc cggtggttac ggtggttaac caggtcaaga taacagagtc 420
aacgtcggta tggactacaa gcagacccag ttgtgtatgg ttggtgtgct tccaccattg 480
ggtaaacatt ggggtaagggt tactcagtg tccaacactt ccgttcagaa cggtgattgt 540
ccacctttgg agttgatcac ctccgttatt caagacgggtg acatggttga cactggttct 600
gggtctatga acttcgtcga cttgcagact aacaagtcag acgttctctt ggacatctgt 660
gtgtactgtc gtaagtaccc agactacttg caaatggctg ctgatccata cggtgacaga 720
cgtgtcttct acctgagaaa agaacagatg ttccgacag acttcttcaa cagagctgggt 780
actgttggtg aaccagttcc agacgacttg ttggtcaaa ggtgtaacaa cagatcctcc 840
gttgcttctc ctatctacgt tcacactcca tccggttctc ttggttcttc tgaagcccag 900
ttgttcaaca ctgtcgaagt gttgcaaaag gccagggtgc acaacaacgg tatttgtttg 960
ggtaaccact tgttcgttac gtttgtcgac actaccagat ccaccaatat gactttgtgt 1020
gcctccgttt ccaagtcgcg tacttacact aactccgact acaagagta catgagacac 1080
gtcgaagagt tcgacctgca gtctcatttc cagctgtggt ccattacttt gtccgcgag 1140
gttatggcct acatccacac tatgaacca tccgttttgg aggactggaa cttcggtttg 1200
tctccaccac ctaacggtac ttggagggat acctacagat acgttcagtc ccaggctatc 1260
acttgccaaa agccaactcc agagaagag aagcaggacc catacaagga catgtctttc 1320
tgggaagtca acctgaagaa aaagtctctc tcagagctgg atcagttccc actgggtaga 1380
aagttcttgt tgcagtcagg ttacagaggt agaacttccg ctagaactgg tatcaagagg 1440
cctgctgttt ctaagccatc tactgcccc aagagaaagc gtaccaagac taagaag 1497

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SEQ ID NO: 63      moltype = DNA length = 1503
FEATURE           Location/Qualifiers
source            1..1503
                  mol_type = other DNA
                  organism = synthetic construct

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SEQUENCE: 63
atgtggagac cttccgacag cacagtgtac gtgccccccc ctaatccagt gagtaagggtg 60
gtcggccaccg atgcctacgt gacaagaacc aatatcttct atcacgcctc aagcagccgc 120
ctgctggctg ttggccaccc ctacttctct attaaaagag ccaacaagac cgtggtgccc 180
aaggtgtccg ggtaccagta ccgctgtgtt aaagtgggtc tgcccagccc caacaagttc 240
gccctgcccc atagctccct gttcgaccgg acccccaga ggctcgtgtg ggcctgcacc 300
ggccttgagg tggggcgagg ccagcctctg ggggtgggag tcagtggcca ccccttctg 360
aataagtatg acgatgtgga gaactcaggg agcggcggca accctggcca ggacaacaga 420
gtgaacgtgg gcattggacta taagcagact cagctgtgta ttggtgggctg cggccccacc 480
ctggcgagcg attggggcaa gggcaagcag tgcaccaata ccccgctgca ggcggcgac 540
tgtctctctc tggagctgat caccagcgtg atccaggacg gggataggt ggataccggt 600
ttcgagacca tgaatttcgc cgacctgcag acgaacaagt ccgatgtgcc tatcgatatc 660
tcgggaacca cctgcaactc ccccgattac ctgcagatgg ccgcgcgacc ttacggcgat 720
gcctgtttct tcttctctcg gaaggagcag atgttcgccc gtcacttttt caatcgggcc 780
ggcgagggtg gcgagcctgt gcccgatata ctgatcatca agggctccgg gaaccggacc 840
agcgtggggt ccagcatcta tgtgaataca cctccggcca gcctgggtgc ttccgaggcc 900
cagctgttca ataagccata ctgggtccag aaagcccagg gccacaataa cggcatctgc 960
tggggcaacc agttgttcgt gacagtgggt gatacaacca ggtccacca catgacactg 1020

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tgcgccagcg	tgaccacaag	cagcacctac	acaaactccg	actataaaga	gtacatgcgc	1080
cacgtggagg	agtagcacct	gcagttcatt	ttccagctgt	gtagcattac	cctgagcgcc	1140
gaagtgtatg	cctacattca	caccatgaac	cccagcgtgc	tcgaagattg	gaacttcggc	1200
ctctctcccc	ccccaaatgg	cacctggaa	gacacctatc	ggtacgtgca	gagccaggca	1260
attacatgtc	agaagcccac	ccccgagaag	gagaagcctg	accatacaa	gaacctgtca	1320
ttctgggagg	tgaatctgaa	ggagaagttt	agcagcgagc	tggaccagta	cccactgggt	1380
aggaaattcc	tgtctcagtc	cggtatcgg	ggccgcagct	ccatccgcac	tggcgtgaag	1440
aggcccgccg	tgtctaaggc	ctccgcggcc	cccaagcgca	aaagggccaa	gaccaagcgc	1500
tga						1503

SEQ ID NO: 64 moltype = DNA length = 1521
 FEATURE Location/Qualifiers
 source 1..1521
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 64

atgagcctgt	ggctgcccag	cgaggccaca	gtttatctgc	ctccagtgc	tgtgtccaag	60
gtgggtgtcca	cogatgagta	cgtggcccgg	accaacatct	actaccacgc	cggcacatct	120
agactgctgg	ccgtgggaca	cccttacttc	cccatcaaga	agcccaacaa	caacaagatc	180
ctgggtgctta	aggtgtccgg	cctgcagtac	cggtgtgtca	gaatccatct	gctgtacct	240
aacaagttcg	gcttccccga	caccagcttc	tacaaccccg	acacacagag	actcgtgtgg	300
gcctgtgtgg	gcgtcgaaag	tggaagagga	cagcctctcg	gcgtgggaat	ctctggacat	360
ccccctgtga	acaagctgga	cgacaccgag	aatgcctctg	cctatgcgcg	taatgctggc	420
gtggacaacc	gcgagtgcac	cagcatggac	tacaagcaga	cccagctgtg	cctgatcgcc	480
tgcaagcctc	ctattggaga	gcactgggga	aagggcagcc	cctgcacaaa	tgtggctgtg	540
aacctctggg	attgccctcc	tctggaaactg	atcaacacgg	tgatccagga	cggcgacatg	600
gtggatacag	gcttcggcgc	catggacttc	acaacccctg	aggccaacaa	gagcagagtg	660
ccctgggata	tctgcaccag	catctgcaag	taccccgact	acatcaagat	ggtgtccgag	720
ccttacgggg	acagcctgtt	cttttacctg	cggcgcgagc	agatgttcgt	gcggcacctg	780
tttaatatag	cggcgctgtg	ggcgcgagaac	gtgcccgatg	atctgtacat	caaaggcagc	840
ggcagcaccc	ccaatctggc	cagcagcaac	tacttcctca	cacctagcgg	ctccatggtc	900
acaagcgacg	cccagatctt	caacaagccc	tactggctgc	agagagccca	gggccacaa	960
aatggcatct	gttggggcaa	ccagctgttc	gtgacagtgg	tggacacaac	ccggtccacc	1020
aaacatgagtc	tgtgtgccgc	catcagcacc	agcgagacaa	cctacaagaa	caccaacttc	1080
aaagagtacc	gtgcgccagg	cgaggaatac	gacctgcagt	tcatctttca	gctgtgcaag	1140
atcacccctga	ccgcccagct	gatgacctac	atccacagca	tgaacagcac	catcctggaa	1200
gattggaact	tcggactgca	gcctccacct	ggcggcactc	tggaaagatac	ctacagattc	1260
gtgaccagcc	aggccattgc	gtgccagaag	cacacacctc	cagctcctaa	agaggacctt	1320
ctgaagaagt	acaccttctg	ggaagtgaac	ctgaaaagaga	agttcagcgc	cgacctggac	1380
cagtttctct	tgggcagaaa	gttctctgtg	caagctggcc	tgaaggccaa	gcctaagttc	1440
accctgggca	agagaaaagg	cacacctacc	accagcagca	caagcacacc	agccaagcgg	1500
aagaaacgga	agctgtgata	a				1521

SEQ ID NO: 65 moltype = DNA length = 1512
 FEATURE Location/Qualifiers
 source 1..1512
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 65

atgccatccg	acaatacagt	gtacctgccc	cctccctcag	tcgccagagt	ggtgaatacc	60
gacgattacg	tgacccaac	ctctatcttt	taccacggcg	gcagctcaag	actgtgaca	120
gtgggaaatc	cttactttag	ggtgccagca	ggcgcgcgca	acaaacagga	catcccaaaa	180
gtgagcgctc	atcagtaccg	ggtgtttagg	gtgcagctgc	ccgaccctaa	caagttcggg	240
ctgccagata	cctctatcta	taatcccgag	accacagccc	tgggtgtggg	ctgtgcccgc	300
gtggagatcg	gcagaggcca	gccccggga	gtgggctga	gcggccatcc	cttttacaac	360
aaactcgatg	acacagaaag	gtcacacgca	gccacctcca	acgtgagcga	ggacgtaagg	420
gataacgtga	gcgtggatta	taagcagaca	cagctgtgta	tcctgggatg	tgcccccgcc	480
atcgcgcaac	attgggccaa	aggcaccgcc	tgcaaatcca	ggccactgag	ccaggggcac	540
tgtccacccc	tggagctgaa	aaatactgtg	ctggaagatg	gagacatggt	ggataccgga	600
tacggggcca	tggacttttc	cacactgcag	gacaccaagt	gtgaagtgcc	cctgggacatc	660
tgccagagca	tctgtaagta	cccagattac	ctgcagatga	gcgcgagacc	ctacggcgac	720
tctatgtttt	tttgtctgag	acggggagcag	ctgttcgccc	gccacttttg	gaatagagcc	780
ggaactatgg	gcgacacagt	gcctcagctc	ctgtacatca	agggcacccg	catgcccggc	840
tccccaggct	cttgtgtcta	ctccccaa	ccatccggca	gcacgtgtgac	ctctgacagc	900
cagctgttca	acaagccata	ctggctgcac	aaggcccgag	ggcacacaaa	cgggggtgtgc	960
tggcacaccc	agctgttcgt	gacagtgggt	gacaccacta	gatctaccaa	cctgaccatt	1020
tgtgcctcca	cacagagtcc	cgtgcctggc	cagtatgacg	ccaccaagtt	caaacagtac	1080
tctagacacg	tggaaagata	cgacctgcag	ttcatcttcc	agctgtgcac	tatcacccctg	1140
actgcagacg	tgatgtctta	cattcacagc	atgaacagct	ctatccttga	ggattggaac	1200
ttcggcgctc	ctccccacc	taccacaagc	ctgggtggaca	cctacagatt	cgtgcagagc	1260
gtcgctatta	catgtcagaa	ggatgccgca	cccgccgaga	acaaggaccc	atacgacaag	1320
ctcaagttct	ggaacgtgga	tctgaaggag	aaattcagtc	tggatctgga	ccagtatcca	1380
ctggggccga	agtttcttgt	gcaggccggg	ctgcgcagga	agcctaccat	tggccctaga	1440
aagcggagcg	ctccatccgc	cactaccagt	agcaagccag	ccaaaagggt	gagagtgcgg	1500
gcacggaaat	ga					1512

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SEQ ID NO: 66 moltype = DNA length = 1515
 FEATURE Location/Qualifiers
 source 1..1515
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 66

atgtctctgt	ggcggcctag	cgaggctact	gtctacttac	cacctgtccc	agtgtctaaa	60
gttgaagca	cggatgaata	tgtaacacga	accaaactat	attatcacgc	aggcagtgct	120
aggctgctta	cagtaggcca	tccatattat	tccataccta	aatctgacaa	tcctaaaaaa	180
atagttgtac	caaagggtgc	aggattacaa	tatagggtat	ttaggggtcg	ttaccagat	240
ccaaacaaat	ttggatttcc	tgatacatct	ttttataatc	ctgaaactca	acgcttagtt	300
tgggcctgtg	ttggtttaga	ggtaggtcgc	gggcagccat	taggtgtagg	tatttagtgt	360
catccattat	taataaaat	tgatgacact	gaaaactcta	atagatatgc	cgggtggtcct	420
ggcactgata	ataggggaatg	tatatcaatg	gattataaac	aaacacaaact	gtgtttactt	480
ggttgcaaac	cacctattgg	agagcattgg	ggtaaaagta	gtcctgttag	taacaatgct	540
attacccctg	gtgattgtcc	tccattagaa	ttaaaaaatt	cagttataca	agatggggat	600
atgggttgata	caggcttttg	agctatggat	tttactgctt	tacaagacac	taaaagttaat	660
gttccttttg	acattttgta	ttctatttgt	aaatatccag	attatcttaa	aatgggttgc	720
gagccatatg	gcgatacatt	atttttttat	ttacgtaggg	aacaaatggt	tgtaaaggcat	780
ttttttaata	gatacaggcac	gggtggtgaa	tcggtcacct	ctgacttata	tattaaaggc	840
tcgggttcaa	cagctacttt	agctaacagt	acatactttc	ctacacctag	cggctccatg	900
gttacttcag	atgcacaaat	ttttaataaa	ccatattgga	tgcaacgtgc	tcaggggacac	960
aataatggta	tttgttgggg	caatcagtta	tttgttactg	tggtagatag	cacacgtagt	1020
accaatatgt	ctgtttgtgc	tgcaattgca	aacagtgata	ctacatttaa	aagttagtaat	1080
tttaaagagt	tttaagaca	tggtgaggaa	tttgatttac	aatttatatt	tcagttatgc	1140
aaaaataacat	tatctgcaga	cataatgaca	tatatccaca	gtatgaatcc	tgctattttg	1200
gaagattgga	attttgatt	gaccacacct	ccctcaggtt	ctttggaggga	tacctatagg	1260
ttgtcacct	cacaggccat	tacatgtcaa	aaaactgccc	ccaaaaagcc	caagggaagat	1320
ccatttaag	atgtgtatt	ttgggaggtt	aatttaaaag	aaaagttttc	tcgagattta	1380
gatcagtttc	cactgggtcg	caaattttta	ttacaggcag	gatatagggc	acgtcctaaa	1440
tttaaagcag	gtaaacgtag	tgcaccctca	gcactctacca	ctacaccagc	aaaacgtaaa	1500
aaaactaaaa	gttaa					1515

SEQ ID NO: 67 moltype = DNA length = 1488
 FEATURE Location/Qualifiers
 source 1..1488
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 67

atgcctctcg	aagccaccgt	gtatctgcca	ccagtgccag	tgagcaagggt	ggtgagcacc	60
gacgagtacg	tgagccggac	ctctattttac	tattacgcgc	gctccagtag	gctgctggcc	120
gtgggacacc	cctacttcag	catcaagaac	cctacaaatg	ccaagaagct	gctggttccc	180
aaagtctccg	gtctgcagta	cagagtgttc	agggtagagac	tgcccgaacc	taataagttc	240
ggctttccag	atacatcttt	ttacaatcct	gacacacaga	gactgggtgtg	ggcctgcgtg	300
ggcctggaga	ttggacgggg	gcagcctctg	ggagtgggga	tcagcgggaca	ccctctcctg	360
aataagttcg	atgatacaga	gaccggaaac	aaatatccag	gacagcctgg	cgccgataac	420
agagagtgcc	tgagcatgga	ctataagcag	accagctgt	gtctgctggg	ctgcaagcca	480
cccaccggag	agcactgggg	aaagggcgctg	gcctgcacaa	atgccgcccc	agccaatgat	540
tgccccccac	tggagctcag	caacaccatc	attgaggacg	gagacatggt	ggacaccggc	600
tttgatgta	tggacttcaa	gacctgtcag	gcaaataagt	ctgacgtgcc	tatcgacatt	660
tgtggcagca	cctgcaaaat	ccctgactac	ctgaagatga	cttctgaacc	ttacgggtgat	720
agcctgttct	tttttctgcg	gcgggaaacag	atgtttgtga	gacacttttt	taacagggct	780
ggcacccctg	gggaggctgt	gcctgatgac	ctgtacatca	agggcgacgg	gacaaaccgt	840
agcattcagt	cctctgcctt	tttcccacc	ccctccggct	ccatggtgac	cagcagagtc	900
cagctgttta	acaagcata	ctggctgcag	cgcgccagg	gccacaataa	tggaatctgc	960
tggggaaacc	agggtgtcgt	gactgtgggtg	gacaccaccc	gcagcacaaa	tatgaccctg	1020
tgtactcagg	tgacctctga	cagcacctac	aagaatgaga	actttaaggga	gtacatccgg	1080
cacgtggagg	agtacgacct	ccagtttgtg	tttcagctgt	gcaaagtgc	actgaccgct	1140
gagggtgatga	cttacatcca	cgccatgaat	ccagatatcc	tgggaagactg	gcagttcggc	1200
ctgacacctc	ccccatctgc	aagcctgcag	gatacctacc	gcttcgtgac	ctctcaggcc	1260
atcacatgcc	agaagacagt	gccccctaag	gagaaagagg	accactggg	caagtacact	1320
ttctgggagg	tggatctgaa	ggaaaagttt	tctgccgacc	tggatcagtt	tcactggggc	1380
cgaaaattcc	tgctgcaggc	cgggctgaag	gccaaaccca	aactgaagag	ggcagcccc	1440
actagtaccc	gcacctcatc	cgccaaaaga	aagaaagtga	agaagtga		1488

SEQ ID NO: 68 moltype = DNA length = 1509
 FEATURE Location/Qualifiers
 source 1..1509
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 68

atgagttctgt	ggcgggtctaa	cgaggccacc	gtgtatctgc	ctcctgtgtc	tgtgagcaag	60
gtgggtgtcta	ccgacgaata	tgtgacacgc	accaaactct	actatcacgc	cggatctagc	120
aggctgtctgg	ccgtgggaca	cccctactac	gccatcaaaa	agcaggatag	caacaagatc	180
gcggttccaa	aggtgagtg	gtcccagtag	cgcgtcttca	gagtgaagct	gctgatccc	240
aaacagtttg	gatttccga	caccagtttt	tacgaccccg	cctcccagag	gctggtctgg	300

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gcttgcacgg gaggtagggt cggccggggc cagccactcg gggtagggcat cagcggggc 360
cctctgctga ataagctgga cgacacagag aattccaaca agtacgtggg caactcggc 420
actgacaatc gagaatgcat cagcatggac tacaagcaga cacagctgtg tctgatcgga 480
tgccggccac caattgggca aactggggc aagggcacc cctgcaacgc caatcagggtg 540
aaggctggcg agtgcccacc cctggagctg ctgaatacag tctgagagga tggcgacatg 600
gtggatactg ggttcggcgc tatggacttc accacactgc agggccaaca gtccgatgtg 660
cccttgaca tctgcagcag catctgcaag tatcctgatt acctgaagat ggtgagtgag 720
ccttatggcg acatgctgtt cttttacctg cgcaggggag agatgttcgt gaggcatctg 780
ttcaaccggg ccggaaccgt gggcgaaaca gtgccagccg acctgtacat caagggcacc 840
accgggaccc tgcccagtag ctcctatttt cccacaccta gggctccat ggtgacctcc 900
gacgcacaga tctttaataa gccctattgg ctgcagagag cccaggggca caacaatggg 960
atctgctggt ctaaccagct gttcgtgacc gtgggggaca cactagaag tacaatatg 1020
tctgtgtgca gcgcggtgag caccagcgat agcacctaca agaagacaa tttcaaggaa 1080
tacctgccc ccgagagaaga gtagacctg cagtttattt tccagctgtg taaaatcacc 1140
ttgaccggcg acgtgatgac atacatccac agcatgaatc catctatcct ggaggactgg 1200
aactttgggc tgaccccccc tcccagcgga aactcgagg atacatatag gtacgtgaca 1260
tcccaggcag acggagagca gaagcccagc gcccccaagc ccaagacga cctctgaaa 1320
aactatacct tctgggaggt ggacctgaaa gagaagtttt ccgcccagct ggatcagttt 1380
cctctgggaa ggaagtctct gctgcaggct ggctgaaag ctagacctaa ttttagactg 1440
ggcaagaggg ccgccccgcg cagtacctcc aagaagtcca gcacaaaag acggaagggtg 1500
aaaaactga 1509

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SEQ ID NO: 69      moltype = DNA length = 1497
FEATURE           Location/Qualifiers
source            1..1497
                  mol_type = other DNA
                  organism = synthetic construct

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SEQUENCE: 69
atgagcaacg aagccacagt gtacctgcca cctgtgtccg tgtcaaaggt ggtgagcaca 60
gacgagtagc tgactaggac caacatctac taccacggcg gttcctcaag gctgctggcc 120
gtgggacacc catactacgc cattaaagaag caggattcta ataagatcgc tgtccctaaa 180
gtgtctggcc tgcagtaccg cgtgtttaga gtgaaactgc ccgaccccaa taagttcggc 240
tttcccga caagtttcta cgaccctgcc tctcagcgcc tgggtgtggg ttgcaccggc 300
gtggaaagtgg gcaggggcca gccctggga gtgggcattt ctgggcatcc tctgctgaac 360
aagctcgacg atacgaaaaa ctctaacaag tacgtgggca atagcggcac cgacaacaga 420
gagtgcatct ccatggacta taagcagaca cagctgtgcc tgatcgggtg tgcaccccc 480
atcggagagc actgggggaa gggaaacccc tghtaatgcca accaggtgaa ggctggtgag 540
tgtccccctc tggagctgct gaataccgtg ctgcaggacg gcgacatggt ggataccggc 600
tttgagcca tggactttac aacactccag gccacaaggt ccgatgtgcc tctggacatc 660
tgacgacga tctgcaagta cctgattat ctgaagatgg tgtctgagcc ttacggggag 720
atgctgtttt tttatctgag gagagagcag atgtttgtgc gccatctggt caatcgccgc 780
ggcacctgg gagagacagt gcccgccgat ctgtacatca aaggcacaac cgggacccctg 840
ccatctacaa cgtatttccc tacacctagc ggctctatgg tgactagcga cgcgcagatt 900
tttaacaagc cctattggct gcagagggcc cagggacaca ataagggcat ctgctggagc 960
aatcagctgt tctgtagctg ggtggacacc accaggagca caaatatgct agtgtgctcc 1020
gcccgtgagc ccagcgacag cactataag aatgacaact tcaaggagta cctgagacac 1080
ggcgaggagt acgacctgca gttcatcttt cagctgtgta aaatcacact gaccgctgac 1140
gtgatgacat atattcatag tatgaacctc agcatcctgg aagactggaa cttcgggctg 1200
actccaccac cctggaggat acctacaggt atgtgacatc tcaggccgtg 1260
acatgccaga agcctagcgc ccctaagccc aaggatgac ccctgaagaa ttacaccttc 1320
tgaggagtg acctgaagga aaagtccagc gccagacctg accagtttcc cctggggcgg 1380
aagttctccc tgcaggccgg cctgaaggcc agggcacaatt tccggctggg gaagagggct 1440
gcccctgcca gcacctccaa gaagtcttca accaagcgca ggaagtgaa gtcttga 1497

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SEQ ID NO: 70      moltype = DNA length = 1530
FEATURE           Location/Qualifiers
source            1..1530
                  mol_type = other DNA
                  organism = synthetic construct

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```

SEQUENCE: 70
atgccaaagc atagtaccgt gtacctgccc cccccaagtg tggctagggt ggtgagcacc 60
gatgactatg tgagccgcac cagcatcttt taccatgccg gcagctctag gctgctgacc 120
gtggggaacc cctactttcg ggtggtgccc aacggcgcag gcaacaaca ggcggtgcca 180
aaggtagcgc cctatcagta cagggtgttc agagtcgctc tgcccagacc caataagttc 240
gggtgcccag atagcaccat ctacaatccc gaaactcagc ggctgggtgt ggctgtgtg 300
gggatggaaa ttggccgggg ccagccactg gggatcgggc tgagcgccca cctttctat 360
aacaactgg acgacacaga gagcgccacc gccgctaccg ccgttatcac ccaggacgtg 420
cgcgacaacg tcagtgtgga ctataagcag acacagctgt gcattctggg ctgctgtgca 480
gcaattggcg aacactgggc taagggggacc ctgtgcaagc ctgctcagct gcagccaggc 540
gactgcccc ctctggagct gaagaatacc atcatcgaa atggagacat ggtggacacc 600
ggctacgggg ctatggattt ctccacactg caggacacaa atgtgaggt gcctctggat 660
atctgccagt ccatttgtta gtacccgat tacctgcaga tgagcgccga cccctacggg 720
gactcaatgt ttttttcct gagaaggag cagctgttcc ccaggcactt ctggaacagg 780
gccggcgtga tgggggacac agtgcccacc gatctgtaca tcaaaggaa cagcgccaac 840
atgagagaga cccaggcgag ctgcgtgtac agccctagcc caagcgggag catcatcacc 900
tccgattccc agctgttcaa caagccttat tggctgcata aagctcagg acacaataat 960

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ggcatttggt ggcataacca gcttttcgtg actgtggtgg acaccacaag atcaacaaac 1020
ctgaccctgt gtgccaagtac acagaatccc gtgccttcta cctacgaccc aactaagttc 1080
aagcagttaca gcagacacgt ggaggaatac gatctgcagt tcatcttcca gctgtgtaca 1140
atcacactga ccgcccaggt gatgagttac atccattcta tgaattcaag catcctggag 1200
aactgggaatt tcggcgtgcc accccctccc accacaagcc tgggtggatac ctatagattc 1260
gtgcagagcg tggccgtgac ctgtcagaag gacacaaccc ctcccagaaa gcaggatcca 1320
tatgacaagc tgaatttttg gaccgtggac ctgaaggaga aatttagctc cgacctggac 1380
cagtatcctc tggggagaaa gtttctggtg caggccggcc tgaggagacg gcccaccatt 1440
ggaccaagaa agaggccagc tgcattccaca tcaaccgcct ctaccgcttc tcggcctgcc 1500
aagaggggtgc ggatcaggtc caagaagtga 1530

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SEQ ID NO: 71      moltype = DNA length = 1500
FEATURE
source            Location/Qualifiers
                  1..1500
                  mol_type = other DNA
                  organism = synthetic construct

```

```

SEQUENCE: 71
atgcctcccg aggccaccgt gtacctgcca cctgtgcccg tgagcaaggt ggtgagcacc 60
gacgagtagc tgagcaggac cagcatctac tactacgccc ggagtccccg cctgctgaca 120
gtggggcacc catatttttc aattagaat accctcctctg gcaacggcaa gaaagtgtctg 180
gtgcccgaag tcagcggcct gcagtagccg gtttttcgaa tcaagctgcc tgaccccaac 240
aagtttggtc tcccagacac ctcttctac aaccagaga cccagagact ggtgtgggct 300
tgcacaggcc tggagattgg caggggtcag ccacttgggg tggggatcag cgccaccct 360
ctgctgaaca agtttgacga caccgaaca tccaataagt acgctgggaa gcccgccatc 420
gacaaccggg agtgctgtgc catggattac aaacagacac agctgtgcat cctcggtgc 480
aagctcccaa tcggcgagca ctggggaaa gggacaccct gcaacaacaa cagcggcaat 540
cctggagact gccctccgct gcagctgac aacagcgtga ttcaggacgg ggacatgggtg 600
gataccgggt tcggtgtgat ggactttaac accctgcagg caagcaagag cgatgtgccg 660
attgacatct gtgttagcgt gtgtaagtag ccagactacc tgcagatggc ctccgagccc 720
tatggggact ctctgttctt ctttctgaga agggagcaga tgttcgtgag acacttcttt 780
aaaccgcccg gcaccctggg cgatccagt cccggcgatc tgtacattca ggggagcaac 840
agcggcaata cggccaccgt gcagtcacgc gcttttttct ctaccctag cggtcccatg 900
gtgacctctg agtctcagct gttcaataag ccaactgggc tgcagcgggc ccagggacac 960
aacaacggca tttgctgggg caaccagctg tttgtgaccg tgggtggacac taccgctct 1020
accaacatga cctgtgcgc cgaagtgaag aaggaatcca cctataaaaa cgagaacttc 1080
aaggaatata tttagacatg ggaggagttt gacctgcagt ttattttcca gctgtgcaag 1140
attacactga cagccgatgt gatgacctac atccacaaga tggatgccac catcctggaa 1200
gactggcagt ttggcctgac tccccccctc agcgccctctc tgggaagatac ctaccgcttc 1260
gtgactagca cagccatcac ttgccagaag aacacccctc ctaaaaggcaa ggaggacccc 1320
ctgaaagact acatgttctg ggaagtggat ctgaaggaga agttcagcgc cgacctggat 1380
cagttcccac tgggcagaaa gtttctgctg caggctgggc tgcaggccag gcccagctg 1440
aaaagacctg ctagcagcgc cctctgcacc tcaaccaaga agaaaaaggt gaagagatga 1500

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SEQ ID NO: 72      moltype = length =
SEQUENCE: 72
000

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SEQ ID NO: 73      moltype = DNA length = 1503
FEATURE
source            Location/Qualifiers
                  1..1503
                  mol_type = other DNA
                  organism = synthetic construct

```

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SEQUENCE: 73
atgtggcggc ctagcgacag cacagtatat gtgcctctct ctaaccctgt atccaaagt 60
gttgccacgg atgcttatgt tactcgacc aacatatttt atcatgccag cagtttctaga 120
cttcttgacg tgggacatcc ttatttttcc ataaaaacggg ctaacaaaaac tgttgtgcca 180
aaggtgtcag gatataata cagggtattt aaggtggtgt taccagatcc taacaaattt 240
gcattgcctg actcgtctct tttcgatccc acaacacaaac gtttagtatg ggcatgcaca 300
ggcctagagg tgggcagggg acagccatta ggtgtgggtg taagtggaca tcctttccta 360
aataaatatg atgatgttga aaattcaggg agtgggtgta accctggaca ggataacagg 420
gttaatgtag gtatggatta taacaaaaca caattatgca tggttggatg tgccccccct 480
ttgggcgagc attggggtaa aggtaaacag tgtactaata cacctgtaca ggctggtgac 540
tgcccgccct tagaacttat taccagtgtt atacaggatg gcgatatggt tgacacaggc 600
tttggtgcta tgaattttgc tgatttgcag accaataaat cagatgttcc tattgacata 660
tgtggcacta catgtataata tcagatttat ttacaaatgg ctgcagaccc atatggtgat 720
agattatttt ttttctacg gaaggaacaa atgttttgcca gacatttttt taacagggct 780
ggcgagggtg gggaacctgt gcctgataca cttataatta agggtagtgg aaatcgcacg 840
tctgtaggga gtatgatata tgttaacacc ccgagcggct ctttgggtgc ctctgaggca 900
caattgttta ataagccata ttggctacaa aaagcccagg gacataacaa tggatattgt 960
tgggttaac cactgtttgt tactgtggta gataccacac gcagtaccaa catgacatta 1020
tgtgcatccg taactacatc ttccacatac accaattctg attataaaga gtacatgcgt 1080
catgtggaag agtatgattt acaatttatt tttcaattat gtagcattac attgtctgct 1140
gaagtaatgg cctatattca cacaatgaat ccctctgttt tggaaagactg gaactttggg 1200
ttatcgctc ccccaaatgg tacattagaa gatacctata ggtatgtgca gtcacaggcc 1260
attacctgtc aaaagcccac tctgaaaaag atccctataa gaaccttagt 1320
ttttgggagg ttaattttaa agaaaagtgt tctagtgaat tggatcagta tcctttggga 1380

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cgcaagtttt tgttacaaag tggatatagg ggacggctct ctattcgtac aggtgttaag 1440
cgccctgctg tttccaaagc ctctgctgcc cctaaacgta agcgcgccaa aactaaaagg 1500
taa                                                1503

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SEQ ID NO: 74      moltype = DNA length = 1518
FEATURE            Location/Qualifiers
source              1..1518
                    mol_type = other DNA
                    organism = synthetic construct

```

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SEQUENCE: 74
atgtctcttt ggctgcctag tgaggccact gtctacttgc ctctgtccc agtatctaag 60
gttgaagca  cggaatgaata tgttgcacgc acaaacatat attatcatgc aggaacatcc 120
agactacttg cagttggaca tccctatttt cctattaaaa aacctaacaa taacaaaata 180
ttagttccta aagtatcagg attacaatac aggggtattta gaatacattt acctgacccc 240
aataagtttg gttttcctga cacctcattt tataatccag atacacagcg gctggtttgg 300
gcctgtgtag gtgttgaggt aggtcgtggg cagccattag gtgtgggcat tagtgccat 360
cctttattaa ataaattgga tgacacagaa aatgctagtg cttatgcagc aaatgcaggt 420
gtggataata gagaatgtat atctatggat tacaacaaa cacaattgtg tttaatcggg 480
tgcaaacacac ctatagggga acactggggc aaaggatccc catgtaccaa tgttcagta 540
aatccagggt attgtccacc attagagtta ataaacacag ttattcagga tggatgatg 600
gttgatactg gctttgtgtg tatggacttt actacattac aggttaacaa aagtgaagtt 660
ccactggata tttgtacatc tatttgcaaa tatccagatt atattaaaat ggtgtcagaa 720
ccatatggcg acagcttatt tttttattta cgaagggaac aaatgtttgt tagacattta 780
tttaataggg ctggtgtgtg tgggtgaaat gtaccagacg atttatacat taaaggctct 840
gggtctactg caaatttagc catgtcaaat tatttccta cacctagtgg ttctatgggt 900
acctctgatg ccaaatattt caataaacct tattggttac aacgagcaca gggccacaat 960
aatggcattt gttggggtaa ccaactattt gttactgttg ttgatactac acgcagtaca 1020
aatatgtcat tatgtgtgct catatctact tcagaaacta catataaaaa tactaacttt 1080
aaggagtagc tacgacatgg ggaggaatat gatttacagt ttatttttca actgtgcaaa 1140
ataaccttaa ctgcagacgt tatgacatac atacattcta tgaattccac tattttggag 1200
gactggaatt ttggtctaca acccccccca ggaggcacac tagaagatag ttatagggtt 1260
gtaacatccc aggaatttgc ttgtcaaaaa catcacctc cagcacctaa agaagatccc 1320
cttaaaaaat acacttttgg ggaagtaaat ttaaaaggaa agttttctgc agacctagat 1380
cagtttcctt taggacgcaa atttttacta caagcaggat tgaaggccaa accaaaattt 1440
acattaggaa aacgaaaagc tacaccacc accctcatcta cctctacaac tgctaaacgc 1500
aaaaaacgta agctgtaa                                     1518

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SEQ ID NO: 75      moltype = DNA length = 1512
FEATURE            Location/Qualifiers
source              1..1512
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 75
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gtttctgcat accaatatag agtatttagg gtgcagttac ctgacccaaa taaatttggg 240
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ttaaagtttt ggaatgtgga tttaaaggaa aagttttctt tagacttaga tcaatatccc 1380
cttgagcgtg aatttttggg tcaggctgga ttgcgtgcga agccaccat aggccctcgc 1440
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SEQ ID NO: 76      moltype = DNA length = 1515
FEATURE            Location/Qualifiers
source              1..1515
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 76

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aggctgtcta cagtaggccca tccatattat tccataccata aatctgacaa tcctaaaaaa 180
atagttgtac caaagggtgc aggattacaa tatagggtat ttaggggttcg tttaccagat 240
ccaaacaaat ttggatttcc tgatacatct ttttataatc ctgaaactca acgcttagtt 300
tgggcctgtg ttggtttaga ggtagggtgc gggcagccat taggtgtagg tattagtgtg 360
catccattat taaataaatt tgatgacact gaaaactcta atagatatgc cgggtgtcct 420
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ggttgcaaac cactatttgg agagcattgg ggtaaaagta gtccctgttag taacaatgct 540
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gttccttttg acatttgtaa ttctatttgt aaatatccag attatcttaa aatgggtgtc 720
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1515

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SEQ ID NO: 77      moltype = DNA length = 1488
FEATURE           Location/Qualifiers
source            1..1488
                  mol_type = other DNA
                  organism = synthetic construct

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SEQUENCE: 77
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gttgccatc catatttttc tattaaaaat cctactaacg ctaaaaaatt attggtaccc 180
aaagtatcag gcttgcaata taggggtttt aggggtccgt taccagatcc taataaattt 240
ggatttctcg acacctcctt ttataacctt gatacacaac gattagtagt ggcatgtgta 300
ggccttgaaa taggtagagg gcagccatta ggcgttgcca taagtgtgca tcctttatta 360
aacaattttg atgacactga aaccggtaac aagtatccgt gacaaccggg tgctgataat 420
agggatgtt tatccatgga ttataaacaa acacagttat gtttacttgg atgtaagcct 480
ccaacagggg aacattgggg taaaggtgtt gctgtacta atgcagcacc tgccaatgat 540
tgtccacctt tagaacctat aaatactatt attgaggtg gtgataggt ggacacagga 600
tttggttgca tggattttta aacattgcag gctaataaaa gtgatgttcc tattgatatt 660
tgtggcagta catgcaataa tccagattat ttaaaaatga ctagtggacc ttatgggtgat 720
agtttatttt tctttcttcg acgtgaacaa atgtttgtta gacacttttt taatagggtc 780
ggtacattag gagaggctgt tcccgatgac ctgtacatta aagggttcagg aactactgcc 840
tctattcaaa gcagtgtctt ttctccact cctagtggat caatgggtac ttccgaatct 900
cagttattta ataagcataa ttggctacaa cgtgcacaag gtcataataa tggattttgt 960
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catgttgaag aatatgattt acagtttgtt ttccaactat gcaaaagtac cttaactgca 1140
gaagttatga catatattca tgcetatgat ccagatatat tagaagattg gcaatttgtt 1200
ttaacacctc ctccatctgc tagtttacag gatacctata ggtttgttac ctctcaggct 1260
attacgtgtc aaaaaacagt acctccaaag gaaaaggaaag accccttagg taaatataca 1320
ttttgggaag tggattttaa ggaataattt tcagcagatt tagatcagtt tcctttggga 1380
cgcaagtttt tattacaggc aggtctttaa gcaaaaccta aacttaaacg tgcagcccc 1440
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SEQ ID NO: 78      moltype = DNA length = 1509
FEATURE           Location/Qualifiers
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                  mol_type = other DNA
                  organism = synthetic construct

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SEQUENCE: 78
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aggctattag ctgtgggtca cccatactat gctattaaaa aacaagattc taataaaaata 180
gcagtaccca aggtatctgg tttgcaatac agagtattta gagtaaaatt accagatcct 240
aataagtttg gatttccaga cacatcattt tatgatccgt cctcccagcg tttagttttg 300
gctgtgacag gagttgaaat aggtcgcggg cagccattag gtgtaggat tagtggtcat 360
cttttattaa ataaattgga tgatactgaa aattctaata aatatgttgg taactctggt 420
acagataaca gggaaatgcat ttctatggat tataaacaaa cacaattgtg tttaataggt 480
tgtaggcctc ctataggtga acattgggga aaaggcacac cttgtaatgc taaccaggta 540
aaagcaggag aatgtcctcc ttggaggtta ctaaacactg tactacaaga cggggacatg 600
gtagacacag gatttggtgc aatggatttt actacattac aagctaataa aagtgtatgt 660

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ccccatagata tatgcagttc catttgcaaa tatcctgatt atctaaaaat ggtttctgag 720
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ttaatatagg ctggaactgt aggtgaaaca gtacctgcag acctatatat taagggtacc 840
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gatgcacaaa tatttaataa accatatagg ttgcaacgtg cacaaggcca taataatggt 960
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tatttaaggc atgggtgaaga atagtattta cagtttattt ttcagttatg taaaataaca 1140
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SEQ ID NO: 79      moltype = DNA length = 1497
FEATURE           Location/Qualifiers
source            1..1497
                  mol_type = other DNA
                  organism = synthetic construct

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SEQUENCE: 79
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gtgggtccacc cactactatgc tattaaaaaa caagattcta ataaaatagc agtaccacag 180
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tgaggaggtg atttaaaagg aaagtgttct gcagacttag atcaatttcc gttgggcccgt 1380
aaatttttgt tacaagcagg actaaaggcc aggcctaatt ttagattagg caagcgtgca 1440
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SEQ ID NO: 80      moltype = DNA length = 1530
FEATURE           Location/Qualifiers
source            1..1530
                  mol_type = other DNA
                  organism = synthetic construct

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SEQUENCE: 80
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aaggatccg catatcagta taggggtgtt agagtacgtt taccgatcc taataaattt 240
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ggacctcgta	agcgtcctgc	tgcttccacg	tctactgcat	ctactgcatc	taggcctgcc	1500
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SEQ ID NO: 81 moltype = DNA length = 1500
 FEATURE Location/Qualifiers
 source 1..1500
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 81

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gtaggacatc	cctatttttc	tattaaaaac	accagtagtg	gtaatggtaa	aaaagtttta	180
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SEQ ID NO: 82 moltype = DNA length = 1485
 FEATURE Location/Qualifiers
 source 1..1485
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 82

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gttggcaatc	catatttttc	catcaaaagt	cccaataaca	ataaaaaagt	attagttccc	180
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tgcactgaag	taactaagga	aggtacatat	aaaaatgata	attttaagga	atatgtacgt	1080
catgttgaa	aatgtgactt	acagtttgtt	tttcagcttt	gcaaaattac	actaactgca	1140
gagataatga	catatataca	tactatggat	tccaatattt	tggaggactg	gcaatttgggt	1200
ttaacacctc	ctccgtctgc	cagttttacag	gacacatata	gatttgttac	ctcccaggct	1260
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ttttgggagg	tttaactaaa	ggaaaagttt	tctgcagatc	tagatcagtt	tcctttggga	1380
cgaaagtttt	tattacaatc	aggccttaaa	gcaaagccca	gactaaaacg	ttcggccctc	1440
actacccgtg	caccatccac	caaacgcaaa	aagggttaaaa	aataa		1485

1. A truncated recombinant human papillomavirus (HPV) L1 protein with at least 2 amino acids deleted from positions 2 to 66, or positions corresponding thereto, of the N-terminus.

2. The protein according to claim 1, wherein the HPV L1 protein has a deletion of amino acids at positions 2-3, 2-5, 2-7, 2-10, 2-15, 2-17, 2-20, 2-22, 2-31, or 2-66, or positions corresponding thereto, of the N-terminus.

3. The protein according to claim 1, wherein the HPV is selected from HPV serotypes 6, 11, 16, 18, 31, 33, 35, 45, 52, and 58.

4. The protein according to claim 3,

wherein the HPV6 L1 protein has a deletion of amino acids at positions 2-3, or positions corresponding thereto, of the N-terminus relative to SEQ ID NO: 1;

wherein the HPV11 L1 protein has a deletion of amino acids at positions 2-3, or positions corresponding thereto, of the N-terminus relative to SEQ ID NO: 10;

wherein the HPV16 L1 protein has a deletion of amino acids at positions 2-5, 2-10, 2-15, or 2-20, or positions corresponding thereto, of the N-terminus relative to SEQ ID NO: 2;

wherein the HPV18 L1 protein has a deletion of amino acids at positions 2-5, 2-7, or 2-66, or positions corresponding thereto, of the N-terminus relative to SEQ ID NO: 3;

wherein the HPV31, HPV33, or HPV35 L1 protein has a deletion of amino acids at positions 2-5, or positions corresponding thereto, of the N-terminus relative to SEQ ID NOs: 4, 5, or 6;

wherein the HPV45 or HPV58 L1 protein has a deletion of amino acids at positions 2-10, 2-17, 2-22, or 2-31, or positions corresponding thereto, of the N-terminus relative to SEQ ID NOs: 7 or 9; or

wherein the HPV52 L1 protein has a deletion of amino acids at positions 2-31, or positions corresponding thereto, of the N-terminus relative to SEQ ID NO: 8.

5. The protein according to claim 1, wherein the HPV L1 protein comprises or consists of an amino acid sequence selected from one of SEQ ID NOs: 21-41, or a variant having at least about 90%, or at least about 95%, or at least about 98%, at least about 99% identity to one of SEQ ID NOs: 21-41.

6. A recombinant polynucleotide encoding the protein according to claim 1.

7. The polynucleotide of claim 6, comprising or consisting of a nucleotide sequence selected from one of SEQ ID NOs: 42-62, or a variant having at least about 60%, at least about 70%, at least about 80%, at least about 90%, or at least about 95%, or at least about 98%, at least about 99% identity to one of SEQ ID NOs: 42-62.

8. A recombinant vector comprising the polynucleotide according to claim 6.

9. An isolated cell comprising the recombinant vector according to claim 8.

10. A composition comprising the protein according to claim 1.

11. A recombinant human papillomavirus (HPV) virus-like particle (VLP) comprising the protein of claim 1.

12. An immunogenic composition comprising the protein, or a virus-like particle (VLP) made therefrom, according to claim 1, and one or more pharmaceutically acceptable carriers, excipients and/or adjuvants.

13. The composition of claim 12, wherein the composition comprises serotypes HPV 6, 11, 18, 16, 31, 33, 35, 45, 52, and 58 of the L1 protein.

14. The composition of claim 13, wherein the composition comprises serotypes HPV 6 at 30 µg/human dose, HPV 11 and 18 at 40 µg/human dose, HPV 16 at 60 µg/human dose, and HPV 31, 33, 35, 45, 52, and 58 at 20 µg/human dose of the L1 protein.

15. The immunogenic composition of claim 12, wherein the adjuvant is aluminum hydroxide adjuvant.

16. The immunogenic composition of claim 14, wherein the immunogenic composition is formulated for the prevention of HPV infection or a disease associated with an HPV infection.

17. A method for inducing an immune response to HPV in an individual, comprising administering the immunogenic composition according to claim 14 to an individual.

18. A method for preventing an HPV infection or a disease associated with an HPV infection, comprising administering the immunogenic composition of claim 14 to a subject in need thereof.

19. The method of claim 18, wherein the administration occurs twice at an interval of two weeks.

20. A method for producing the HPV L1 protein according to claim 1, comprising expressing a polynucleotide encoding the HPV L1 protein in a yeast cell.

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